

Biochemistry

retake review notes

Foundations

Biochemical foundations · Water, acids, bases and buffers

Proteins

Amino acids · Protein structure and folding · Hemoglobin and globular proteins · Collagen, elastin and fibrous proteins

Enzymes and energy

Enzymes, kinetics and regulation · Bioenergetics and oxidative phosphorylation

carbohydrates

Carbohydrate structure and stereochemistry · Glycoconjugates

Lipids

Dietary lipids and absorption · Fatty acids, triacylglycerols and ketones · Complex lipids and eicosanoids · Cholesterol, lipoproteins and steroids

Metabolic integration

Insulin and glucagon · Fed-fast integration · Diabetes mellitus · Obesity and energy balance

Nutrition and vitamins

Nutrition · Vitamins

Molecular biology

DNA structure, replication and repair · RNA, transcription and processing · Translation · Gene regulation · Biotechnology

Laboratory

Laboratory principles and tests

coral headings, numbered key points, things to learn

blue clinical links

grey box a past paper asked this, with the corrected reasoning



Test yourself: three practice MCQs per topic, answers upside down

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Biochemical foundations

Teacher slides and laboratory material · teacher slides

What biochemistry explains in medicine

- Biochemistry studies the chemistry of living systems.
- Molecular abnormalities can explain disease mechanisms and laboratory findings.
- Mechanistic understanding supports diagnosis, prevention, and rational treatment.

→ A pathogenic enzyme variant can change a metabolite concentration and create a recognizable disease pattern.

Elements and major biomolecule classes

- CHNOPS denotes carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur.
- Carbohydrates commonly provide fuel and participate in structure and recognition.
- Lipids store energy, form membranes, insulate, and generate signaling molecules.
- Proteins act as enzymes, receptors, transporters, antibodies, and structural elements.
- Nucleic acids store and express genetic information.

→ Different biomolecule classes require different laboratory measurements and therapeutic approaches.

central dogma and genotype-to-phenotype reasoning

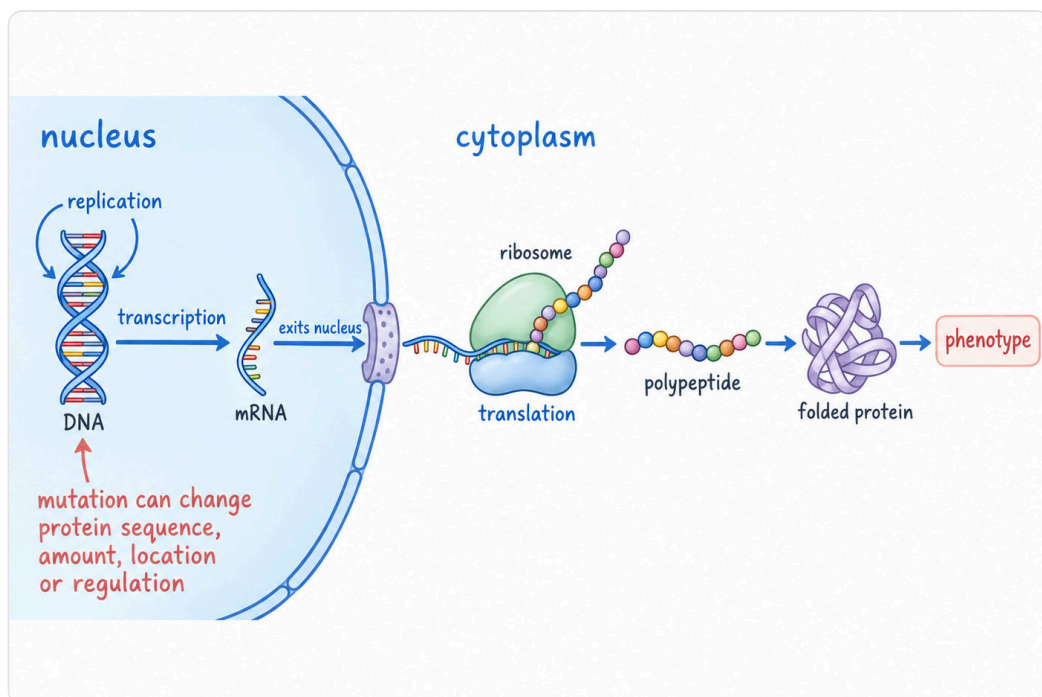


fig. 1 DNA information is transcribed into mRNA, translated, and expressed as protein and phenotype.

Anabolism, catabolism, and ATP coupling

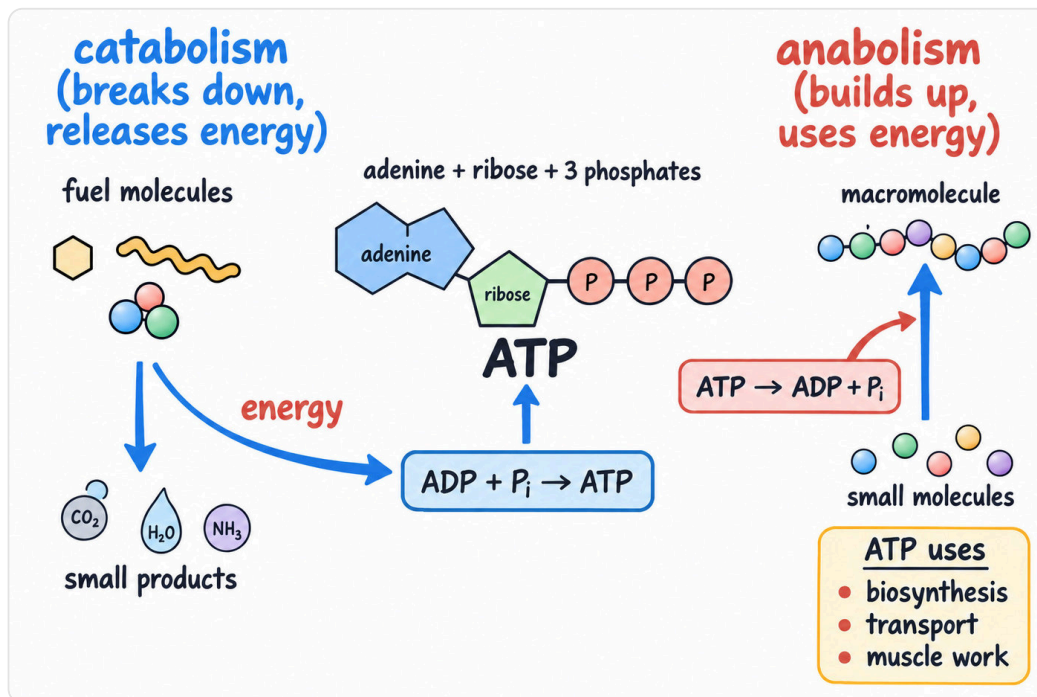


fig. 2 ATP couples energy released by catabolism to biosynthesis, transport, and muscle work.

Functional groups determine biochemical behavior

- Hydroxyl groups support polarity and hydrogen bonding.
- Carboxyl groups can donate protons; amino groups can accept protons.
- Carbonyl groups occur in aldehydes and ketones.
- Phosphate groups contribute negative charge and participate in energy and signal transfer.
- Two oxidized cysteine sulfhydryls can form a disulfide bond.

→ Ionization of functional groups affects drug solubility, membrane passage, and protein binding.

covalent and noncovalent forces in biomolecules

- Ionic interactions are attractions between opposite charges.
- Polar covalent bonds share electrons unequally; nonpolar covalent bonds share them more evenly.
- Hydrogen bonds are individually weak but collectively important.
- Disulfide bonds are covalent links between cysteine residues.
- The hydrophobic effect drives nonpolar groups away from water.

→ Heat, extreme pH, and detergents can disrupt noncovalent forces and denature proteins.

past papers asked

question

Which substance is produced in every dehydration synthesis reaction?

answer and why

Water is the best answer. Condensation joins monomers while releasing water.

question

answer and why

Formic acid is also called

Methanoic acid is the best answer. The IUPAC name for one-carbon formic acid is methanoic acid.

The general formula of amines is

R-NH₂ is the best answer. A primary amine has the general form R-NH₂.

If you replace the oxygen of the hydroxyl group with sulfur in R-OH, which functional group will you get?

Thiol is the best answer. Replacing O in an alcohol with S gives a thiol, R-SH.

Test yourself

1. Which set contains the six elements emphasized as CHNOPS?

- A. Calcium, hydrogen, neon, oxygen, potassium, and sodium
- B. Carbon, helium, nitrogen, osmium, phosphorus, and selenium
- C. Carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur
- D. Copper, hydrogen, nickel, oxygen, potassium, and sulfur

2. ATP hydrolysis is linked to pumping Na⁺ against its electrochemical gradient. What biochemical principle makes this possible?

- A. The favorable and unfavorable processes are mechanistically coupled
- B. Each component process must independently release free energy
- C. Each process can occur only at its own equilibrium
- D. ATP reverses the ion gradient without a coupled protein cycle

3. What makes a covalent bond polar?

- A. Complete loss of both electrons
- B. Absence of nuclei
- C. Unequal electron sharing caused by an electronegativity difference
- D. Equal attraction by identical atoms

Answers: 1 C · 2 A · 3 C

water, acids, bases and buffers

Teacher slides and laboratory material · teacher slides

calculation toolkit

Relationship

use / trap

$\text{pH} = -\log_{10}[\text{H}^+]$	One pH unit is a tenfold change in hydrogen-ion activity (concentration approximation in dilute solutions).
$\text{pH} = \text{pK}_a + \log_{10}([\text{A}^-]/[\text{HA}])$	At equal conjugate base and weak acid, $\text{pH} = \text{pK}_a$. Use concentrations after any strong acid/base neutralization.
Buffering range $\approx \text{pK}_a \pm 1$	A useful rule of thumb, not an abrupt cutoff. Higher total concentration increases buffer capacity at the same ratio.
0.070 M lactate / 0.080 M lactic acid; pK_a 3.74	$3.74 + \log_{10}(0.875) = 3.68$. Concentration ratio is base/acid, not acid/base.
pI for an amino acid	Average the two pK_a values that bracket the electrically neutral species. Do not blindly average all pK_a values.

Body-water compartments and ionic composition

- About two thirds of total body water is intracellular.
- ECF comprises plasma, interstitial fluid, and small transcellular compartments.
- Sodium is the major ECF cation; potassium is the major ICF cation.
- Solute distribution and membrane permeability govern water movement.
- Fluid therapy changes ECF first and then redistributes according to tonicity and permeability.

water polarity, hydrogen bonding, and solubility

- Water is a dipole with partial negative charge on oxygen and partial positive charge on hydrogen.
- Hydrogen bonding explains much of water's solvent behavior.
- Hydrophilic solutes interact favorably with water; hydrophobic solutes do not.
- Amphipathic molecules contain both polar and nonpolar regions.
- Amphipathic bile salts emulsify dietary lipids.

pH, pOH, and logarithmic concentration reasoning

- $\text{pH} = -\log_{10}[\text{H}^+]$.
- At 25°C, $\text{pH} + \text{pOH} = 14$.
- A fall of one pH unit means a tenfold increase in $[\text{H}^+]$.
- Acidemia and alkalemia refer to measured blood pH, not the underlying process.
- Arterial pH below the reference range is acidemia; above it is alkalemia.

Acids, bases, conjugate pairs, K_a , and pK_a

- Acid and conjugate base differ by one proton.
- K_a measures the extent of acid dissociation.
- $pK_a = -\log_{10} K_a$.
- Lower pK_a means stronger acid.
- Weak does not mean dilute; it describes incomplete dissociation.

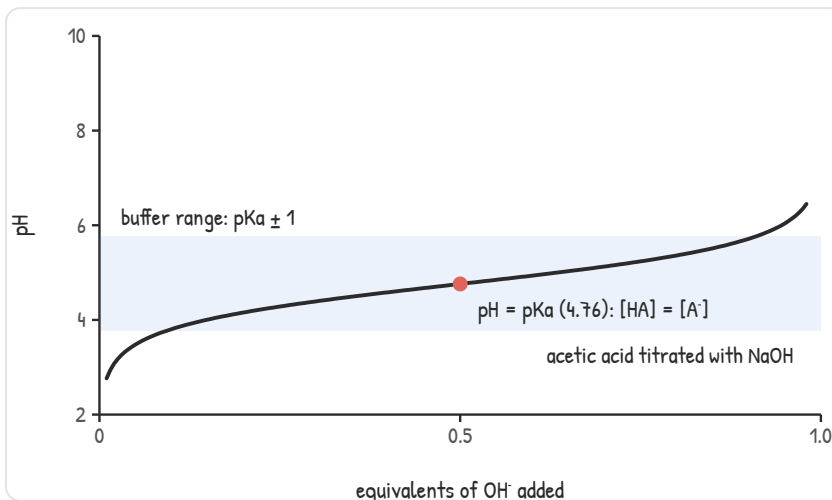
→ The pK_a of a drug helps predict its charge state and membrane permeability at a given pH.

Henderson-Hasselbalch equation and ionization ratios

- $pH - pK_a = \log([A^-]/[HA])$.
- $pH = pK_a$ gives $[A^-] = [HA]$.
- Each unit of $pH - pK_a$ changes the ratio tenfold.
- Above pK_a the deprotonated form predominates; below pK_a the protonated form predominates.

→ Ion trapping of weak acids and bases follows the same $pH - pK_a$ logic.

Titration curves and how buffers resist pH change

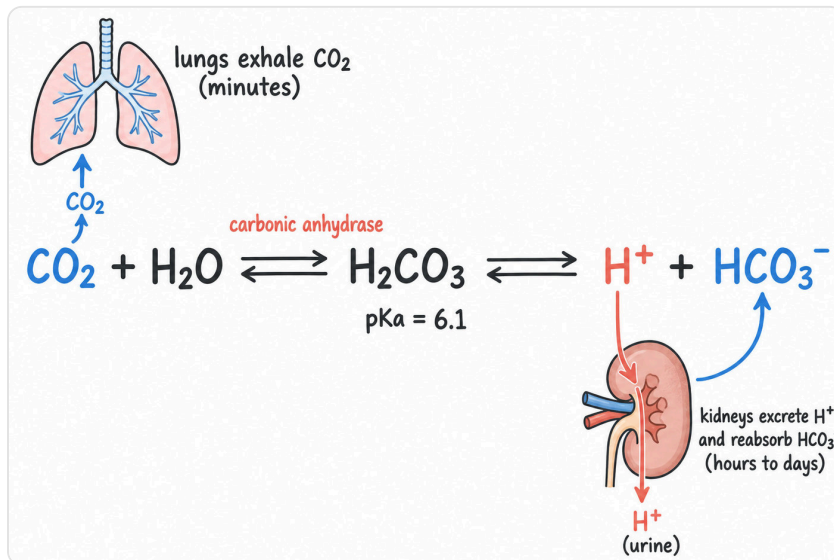


- At half-equivalence, $pH = pK_a$ and buffering is maximal.
- The useful buffer range is approximately $pK_a \pm 1$.
- The conjugate base consumes added H^+ .
- Higher total buffer concentration gives greater capacity at the same pH and ratio.

→ A patient's ability to buffer an acid load depends on available buffer stores as well as ventilation and renal response.

fig. 3 Weak-acid titration curve, computed from Henderson-Hasselbalch for acetic acid (pK_a 4.76). The flat middle is where the buffer resists pH change.

Bicarbonate, phosphate, proteins, and hemoglobin in acid-base control



- Bicarbonate couples blood chemistry to pulmonary CO_2 elimination.
- Phosphate is important intracellularly and in urine.
- Ionizable protein side chains buffer H^+ .
- Deoxyhemoglobin binds H^+ more readily than oxyhemoglobin.
- Respiratory compensation for metabolic acidosis lowers PaCO_2 .

→ Kussmaul breathing in ketoacidosis lowers CO_2 and partially raises pH.

fig. 4 The bicarbonate buffer couples rapid pulmonary CO_2 removal to slower renal H^+ excretion and bicarbonate reabsorption.

past papers asked

question	answer and why
A buffer solution is 0.080 M in lactic acid ($\text{pK}_a = 3.74$) and 0.070 M in sodium lactate. The pH of the solution is	3.68 is the best answer. Henderson-Hasselbalch gives $3.74 + \log(0.070/0.080) \approx 3.68$.
The chemical reactions in the small intestine take place in a basic (alkaline) environment. Which number indicates this basic pH?	8.5 is the best answer. pH 8.5 is clearly alkaline; values below 7 are acidic.
For octanoic acid, $\text{pK}_a = 4.89$. Which pH range would be the best buffering region for this acid?	3.89–5.89 is the best answer. A weak acid buffers effectively within about one pH unit of its pK_a .
Hydroxyapatite has a very small dissolution equilibrium constant (approximately 10^{-26} in the source). Which description is correct under physiological conditions?	A very small dissolution equilibrium constant favors solid hydroxyapatite. Acid promotes dissolution; added calcium/phosphate can suppress it by the common-ion effect. Source correction: hydroxyapatite is the inorganic mineral component of bone, not the organic matrix.
As the pK_a of an acid increase, the acid will be:	Weaker is the best answer. A larger pK_a means a smaller K_a and therefore a weaker acid.
Which of the following is not an example of a weak acid?	Sulfuric acid is the best answer. Sulfuric acid is treated as a strong acid, whereas lactic, pyruvic, and carbonic acids are weak acids.
The strength of an organic acid depends on:	Electronegativity is the best answer. Electron-withdrawing electronegativity stabilizes the conjugate base and increases organic-acid strength.
Polyprotic acids such as H_3PO_4 can act as acid-base buffers:	At pH values around any of their pK_a 's is the best answer. Each ionizable proton provides a separate buffer region centered near its own pK_a .
A buffer solution contains propionic acid and its conjugate base; the pK_a of propionic acid is 4.88. At what pH would the solution buffer best?	5.0 is the best answer. A buffer works best near its pK_a ; pH 5.0 is closest to pK_a 4.88.

Test yourself

1. Which ions are most characteristic of extracellular fluid?
 - A. Potassium and magnesium
 - B. Phosphate and protein
 - C. Sodium and chloride
 - D. Iron and copper
2. An ideal acetate buffer is diluted tenfold with water without changing the acid-to-base ratio. What happens approximately?
 - A. pH decreases by exactly 1 and capacity rises
 - B. pH rises by exactly 1 and capacity is unchanged
 - C. pH stays similar while buffer capacity decreases
 - D. Both pH and capacity remain exactly unchanged
3. A monoprotic weak acid has pK_a 5.0. At pH 5.0, approximately what fraction is in its deprotonated form?
 - A. 50%
 - B. 1%
 - C. 10%
 - D. 99%

Answers: 1 C · 2 C · 3 A

Amino acids

Lippincott 6e, Chapter 1 · teacher slides

common amino-acid structure and side-chain classification

- The R group distinguishes one amino acid from another.
- Nonpolar side chains favor hydrophobic interactions.
- Uncharged polar side chains can hydrogen bond.
- Acidic side chains are usually negative and basic side chains often positive at physiological pH.
- Proline has a secondary amino group; glycine has two alpha-carbon hydrogens.

→ A single side-chain substitution can alter folding, solubility, charge, or catalytic activity.

Nonpolar residues and the hydrophobic effect

- Nonpolar residues usually do not form ionic or hydrogen bonds through their side chains.
- They pack inside soluble proteins but can face membrane lipids.
- Hydrophobic clustering stabilizes a protein's three-dimensional structure.

→ The beta-globin Glu-to-Val substitution in HbS creates a hydrophobic interaction that promotes deoxyhemoglobin polymerization.

Polar residues, phosphorylation, and disulfide bonds

- Serine, threonine, and tyrosine are common phosphorylation sites.
- Asparagine and glutamine have polar amide side chains.
- Two cysteine thiols can oxidize to a disulfide.
- Disulfides are especially common in extracellular and secreted proteins.

→ Kinases regulate many enzymes and receptors through reversible phosphorylation.

Acidic and basic side chains, especially histidine

- Aspartate and glutamate are acidic and usually negative.
- Lysine and arginine are basic and usually positive.
- Histidine can donate or accept protons near physiological pH.
- Local electrostatic environment can shift a side-chain pKa.

→ Histidines contribute to hemoglobin buffering and the Bohr effect.

Amino-acid stereochemistry and the glycine exception

- Chiral amino acids occur as D and L enantiomers.
- Human proteins use L-amino acids.
- Glycine is achiral.
- D-amino acids occur in some bacterial cell walls and antibiotics.

→ Stereochemistry affects enzyme recognition and drug design.

Amino-acid ionization, titration, and isoelectric point

- At pI, average net charge is zero.
- pH below pI gives a net positive charge.
- pH above pI gives a net negative charge.
- Titration curves reveal pKa values and buffering regions.
- Ionizable side chains add extra titration steps.

→ Electrophoresis separates proteins according to net charge at the buffer pH.

Aromatic amino acids and analytical recognition

- Tryptophan is the strongest common contributor to A280.
- Tyrosine also absorbs near 280 nm; phenylalanine contributes less.
- Tyrosine is aromatic yet has a polar hydroxyl group.
- The xanthoproteic reaction detects aromatic residues by a yellow-to-orange color.

→ A280 is commonly used to estimate protein concentration, but nucleic acids and other chromophores can interfere.

Past papers asked

question

answer and why

Which of the following statements about solutions of amino acids at physiologic pH is true?

All amino acids contain both positive and negative charges is the best answer. At physiological pH, free amino acids generally contain both NH_3^+ and COO^- groups as zwitterions.

Choose the incorrect statement about amino acids.

D-amino acids occur, for example, in bacterial peptidoglycan and as D-serine in humans. Most ribosomally incorporated amino acids are L, with achiral glycine. The vague source phrase "tyrosine is a modified amino acid" is clarified here.

An amino acid not found in proteins is ...

β -Alanine is the best answer. β -Alanine is not one of the standard α -amino acids incorporated into proteins.

Test yourself

1. Which atoms are attached to the alpha carbon of a standard alpha-amino acid?

- A. Two phosphate groups and a base
- C. Three fatty acids and glycerol

- B. An amino group, carboxyl group, hydrogen, and side chain
- D. A sugar, base, and phosphate

2. A free amino acid with pI 6.0 is electrophoresed at pH 9.0. Toward which electrode does it tend to migrate?

- A. Neither, because all amino acids are always neutral
- C. The positive anode

- B. The negative cathode
- D. Both electrodes at equal speed

3. A serine in a protein is replaced by alanine. Which potential side-chain modification is directly lost at that residue?

- A. Phosphorylation of its hydroxyl group
- C. Disulfide formation by its thiol

- B. Protonation of its side-chain imidazole
- D. Formation of its backbone peptide bond

Answers: 1 B · 2 C · 3 A

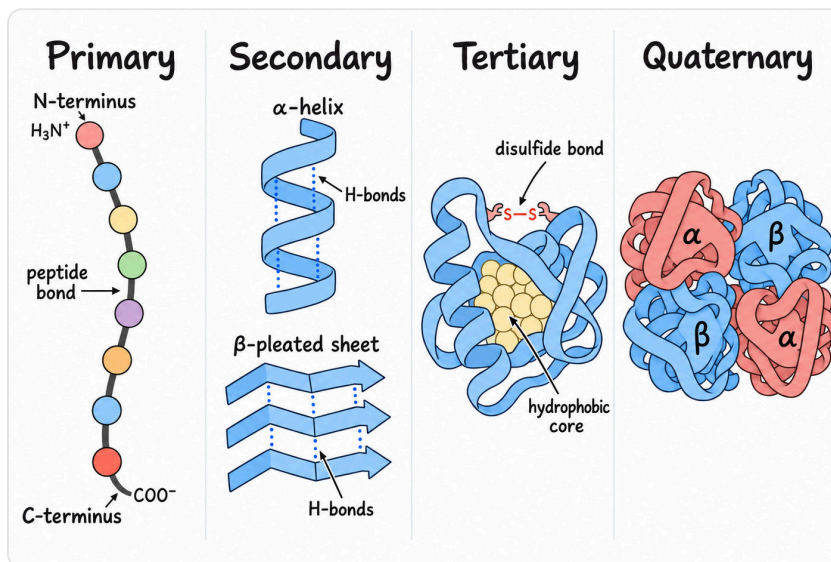
Protein structure and folding

Lippincott 6e, Chapter 2 · teacher slides

Primary structure encodes folding and function

- Primary structure is amino-acid sequence.
 - Sequences are written N terminus to C terminus.
 - The sequence contains the information needed for native folding, although cells often require chaperones.
 - A changed sequence can produce a changed phenotype.
- Missense variants can cause sickle cell disease or destabilize collagen.

Peptide-bond formation, resonance, and planarity



- Peptide formation is a condensation reaction.
- The peptide C-N bond has partial double-bond character.
- Restricted rotation makes the peptide unit planar.
- Most peptide bonds adopt the trans configuration.
- Denaturation usually does not hydrolyze peptide bonds.

→ Proteases hydrolyze selected peptide bonds during digestion, signaling, and zymogen activation.

fig. 5 Protein structure progresses from amino-acid sequence through local secondary structure and a folded chain to assemblies of multiple subunits.

Alpha helices, beta sheets, and beta turns

- An alpha helix has about 3.6 residues per turn and i-to-i+4 backbone hydrogen bonds.
- Proline often disrupts an alpha helix.
- Beta sheets are stabilized by backbone hydrogen bonds between extended strands.
- Beta turns permit compact folding and often contain glycine or proline.

→ Amyloid fibrils are enriched in ordered beta-sheet structure.

Tertiary structure, motifs, domains, and side-chain forces

- Motifs are recurring combinations of secondary structures.
 - Domains are modular structural and functional units.
 - Hydrophobic burial is a major folding force in aqueous solution.
 - Ionic and hydrogen bonds are sensitive to pH and solvent.
 - Disulfides provide covalent stabilization, especially extracellularly.
- A domain-specific mutation can selectively abolish binding or catalysis while leaving other domains intact.

Quaternary structure requires multiple subunits

- One polypeptide has no quaternary structure.
 - Two or more interacting polypeptides create quaternary structure.
 - Hydrophobic, ionic, and hydrogen-bond interactions usually dominate subunit association.
 - Hemoglobin is an $\alpha_2\beta_2$ cooperative tetramer.
- A variant at a subunit interface can alter cooperativity without directly changing an active or ligand-binding site.

Protein folding and molecular chaperones

- Folding is ordered rather than a random search of all conformations.
 - Hydrophobic collapse helps organize early folding intermediates.
 - Hsp70 protects exposed hydrophobic regions of nascent chains.
 - Chaperonins provide a protected folding chamber.
 - Many chaperone cycles require ATP.
- Cell stress increases heat-shock proteins.

Denaturation disrupts higher structure, not primary sequence

- Denaturation causes loss of native conformation and usually function.
 - Peptide bonds are generally preserved.
 - Hydrophobic exposure can cause precipitation and aggregation.
 - Reversibility depends on sequence, conditions, and avoidance of aggregation.
- Fever and severe acidosis can impair enzyme function, although ordinary fever is not equivalent to wholesale protein denaturation.

Protein misfolding, amyloid, and prion propagation

- Amyloid fibrils are ordered and beta-sheet rich.
 - Alzheimer plaques contain A beta fragments produced from APP.
 - PrP^{Sc} promotes conformational conversion of PrP^C.
 - Sequence mutation is not required for every misfolding disease.
- Amyloid disorders include Alzheimer disease and systemic amyloidoses.

Past papers asked

question

answer and why

A tripeptide has

3 amino acids and 2 peptide bonds is the best answer. Three amino acids are linked by two peptide bonds.

Peptide bond is a

Covalent bond is the best answer. A peptide bond is a covalent amide bond.

question

answer and why

The oxygen carrier of muscle is the globular protein myoglobin. Which one of the following amino acids is highly likely to be localized within the interior of the molecule?

Valine is the best answer. Valine is nonpolar and is therefore favored in the hydrophobic interior of a globular protein.

A coiled peptide chain held in place by hydrogen bonding between peptide bonds in the same chain (not between two chains) is?

α -helix is the best answer. An α -helix is stabilized by intrachain backbone hydrogen bonds.

The secondary structure of protein is held together by which type of bonding?

Hydrogen bonding is the best answer. Backbone hydrogen bonds stabilize α -helices and β -sheets.

Test yourself

1. Which statement about a peptide bond is correct?

- A. It is a phosphodiester bond
- B. It is freely rotating like a single carbon-carbon bond
- C. It is an amide linkage between an alpha-carboxyl and an alpha-amino group
- D. It is disrupted by every noncovalent denaturant

2. List the four main types of interactions that stabilize the tertiary structure of globular proteins.

- A. Disulfide bonds, hydrophobic interactions, hydrogen bonds and ionic interactions
- B. Peptide bonds, glycosidic bonds, phosphodiester bonds and ester bonds
- C. Hydrophobic interactions alone, without any polar contributions
- D. Hydrogen bonds, ionic interactions, DNA base pairing and codon recognition

3. Which structural level usually remains intact during reversible protein denaturation?

- A. Primary structure
- B. Secondary structure
- C. Tertiary structure
- D. Quaternary structure

Answers: 1 C · 2 A · 3 A

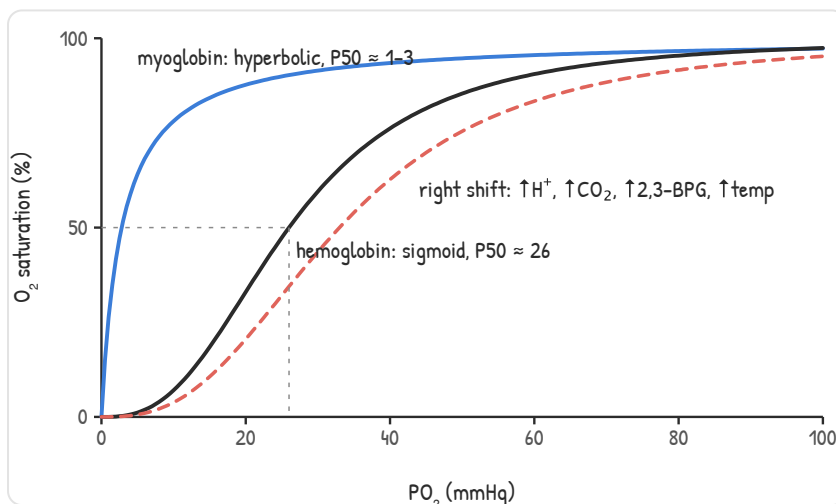
Hemoglobin and globular proteins

Lippincott 6e, Chapter 3 · teacher slides

Heme chemistry and HbA architecture

- Functional heme oxygen binding requires Fe^{2+} .
 - HbA contains two alpha and two beta chains.
 - Each globin subunit contains one heme.
 - Hemoglobin subunits associate mainly through noncovalent forces.
- Oxidant drugs can convert Fe^{2+} to Fe^{3+} and cause methemoglobinemia.

Myoglobin stores oxygen; hemoglobin transports it

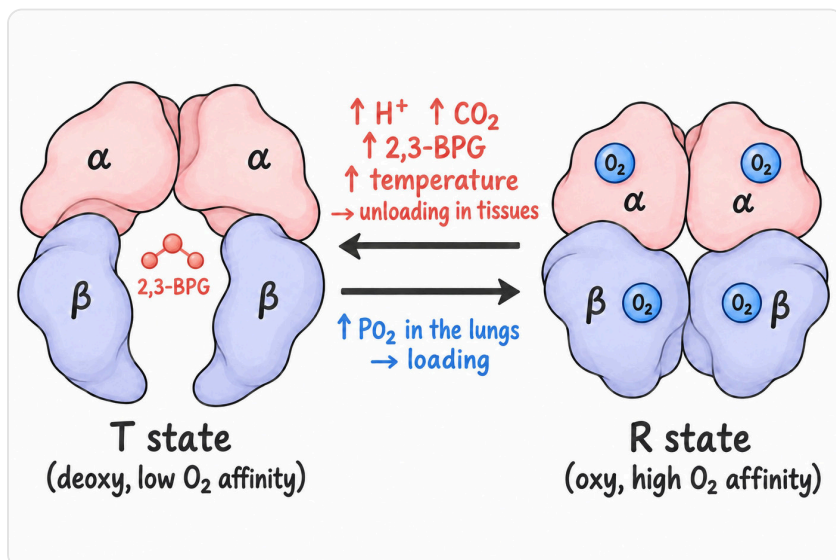


- Myoglobin is monomeric and found in cardiac and skeletal muscle.
- Hemoglobin is tetrameric and found in erythrocytes.
- Myoglobin has a hyperbolic oxygen-binding curve.
- Hemoglobin has a sigmoidal oxygen-binding curve.

→ Myoglobin rises early after muscle injury but is less cardiac-specific than troponin.

fig. 6 Oxygen-binding curves computed with the Hill equation (Hb $n = 2.7$). A right shift raises P₅₀ and unloads more O₂ in tissues.

cooperativity, T/R states, and P50



- T state is tense, deoxygenated, and lower affinity.
- R state is relaxed, oxygenated, and higher affinity.
- Positive cooperativity creates a sigmoidal curve.
- Higher P₅₀ means lower oxygen affinity.

→ Right shifts improve unloading but can impair loading if extreme; left shifts do the reverse.

fig. 7 Hemoglobin shifts between the low-affinity T state favored by tissue conditions and the high-affinity R state favored by oxygen loading in the lungs.

Bohr effect: active tissues promote oxygen unloading

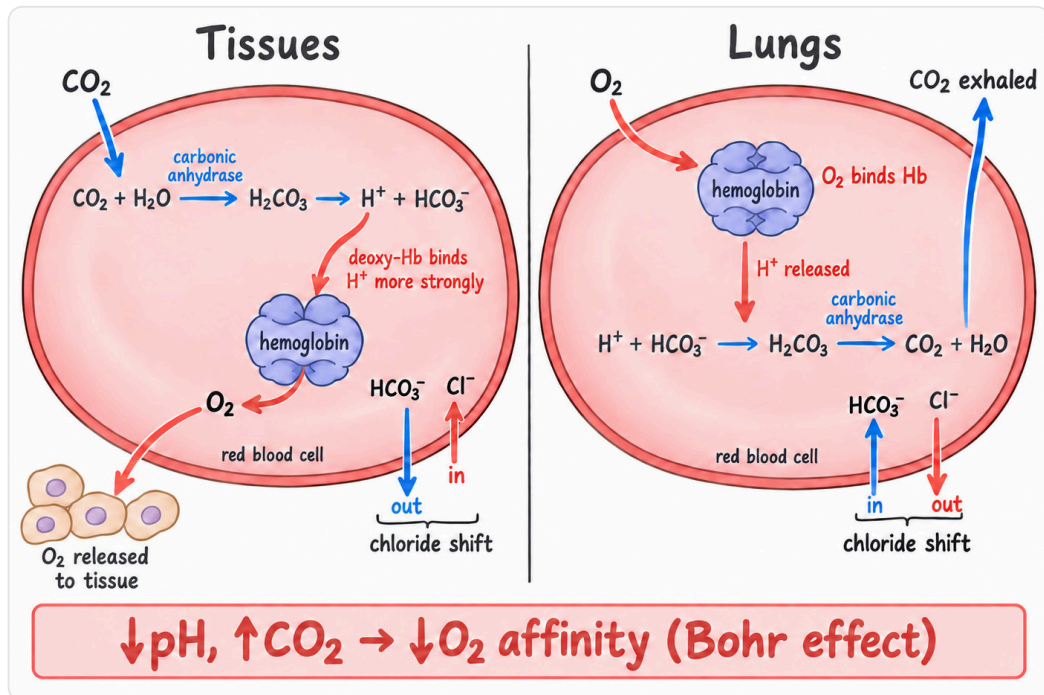


Fig. 8 Tissue CO₂ transport and the chloride shift reverse in the lungs; lower pH and higher CO₂ reduce hemoglobin oxygen affinity.

2,3-BPG stabilizes deoxyhemoglobin

- One 2,3-BPG binds per deoxyhemoglobin tetramer.
 - It binds chiefly between the beta chains.
 - It lowers oxygen affinity and raises P₅₀.
 - Hypoxia and chronic anemia increase its concentration.
- High-altitude adaptation includes increased 2,3-BPG.

Fetal hemoglobin and placental oxygen transfer

- HbF composition is $\alpha_2\gamma_2$.
 - HbF binds 2,3-BPG weakly.
 - Weaker BPG binding increases oxygen affinity.
 - HbF predominates before birth and falls after birth.
- Increasing HbF with hydroxyurea reduces HbS polymerization and sickle crises.

carbon monoxide poisoning versus methemoglobinemia

- CO reduces available sites and left-shifts remaining oxygen binding.
- Methemoglobin contains Fe³⁺ rather than Fe²⁺.
- Chocolate-brown blood suggests methemoglobinemia.
- Oxidant drugs such as nitrates or benzocaine can cause acquired methemoglobinemia.

→ CO poisoning is treated with high-concentration oxygen and sometimes hyperbaric oxygen.

HbA1c records cumulative glucose exposure

- Glycation is nonenzymatic and differs from enzyme-directed glycosylation.
 - The extent of HbA1c formation rises with ambient glucose concentration and time.
 - HbA1c reflects roughly the preceding two to three months with greater weighting of recent weeks.
 - Conditions changing erythrocyte lifespan can bias the result.
- HbA1c is used to diagnose and monitor diabetes when the assay is reliable.

HbS: a primary-structure change causes polymerization

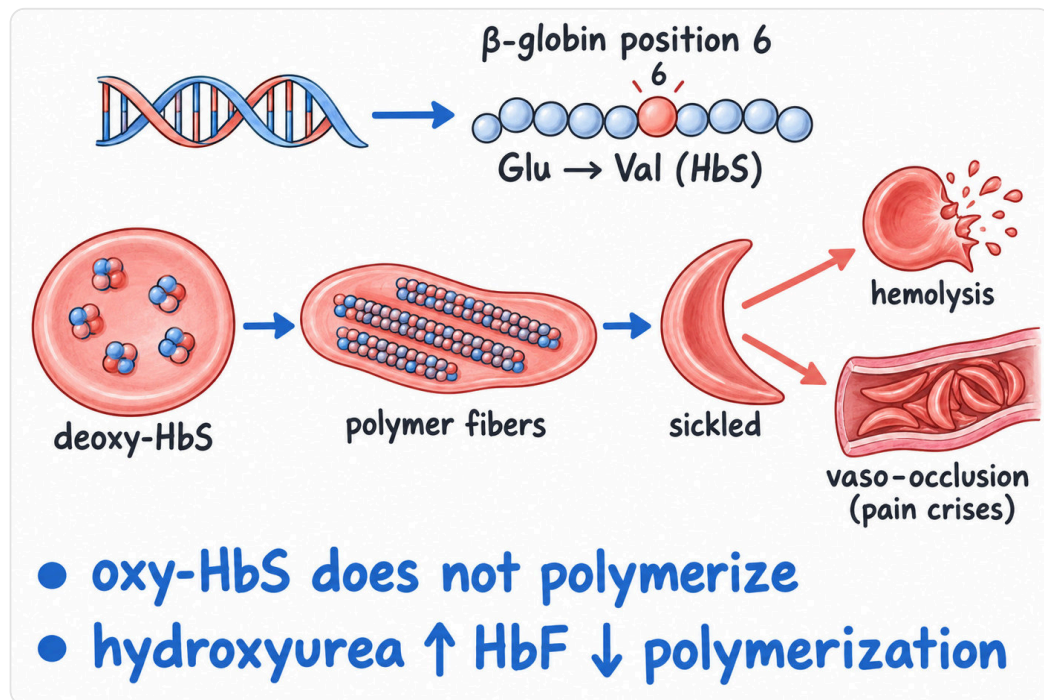


fig. 9 The β-globin Glu-to-Val substitution promotes deoxy-HbS polymerization, sickling, hemolysis, and vaso-occlusion.

Thalassemia is impaired globin-chain production

- Thalassemia is primarily a quantitative globin defect.
 - Beta-thalassemia leaves excess alpha chains that precipitate.
 - Severity depends on how many genes are affected and whether production is reduced or absent.
 - Beta-thalassemia often raises HbA2 and HbF.
- Severe disease produces anemia, marrow expansion, transfusion dependence, and iron overload.

past papers asked

question

Which of the following mutations would produce a minor thalassemia?

answer and why

Deletion of one β-globin locus is the best answer. Loss of one β-globin locus can produce β-thalassemia trait/minor.

question

answer and why

The ability of hemoglobin to serve as an effective transporter of oxygen and carbon dioxide between lungs and tissues is explained by which of the following properties?

In contrast to myoglobin, hemoglobin exhibits greater changes in secondary and tertiary structure after oxygen binding is the best answer. Oxygen binding produces cooperative conformational changes in hemoglobin, unlike monomeric myoglobin.

Which of the following would produce Methemoglobinemia?

Converting Ferrous iron (Fe^{2+}) to Ferric (Fe^{3+}) is the best answer. Oxidation of heme iron from Fe^{2+} to Fe^{3+} forms methemoglobin, which cannot bind oxygen normally.

Blood is drawn from a child with severe anemia and the hemoglobin protein is degraded for the analysis of peptide and amino acids. Of the results below, which change in hemoglobin primary structure is most likely to correlate with the clinical phenotype of anemia?

Leu-Glu-Ile to Leu-Val-Ile is the best answer. Replacing glutamate with valine reproduces the key nonconservative change of sickle hemoglobin.

All of the following statements correctly describes transport of O_2 by hemoglobin except

Increased CO_2 concentrations increase O_2 affinity is the best answer. Higher CO_2 lowers hemoglobin oxygen affinity through the Bohr effect rather than increasing it.

In sickle cell anemia, the basis of the malfunction of the hemoglobin molecule is ...

Substitution of a single amino acid is the best answer. Sickle hemoglobin results from a single β -globin substitution, Glu6Val, changing primary structure.

Test yourself

1. What specific amino acid substitution occurs in HbS and at which position of the beta chain?

- A. Alpha-chain position 6: valine replaced by glutamate
- B. Beta-chain position 6: glutamate replaced by valine
- C. Beta-chain position 6: glutamate replaced by lysine
- D. Beta-chain position 26: glutamate replaced by lysine

2. In severe beta-thalassemia, why can erythroid precursors be damaged even though alpha chains have normal amino-acid sequences?

- A. Normal alpha chains cannot bind any heme
- B. Alpha chains lose their peptide backbones as beta synthesis falls
- C. Excess unmatched alpha chains precipitate
- D. Beta-chain deficiency necessarily produces the HbS point mutation

3. Stored erythrocytes have depleted 2,3-BPG. Before 2,3-BPG is replenished, their hemoglobin would tend to show which change?

- A. A left-shifted oxygen-dissociation curve
- B. An increased P_{50} from stronger T-state stabilization
- C. A right shift caused by tighter 2,3-BPG binding
- D. Unchanged affinity because 2,3-BPG acts only on dissolved oxygen

Answers: 1 B · 2 C · 3 A

collagen, elastin and fibrous proteins

Lippincott 6e, Chapter 4 · teacher slides

collagen triple helix, Gly-X-Y, and major types

- Collagen is the most abundant protein in the body.
 - Gly-X-Y repeats permit triple-helix packing.
 - Type I predominates in bone, skin, tendon, and scar.
 - Type II predominates in cartilage and vitreous body.
 - Type III forms reticular fibers; type IV forms basement-membrane networks.
- Tissue distribution helps predict the phenotype of a collagen defect.

Intracellular collagen synthesis and vitamin c

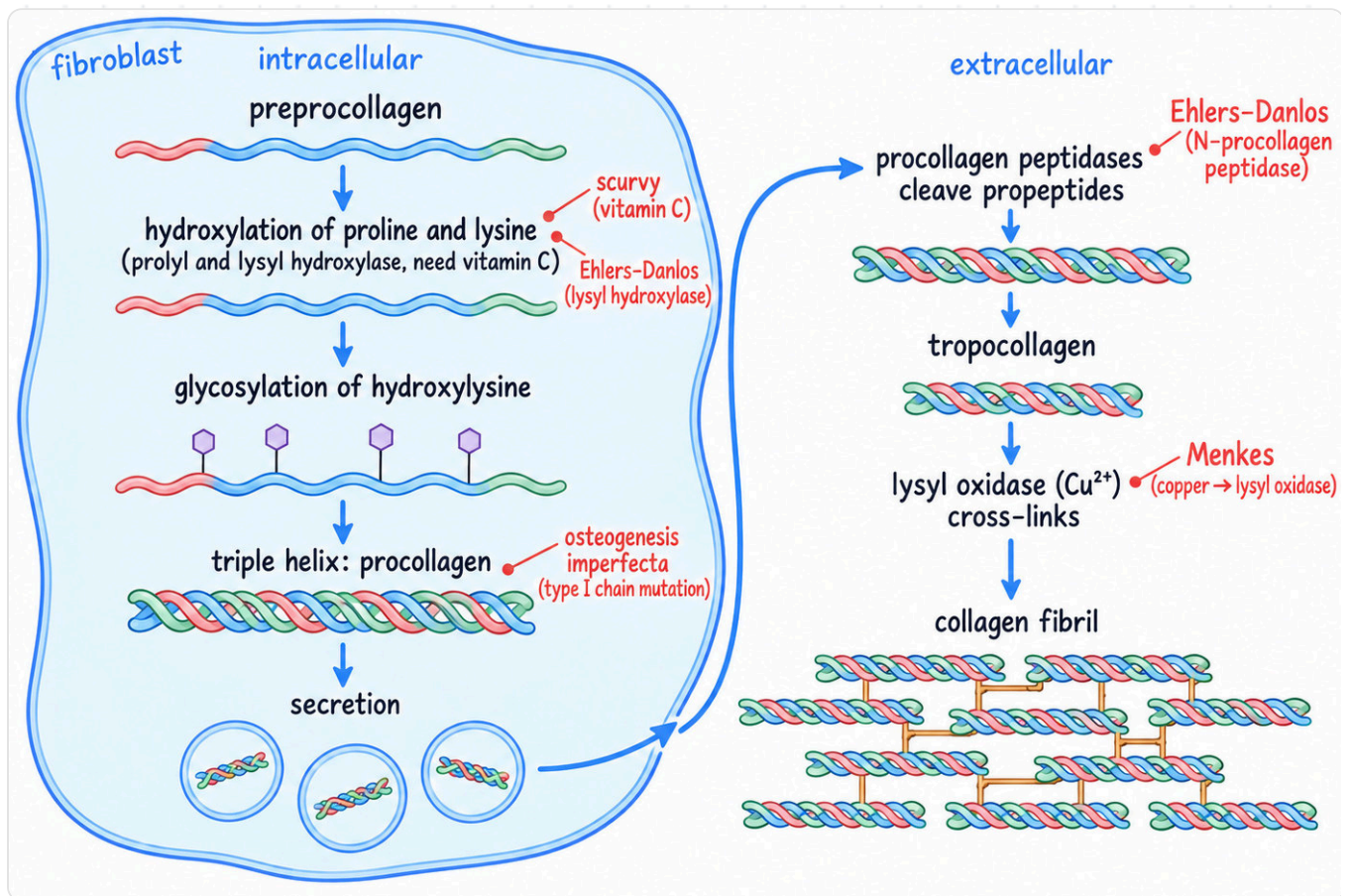


Fig. 10 Collagen is modified and assembled as procollagen inside fibroblasts, then cleaved and cross-linked extracellularly into fibrils.

- Hydroxylation occurs in rough ER after residues enter the chain.
 - Vitamin C keeps iron in the reduced state needed by hydroxylases.
 - Hydroxyproline stabilizes the triple helix through hydrogen bonding.
 - Propeptides align chains and keep procollagen soluble inside the cell.
- Vitamin C deficiency causes weak collagen, bleeding, and poor wound healing.

Extracellular procollagen cleavage and copper-dependent cross-linking

- Procollagen cleavage occurs extracellularly.
 - Tropocollagen self-assembles into staggered fibrils.
 - Lysyl oxidase requires copper.
 - Cross-link density increases tensile strength with maturation.
- Copper deficiency or lysyl-oxidase inhibition weakens collagen cross-linking.

collagen degradation and matrix remodeling

- Collagenases are matrix metalloproteinases.
 - Initial collagenase cleavage enables further proteolysis.
 - MMP activity is regulated by activation and tissue inhibitors.
 - Excess breakdown weakens matrix; excess deposition causes fibrosis.
- MMP activity contributes to arthritis, tumor invasion, and vascular remodeling.

Scurvy, osteogenesis imperfecta, and Ehlers-Danlos syndrome

- Scurvy impairs proline and lysine hydroxylation.
 - Osteogenesis imperfecta commonly involves type I collagen.
 - Glycine substitution can disrupt the entire triple helix.
 - Classic EDS often involves type V; vascular EDS involves type III.
 - Dominant-negative chains can damage a multichain collagen molecule.
- Bleeding plus dietary deficiency suggests scurvy; congenital recurrent fractures suggest OI.

Elastin structure permits stretch and recoil

- Elastin provides elasticity rather than high tensile strength.
 - Tropoelastin is the soluble precursor.
 - Fibrillin microfibrils guide elastic-fiber assembly.
 - Desmosine cross-links connect tropoelastin chains.
 - Elastic recoil is crucial in alveoli and large arteries.
- Loss of alveolar elastin reduces recoil and promotes expiratory airway collapse.

Fibrillin defects and unopposed elastase injury

- Fibrillin-1 mutation causes Marfan syndrome.
 - Alpha1-antitrypsin inhibits neutrophil elastase.
 - Unopposed elastase destroys alveolar walls and recoil.
 - Misfolded Z-variant AAT can accumulate in hepatocytes.
- Marfan syndrome raises risk of aortic root disease and lens dislocation.

Past papers asked

question	answer and why
All of the following statements regarding fibrous proteins are true except	Hemoglobin is the best example is the best answer. Hemoglobin is globular, whereas collagen and keratin are fibrous structural proteins.
Which of the substances below is primarily found in tendons?	Collagen is the best answer. Protein function follows primary sequence and higher-order folding interactions.

question

answer and why

During synthesis of mature collagen fiber, which one of the following steps would occur within the fibroblast?

Formation of a triple helix is the best answer. Protein function follows primary sequence and higher-order folding interactions.

Deficiency of lysyl hydroxylase causes syndrome characterized by hypermobile joints and skin abnormalities.

Ehlers-Danlos is the best answer. Protein function follows primary sequence and higher-order folding interactions.

A child with tall stature, loose joints, and detached retinas is found to have a mutation in type II collagen. Recall that collagen consists of a repeating tripeptide motif where the first amino acid of each tripeptide is the same. Which of the following amino acids is the recurring amino acid most likely to be altered in mutations that distort collagen molecules?

Glycine is the best answer. Protein function follows primary sequence and higher-order folding interactions.

The highest concentration of cystine can be found in?

Keratin is the best answer. Keratin is rich in cystine disulfide bonds, which strengthen hair and nails.

Which one of the following results in Ehlers-Danlos syndrome?

Deficiency of lysyl hydroxylase is the best answer. Lysyl hydroxylase defects impair collagen maturation and can cause hyperextensible skin and hypermobile joints.

Glycine and proline are the most abundant amino acids in the structure of

Collagen is the best answer. Collagen is rich in glycine and proline/hydroxyproline, supporting its triple helix.

Test yourself

1. What class of enzymes degrades collagen, and to which larger enzyme family do they belong?

- A. Lysyl oxidases that initiate covalent cross-linking
- B. Prolyl hydroxylases that modify procollagen in ER
- C. Procollagen peptidases that remove terminal propeptides during maturation
- D. Collagenases within the matrix metalloproteinase family

2. Hydroxylation of selected proline and lysine residues during collagen synthesis requires which vitamin?

- A. Vitamin K
- B. Vitamin B12
- C. Vitamin A
- D. Vitamin C

3. Where do proline and lysine hydroxylation of collagen chains occur?

- A. Inside lysosomes after collagen degradation
- B. Only in the nucleus
- C. In the rough endoplasmic reticulum before secretion
- D. In the extracellular matrix after fibril assembly

Answers: 1 D · 2 D · 3 C

Enzymes, kinetics and regulation

Lippincott 6e, Chapter 5 · teacher slides

Enzyme kinetics at a glance

Pattern	K_m	v_{max} / distinguishing point
Michaelis-Menten	$[S]$ at $v = v_{max}/2$	$v = v_{max}[S]/(K_m + [S])$; at $[S] = 3K_m$, $v = 0.75 v_{max}$.
More enzyme	Unchanged	v_{max} rises proportionally when other conditions are fixed.
Competitive inhibition	Increased apparent K_m	v_{max} unchanged; high substrate can overcome reversible competition.
Pure noncompetitive inhibition	Unchanged	v_{max} decreased. Mixed inhibition need not leave K_m unchanged.
Uncompetitive inhibition	Decreased apparent K_m	v_{max} decreased in the same ratio; inhibitor binds ES.
Irreversible inactivation	No universal K_m prediction	Active enzyme falls; removing free inhibitor does not restore already-inactivated enzyme.
Catalytic performance	K_m alone is insufficient	$k_{cat} = v_{max}/[E]_{total}$; k_{cat}/K_m is the specificity constant at low $[S]$.

Active sites create specificity and induced fit

- The active site is created by tertiary structure.
- Specificity can be absolute, group, bond, or stereochemical.
- Induced fit is conformational adjustment after binding.
- Catalytic residues need not be adjacent in sequence.

→ Many drugs work by occupying an enzyme active site or an allosteric pocket.

Enzymes lower activation energy without changing equilibrium

- Activation energy is the barrier to reaching the transition state.
- Enzymes lower activation energy.
- Enzymes do not change reaction ΔG .
- Enzymes do not change the equilibrium constant.
- Transition-state stabilization is central to catalysis.

→ Transition-state analogs can be potent enzyme inhibitors and useful drugs.

Apoenzyme, holoenzyme, cofactor, and coenzyme

- Apoenzyme plus required nonprotein component equals holoenzyme.
- Metal ions such as Zn^{2+} or Fe^{2+} can be cofactors.
- NAD^+ is a vitamin-derived coenzyme and cosubstrate.
- Tightly bound FAD can function as a prosthetic group.

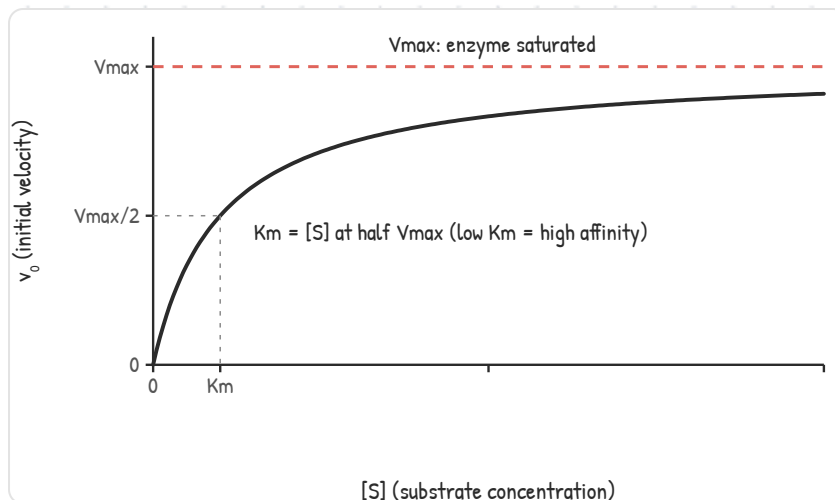
→ Vitamin deficiencies can reduce the activity of all enzymes requiring the corresponding coenzyme.

Temperature and pH alter enzyme rate

- Rate rises with temperature until structural instability dominates.
- High heat decreases activity through denaturation.
- pH controls active-site residue ionization.
- Extreme pH can disrupt tertiary structure.
- Pepsin and cytosolic enzymes have different pH optima.

→ Specimen handling temperature and pH can change measured enzyme activity.

Michaelis-Menten saturation, K_m , and V_{max}



- $v = V_{max}[S]/(K_m + [S])$.
- At $[S] = K_m$, $v = V_{max}/2$.
- Lower K_m commonly means higher apparent affinity.
- At saturation, adding substrate does not raise velocity further.
- Increasing enzyme concentration increases V_{max} .

→ K_m and V_{max} help distinguish inhibitor mechanisms and compare enzyme variants.

Fig. 11 Michaelis-Menten curve $v_0 = V_{max}[S]/(K_m + [S])$, computed with $K_m = 2$ and $V_{max} = 1$.

Lineweaver-Burk plot interpretation

- y-intercept = $1/V_{max}$.
- x-intercept = $-1/K_m$.
- slope = K_m/V_{max} .
- Competitive and noncompetitive inhibitors produce different intersection patterns.

competitive inhibition raises apparent K_m

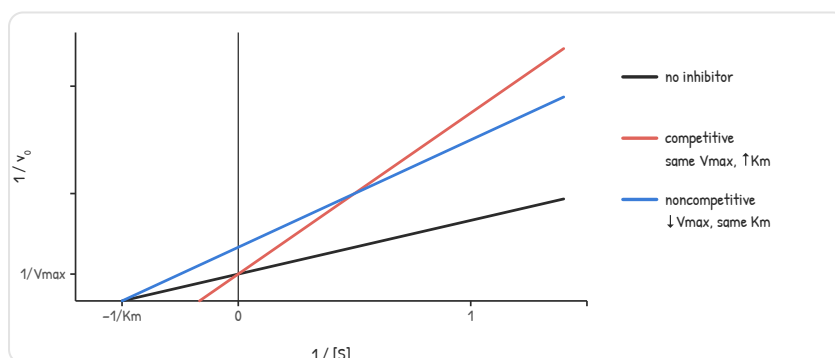


Fig. 12 Lineweaver-Burk lines computed from $1/v_0 = (K_m/V_{max})(1/[S]) + 1/V_{max}$: competitive inhibitors share the y-intercept, noncompetitive inhibitors share the x-intercept.

- Competitive inhibitor binds the active site.
- Apparent K_m increases.
- V_{max} is unchanged.
- Increasing substrate can overcome reversible competitive inhibition.

→ Fomepizole or ethanol competitively limits toxic alcohol metabolism by alcohol dehydrogenase.

Noncompetitive versus irreversible inhibition

- Pure noncompetitive inhibition decreases V_{max} and leaves K_m unchanged.
- Irreversible inhibition permanently reduces active enzyme concentration.
- Increasing substrate does not restore the original V_{max} .
- Real mixed inhibition can change both K_m and V_{max} .

→ Aspirin irreversibly acetylates cyclooxygenase.

Allosteric enzymes control pathway flux

- Allosteric sites are distinct from catalytic sites.
- Allosteric enzymes often show cooperative sigmoidal kinetics.
- Regulatory enzymes commonly occur at irreversible pathway-control steps.
- End-product feedback conserves energy and precursors.

→ Allosteric drugs can modulate activity without competing with substrate.

Fast and slow modes of enzyme regulation

- Kinases add phosphate; phosphatases remove it.
- Phosphorylation can activate one enzyme and inhibit another.
- Zymogen activation is proteolytic and irreversible for that molecule.
- Induction increases enzyme synthesis over hours to days.
- Multiple regulation modes can act on the same pathway.

→ Digestive proteases and coagulation factors are synthesized as zymogens.

Plasma enzymes, isoenzymes, and tissue injury

- Plasma enzyme elevation can indicate cell injury rather than increased pathway activity in plasma.
- Isoenzymes differ in sequence, charge, kinetics, or tissue distribution.
- ALT is relatively useful for hepatocellular injury.
- Cardiac troponin is the preferred myocardial-injury biomarker.
- A result must be interpreted with symptoms, timing, and reference intervals.

→ Serial troponin measurements help detect an evolving myocardial injury pattern.

Phosphorylation changes enzyme conformation and activity

- Reversible phosphorylation changes the activity of an existing protein.
- A phosphate group can alter charge interactions and protein conformation.
- Phosphorylation can activate or inhibit, depending on the enzyme.
- Protein kinases add phosphate; protein phosphatases remove it.

creatinine kinase, alkaline phosphatase, and urinary amylase

- Creatine kinase is a sensitive biochemical marker of skeletal-muscle injury.
- CK-MM predominates in skeletal muscle.
- Serum alkaline phosphatase commonly originates from liver/biliary tissue or bone.
- Amylase can be detected in urine after renal filtration.
- Macroamylase is poorly filtered because the complex is large.

→ Rhabdomyolysis elevates CK and may cause myoglobin-associated acute kidney injury.

past papers asked

question	answer and why
Which of the following statements about Michaelis-Menten constant is correct?	A low K_m indicates a high affinity of an enzyme for its substrate. is the best answer. Michaelis-Menten parameters relate substrate concentration to reaction velocity. K_m is independent of total enzyme concentration in the Michaelis-Menten model. Low K_m is often used as an affinity shorthand, but K_m is not universally a dissociation constant. The original wording permits both A and the customary textbook interpretation of B.
When $K_m = S$, reaction velocity (v) would be equal to.....	$1/2 V_{max}$ is the best answer. Michaelis-Menten parameters relate substrate concentration to reaction velocity.
Which of the following can act as a feedback inhibitor of an enzyme?	End product is the best answer. An end product commonly inhibits an early committed enzyme in its own biosynthetic pathway.
Zymogen or proenzyme is a	Enzyme precursor is the best answer. A zymogen is an inactive enzyme precursor activated by a later step such as proteolysis.
The effect of reversible competitive inhibitor can be canceled out by	Increasing the substrate concentration is the best answer. High substrate can outcompete a reversible competitive inhibitor for the active site.
Which enzyme class catalyzes the reaction shown?	Lactate dehydrogenase couples lactate oxidation to NAD^+ reduction. It is an oxidoreductase.
When $S=3K_m$, the V is equal to..... ?	$3/4 V_{max}$ is the best answer. Michaelis-Menten parameters relate substrate concentration to reaction velocity.
Ibuprofen is a reversible inhibitor of the COX-1 enzyme which is involved in Arachidonic Acid metabolism. Blocking of enzyme action by blocking its active site is called as	Competitive inhibition is the best answer. Ibuprofen reversibly competes at the cyclooxygenase active site; aspirin inhibits COX irreversibly by acetylation.
The catalytic efficiency of two enzymes acting on a substrate is commonly compared using which parameter?	Editorial repair: the source offered K_m alone. k_{cat}/K_m , not K_m alone, is the specificity constant used for low-substrate catalytic performance.
Which one of the following is a reversible inhibitor of the COX-1 enzyme that is involved in prostaglandin synthesis?	Ibuprofen is the best answer. Ibuprofen reversibly competes at the cyclooxygenase active site; aspirin inhibits COX irreversibly by acetylation.
A non-competitive inhibitor of an enzyme ...	Decreases V_{max} with no change in K_m is the best answer. Pure noncompetitive inhibition lowers V_{max} without changing K_m .
Which of the following can restore the activity of an enzyme already irreversibly inhibited?	The original four options cannot reverse irreversible enzyme inactivation. Adding E prevents teaching an incorrect source key.

Test yourself

1. What is feedback inhibition?

- A. An end product inhibits an earlier regulatory enzyme in its pathway
- B. A substrate permanently destroys its enzyme
- C. Every enzyme activates the next one
- D. A ribosome stops at a stop codon

2. A lower K_m generally indicates what about enzyme-substrate interaction?

- A. Higher apparent substrate affinity
- B. Lower V_{max} in every case
- C. Irreversible inhibition
- D. Loss of the active site

3. What biochemical principle explains the historical use of ethanol in ethylene-glycol poisoning?

- A. Ethanol irreversibly hydrolyzes ethylene glycol in plasma
- B. Ethanol activates the toxic-metabolite pathway by increasing its affinity
- C. Ethanol competes for alcohol dehydrogenase, reducing formation of toxic ethylene-glycol metabolites
- D. Ethanol covalently destroys every alcohol dehydrogenase molecule

Answers: 1 A · 2 A · 3 C

Bioenergetics and oxidative phosphorylation

Lippincott 6e, Chapter 6 · teacher slides

Free energy, reaction direction, and energetic coupling

- ΔG predicts feasibility, not reaction rate
 - Actual concentrations can change the sign of ΔG
 - Free-energy changes of coupled reactions are additive
 - ATP is kinetically stable but has high phosphate-transfer potential
- Energy failure during hypoxia

Electron-transport chain organization and electron flow

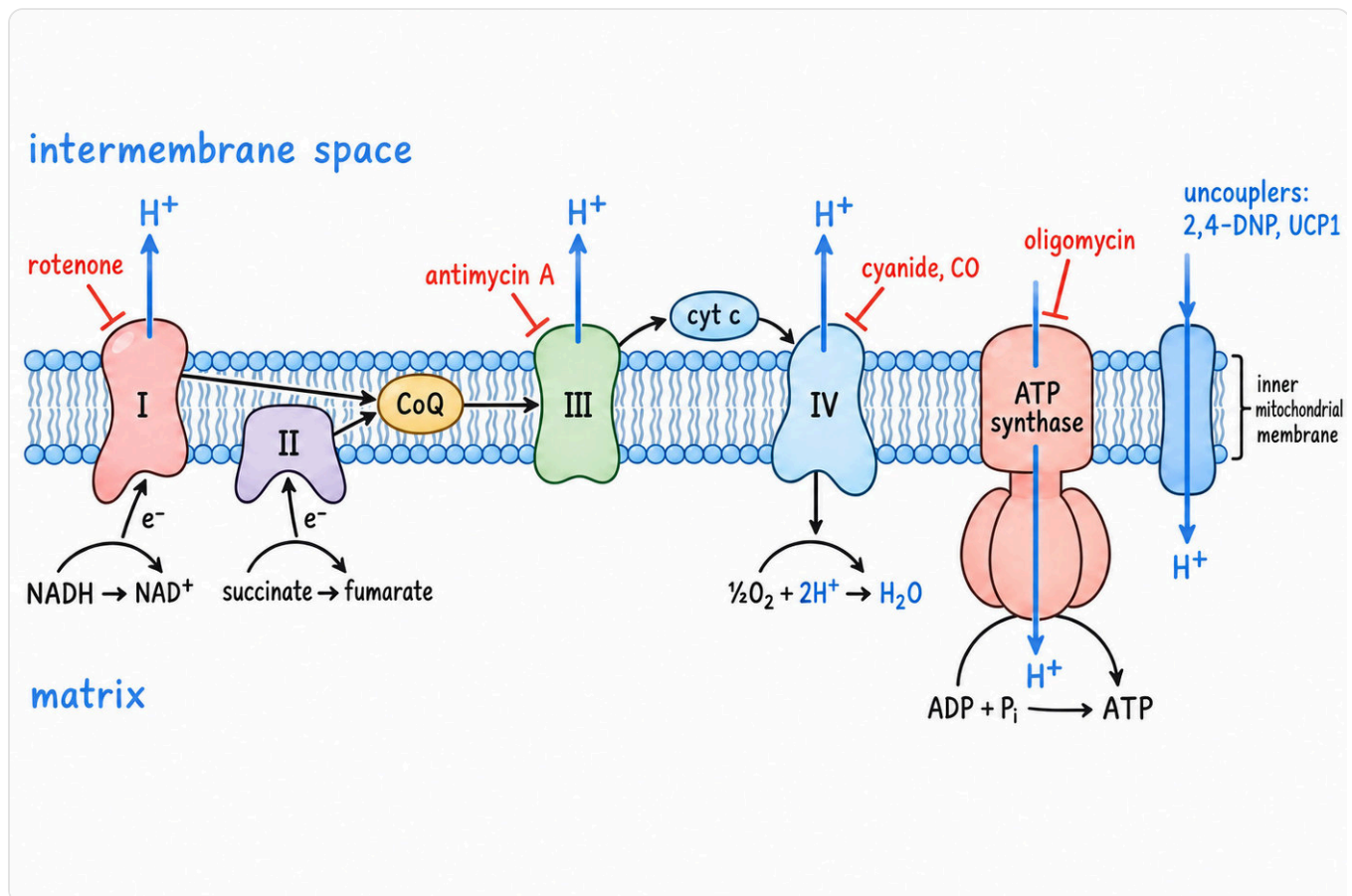


fig. 13 Electron transport across the inner mitochondrial membrane, proton pumping, ATP synthesis, inhibitors, and uncoupling.

- ETC is in the inner mitochondrial membrane
- O_2 accepts electrons at Complex IV
- CoQ and cytochrome c are mobile carriers
- Cyanide and carbon-monoxide poisoning
- Complex II is also succinate dehydrogenase

chemiosmosis, ATP synthase, and oxidative phosphorylation

- Proton-motive force is the coupling intermediate
- F_0 conducts protons
- F_1 catalyzes ATP formation
- ETC blockade prevents maintenance of the gradient

→ Oligomycin inhibition of ATP synthase

Respiratory-chain inhibitors versus uncouplers

- Inhibitor: electron flow and oxygen consumption fall
- Uncoupler: oxygen consumption rises while ATP falls
- UCP1 supports physiologic thermogenesis
- DNP is a dangerous synthetic uncoupler

→ Cyanide poisoning

cytosolic NADH shuttles and reactive oxygen species

- Shuttles move reducing equivalents, not NADH itself
- Malate-aspartate is more energy efficient
- ROS damage lipids, proteins, and DNA
- Antioxidant enzymes limit mitochondrial oxidant injury

→ Oxidative stress

Dehydrogenases, flavins, and cytochrome a/a₃

- Complex I uses FMN; succinate dehydrogenase/Complex II uses FAD.
- Complex II transfers electrons to coenzyme Q without pumping protons.
- Cytochromes a and a₃ belong to Complex IV, cytochrome c oxidase.
- Cytochrome c transfers single electrons between Complexes III and IV.
- Dehydrogenases transfer reducing equivalents without directly requiring O₂ as their acceptor.

ADP/ATP exchange and atractyloside inhibition

- Adenine nucleotide translocase is also called the ADP/ATP antiporter.
- One ADP enters the matrix as one ATP leaves.
- Atractyloside blocks the adenine nucleotide carrier.
- Phosphate transport and carnitine-acylcarnitine transport use different proteins.

cytochrome c release initiates intrinsic apoptosis

- The relevant redistribution is from the intermembrane space into the cytosol.
- Cytosolic cytochrome c promotes apoptosome assembly with Apaf-1.
- Caspase activation drives proteolysis in intrinsic apoptosis.

→ Mitochondrial outer-membrane permeabilization is an important control point in cell death.

High-energy phosphate compounds versus glucose 6-phosphate

- Phosphoenolpyruvate is a high-energy enol phosphate.
- Phosphocreatine provides a rapidly transferable phosphate reserve.
- Carbamoyl phosphate contains a high-energy mixed anhydride.
- Glucose 6-phosphate is the low-energy phosphate ester in this comparison.

ATP accounting: glycolysis, shuttles, and complete glucose oxidation

- Direct glycolytic yield: two net ATP per glucose.
- Two glycolytic NADH: historical maximum six ATP; modern maximum five ATP.
- Modern complete glucose oxidation: approximately 30–32 ATP.
- Modern total includes four substrate-level ATP equivalents plus oxidative phosphorylation.
- Always state the P/O ratio and cytosolic NADH shuttle when comparing totals.

Test yourself

1. Why does the malate-aspartate shuttle generally support a higher ATP yield than the glycerol-phosphate shuttle?
 - A. It pumps protons through Complex II, unlike the glycerol-phosphate shuttle
 - B. It creates an extra molecule of oxygen per electron pair
 - C. It regenerates matrix NADH entering at I, rather than bypassing I at coenzyme Q
 - D. It converts one NADH into two ATP directly in the cytosol
2. Two obligatorily coupled steps have actual free-energy changes of +12 and -20 kJ/mol. What is the combined change?
 - A. -8 kJ/mol
 - B. +32 kJ/mol
 - C. +8 kJ/mol
 - D. -32 kJ/mol
3. Cyanide and carbon monoxide can rapidly stop oxidative phosphorylation by inhibiting which complex?
 - A. ATP-citrate lyase
 - B. Complex IV
 - C. Complex I
 - D. Complex II

Answers: 1 C · 2 A · 3 B

carbohydrate structure and stereochemistry

Lippincott 6e, Chapter 7 · teacher slides

Disaccharides and bonds

Sugar	Building blocks / linkage	Reducing?
Maltose	Glucose- $\alpha(1\rightarrow4)$ -glucose	Yes
Cellobiose	Glucose- $\beta(1\rightarrow4)$ -glucose	Yes
Lactose	Galactose- $\beta(1\rightarrow4)$ -glucose	Yes
Sucrose	Glucose- $\alpha(1\leftrightarrow2)$ - β -fructose	No: both anomeric carbons linked
Trehalose	Glucose- $\alpha(1\leftrightarrow1)$ - α -glucose	No: both anomeric carbons linked
Amylopectin / glycogen	$\alpha(1\rightarrow4)$ chains, $\alpha(1\rightarrow6)$ branches	Glycogen is more highly branched
Cellulose / chitin	$\beta(1\rightarrow4)$ glucose / N-acetylglucosamine	Structural homopolymers

Monosaccharide classification and stereochemistry

- Aldose versus ketose
- Epimer differs at one chiral center
- Anomer differs at the anomeric carbon
- D/L does not specify optical rotation

→ Galactose and fructose metabolic disorders

Glycosidic bonds and reducing sugars

- Free anomeric carbon permits reduction tests
- Glycosidic linkages are named by alpha/beta and carbon positions
- Sucrose is nonreducing
- Benedict reagent detects reducing sugars

→ Reducing substances in urine

Dietary disaccharides and carbohydrate digestion

- Lactose = glucose + galactose
- Sucrose = glucose + fructose
- Brush-border enzymes finish digestion
- Lactase is commonly deficient

→ Lactose intolerance

Intestinal monosaccharide absorption

- SGLT1 transports glucose and galactose
- GLUT5 transports fructose
- GLUT2 supports basolateral exit
- Na⁺/K⁺-ATPase maintains the driving gradient

→ Oral rehydration therapy

cyclic sugars, hemiacetals, and the ring oxygen

- Glucopyranose is a cyclic hemiacetal.
- A pyranose ring has six ring atoms: five C and one O.
- The ring oxygen forms a C-O-C linkage.
- The former carbonyl carbon becomes the anomeric carbon.

Starch, glycogen, and cellulose

- Plant glucose storage: starch.
- Animal glucose storage: glycogen.
- Amylose: alpha(1→4) bonds without branches.
- Amylopectin and glycogen: alpha(1→4) chains with alpha(1→6) branches.
- Cellulose: beta(1→4)-linked glucose; indigestible to human enzymes.

Past papers asked

question	answer and why
A patient gets abdominal pains and cramps, diarrhea and gas in intestines after consuming milk and dairy products. His problem is related to a defect in which sugar and enzyme below?	Lactose-lactase is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior.
All the following compounds possess a chiral carbon, except	Dihydroxyacetone is the best answer. Dihydroxyacetone has no stereogenic carbon.
Which of the following are anomeric isomers?	α -D-Glucose and β -D-glucose is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior.
Which of the following is a homo-polysaccharide?	Cellulose is the best answer. Cellulose is a homopolymer of glucose.
The enzyme which hydrolyses starch to maltose is	Amylase is the best answer. Amylase cleaves $\alpha(1\rightarrow4)$ bonds in starch, producing maltose and oligosaccharides.
Toothpaste and sugar-free gum are probably sweetened with a sugar alcohol made by reducing the aldehyde carbon of xylose to an alcohol. The resulting compound would be named	Xylitol is the best answer. Reduction of xylose yields the sugar alcohol xylitol.
In which of the following disaccharides the glycosidic linkage is in between anomeric carbons of both monosaccharides?	Sucrose is the best answer. Sucrose links both anomeric carbons, making it nonreducing.
The phenomenon of mutarotation is due to which of the following?	Conversion of α -D-Glucose to β -D-Glucose is the best answer. Mutarotation is interconversion of α and β anomers through the open-chain form.
Which of the following is a deoxy sugar?	L-Fucose is the best answer. L-Fucose is 6-deoxy-L-galactose.

question

answer and why

Galactonic acid is caused by oxidation of which factor in galactose?

Aldehyde group is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior. Oxidation of the aldehyde group gives the aldonic acid. An aldehyde contains a carbonyl group, so both the broad and specific choices describe that same site.

The enzyme which hydrolyses Galactose -Glucose linkage is

Lactase is the best answer. Lactase hydrolyzes the $\beta(1\rightarrow4)$ galactose-glucose bond of lactose.

Which option is correct for Cellobiose?

It has $\beta 1 \rightarrow 4$ glycosidic bond is the best answer. Cellobiose consists of glucose residues joined by a $\beta(1\rightarrow4)$ glycosidic bond.

What units make up Lactose with what bonds?

β -galactose+glucose, $\beta 1 \rightarrow 4$ is the best answer. Lactose is β -galactose linked $\beta(1\rightarrow4)$ to glucose.

How many spatial isomers does the linear form of aldohexoses have?

16 is the best answer. An aldohexose has four stereocenters, giving $2^4 = 16$ stereoisomers.

Which of the following produces N-acetyl glucosamine upon hydrolysis?

Chitin is the best answer. Chitin is a polymer of $\beta(1\rightarrow4)$ -linked N-acetylglucosamine.

Amylopectin has:

$\alpha(1\rightarrow4)$ and $\alpha(1\rightarrow6)$ linkage is the best answer. Amylopectin contains $\alpha(1\rightarrow4)$ glucose chains with $\alpha(1\rightarrow6)$ branch points.

Which of the following is a pair of non-reducing carbohydrate?

Sucrose and trehalose is the best answer. Sucrose and trehalose have both anomeric carbons tied up in their glycosidic bonds, so neither is reducing.

Which of the following disaccharides contains a beta-glycosidic bond?

Lactose and cellobiose is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior. The intended comparison is lactose/cellobiose (both beta-1,4). Sucrose also has a beta-D-fructofuranosyl anomeric linkage, so the literal wording also allows lactose/sucrose.

In polysaccharides, monosaccharides are joined by:

Glycosidic bonds is the best answer. Monosaccharides are joined to one another by glycosidic bonds.

The enzyme which hydrolyses Glucose-Fructose linkage is....

Sucrase is the best answer. Sucrase hydrolyzes the glucose-fructose glycosidic bond of sucrose.

Structural polysaccharides include:

Cellulose and chitin is the best answer. Cellulose and chitin are structural polysaccharides; glycogen and starch are storage polysaccharides.

Test yourself

1. Why can an unabsorbed disaccharide produce diarrhea after a meal?

- A. It retains water in the intestinal lumen
- B. It removes all luminal solutes by active transport
- C. It is converted entirely to absorbed glucose before reaching bacteria
- D. It makes the intestinal lumen effectively hypotonic

2. What is a glycosidic bond?

- A. A phosphodiester bond linking two nucleotide sugars
- B. A covalent linkage from a sugar's anomeric carbon to an O or N group
- C. An ester between a sugar hydroxyl and a fatty-acid carboxyl
- D. A hydrogen bond joining two intact sugar rings without covalent change

3. D-glucose and D-galactose differ in configuration at carbon 4 only. What is their relationship?

- A. C4 epimers
- B. Anomers
- C. Enantiomers
- D. Identical molecules

Answers: 1 A · 2 B · 3 A

Glycoconjugates

Lippincott 6e, Chapter 14 · teacher slides

Glycosaminoglycan structure and major examples

- Most GAGs are highly negatively charged
- Hydration provides resilience
- Hyaluronan is unsulfated
- Keratan sulfate lacks uronic acid

→ Cartilage mechanics

Proteoglycans versus glycoproteins

- Proteoglycan = core protein plus GAG chains
- Glycoprotein glycans are shorter and branched
- Proteoglycans are carbohydrate-rich
- Both are formed through specific glycosylation pathways

→ Basement membranes

N-linked glycosylation and mannose-6-phosphate targeting

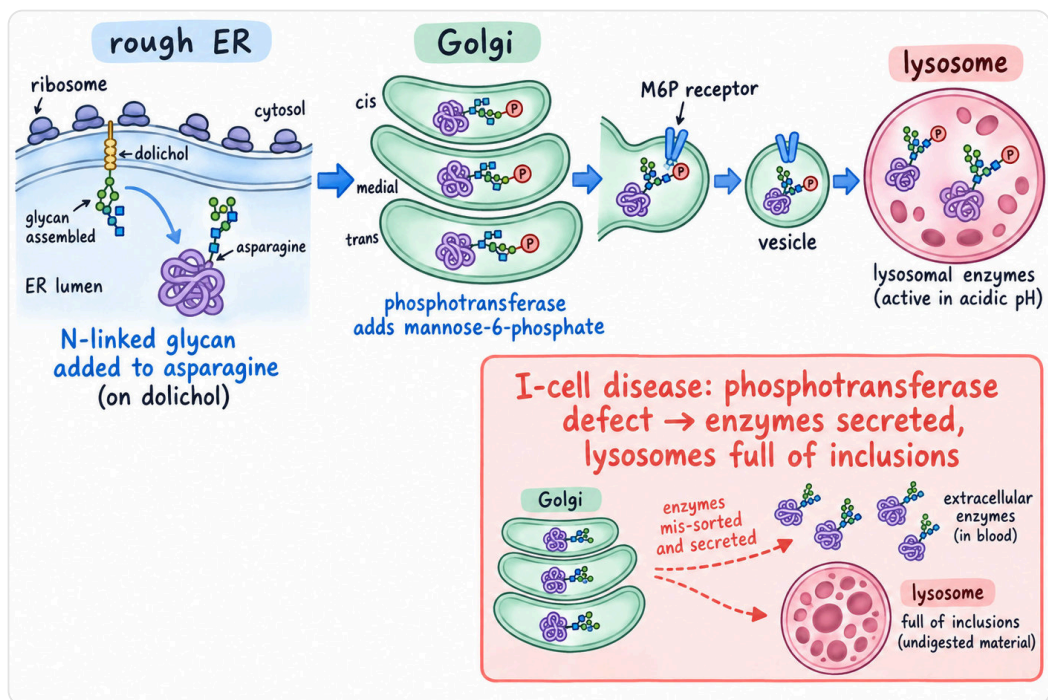


fig. 14 N-Linked glycosylation and mannose-6-phosphate tagging direct lysosomal enzymes from the rough ER through the Golgi to lysosomes.

Mucopolysaccharidoses: Hurler versus Hunter

- Both accumulate dermatan and heparan sulfate
- Hurler is autosomal recessive
- Hunter is X-linked
- Corneal clouding favors Hurler

→ Hurler syndrome

Past papers asked

question	answer and why
Which amino acid is involved in N-glycosidic bond formation?	Asparagine is the best answer. N-linked oligosaccharides attach to the amide nitrogen of asparagine.
All of these polysaccharides are homopolysaccharide except	Heparin is the best answer. Heparin is a heteropolysaccharide with repeating disaccharide units.
Oligosaccharides linked to proteins are called:	Glycoproteins is the best answer. Proteins bearing covalently attached oligosaccharides are glycoproteins.
Which of the following has more than one type of sugar in its structure?	Hyaluronic acid is the best answer. Hyaluronic acid is a repeating disaccharide of glucuronic acid and N-acetylglucosamine.

Test yourself

- Why do most glycosaminoglycans bind large amounts of water?
 - They contain only neutral triglycerides
 - They form peptide bonds with oxygen
 - They contain many negatively charged sulfate and carboxyl groups
 - They are entirely hydrophobic
- Why can deficiency of a single lysosomal exoglycosidase cause accumulation of a large partially degraded GAG?
 - GAG synthesis must stop as soon as degradation slows
 - All other hydrolases require the deficient enzyme as a structural subunit
 - Sequential degradation stops at the residue requiring that enzyme
 - Lysosomal degradation uses no order or substrate specificity
- Before transfer to an asparagine residue, the core N-linked oligosaccharide is assembled on which carrier?
 - Carnitine in the mitochondrial matrix
 - Dolichol pyrophosphate in the ER membrane
 - Albumin in plasma
 - Biotin on acetyl-CoA carboxylase

Answers: 1 C · 2 C · 3 B

Dietary lipids and absorption

Lippincott 6e, Chapter 15 · teacher slides

Lipid classes and fatty-acid structure

- Saturated fatty acids contain no carbon-carbon double bonds
- Cis double bonds disrupt packing and lower melting point
- Longer and more saturated chains generally melt at higher temperatures
- Triacylglycerol is a concentrated hydrophobic energy store
- Complex lipids include additional phosphate, carbohydrate, or nitrogen-containing groups

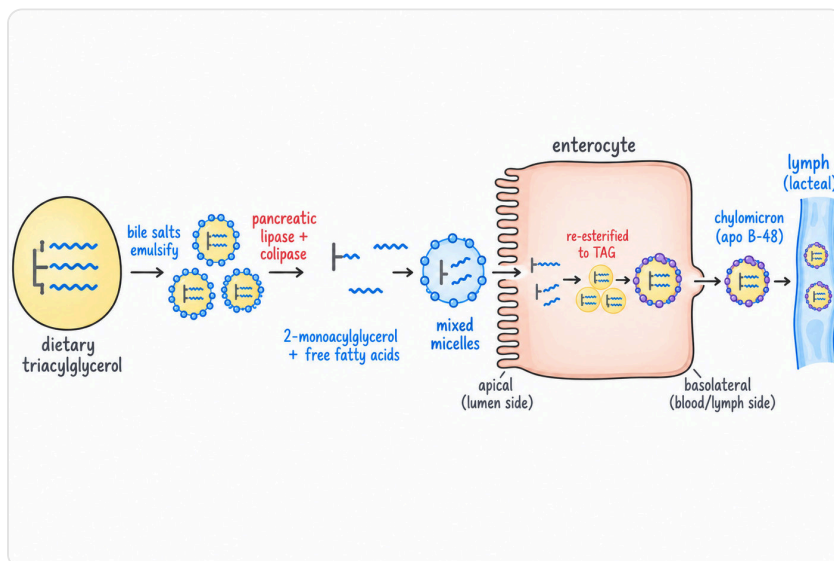
→ Membrane fluidity depends partly on fatty-acid chain length and unsaturation

Enzymatic digestion of dietary lipids

- Acid lipases start digestion
- Pancreatic lipase is the main adult TAG enzyme
- Colipase anchors lipase at the lipid-water interface
- TAG digestion yields 2-MAG plus free fatty acids

→ Pancreatic insufficiency

Bile salts and emulsification



- Bile salts are made in liver and stored in gallbladder
- Conjugation increases amphipathic behavior
- Emulsification increases enzyme-accessible area
- Bile flow is required for fat-soluble vitamin absorption

→ Cholestasis

fig. 15 Dietary triacylglycerol is digested, absorbed, re-esterified, and exported in chylomicrons through intestinal lymph.

Mixed micelles, enterocyte uptake, and re-esterification

- Micelles cross the unstirred water layer
- Micelles are not endocytosed intact
- Long-chain lipids are re-esterified
- Short/medium chains can use portal blood directly

→ Fat malabsorption

chylomicron assembly, clearance, and steatorrhea

- ApoB-48 and MTP are required for assembly
- Chylomicrons travel through lymph
- ApoC-II activates LPL
- ApoE supports remnant uptake

→ Familial chylomicronemia

Drug targets in intestinal lipid handling

- Orlistat inhibits gastric and pancreatic lipases.
- Less TAG digestion means less formation of absorbable fatty acids and 2-monoacylglycerol.
- Ezetimibe inhibits NPC1L1-mediated intestinal cholesterol uptake.
- NPC1L1 inhibition is distinct from HMG-CoA reductase inhibition by statins.

→ Orlistat can produce steatorrhea.

Test yourself

1. After absorption of a meal containing long-chain fatty acids, which route do newly assembled chylomicrons initially use?

- | | |
|--|--|
| A. Excretion into bile before any entry into blood | B. Intestinal lymph before entering systemic blood |
| C. Direct passage through the portal vein as intact chylomicrons | D. Direct secretion into intestinal blood as albumin-bound TAG |

2. Where are chylomicrons assembled and what protein is essential for assembly?

- | | |
|---|---|
| A. Enterocyte endoplasmic reticulum; microsomal triglyceride transfer protein | B. Enterocyte lysosomes; hormone-sensitive lipase |
| C. Liver ER; apoB-100 as the defining structural apoprotein | D. Intestinal lumen; pancreatic lipase as the structural particle component |

3. Why do cis double bonds generally lower the melting point of a fatty acid?

- | | |
|---|-----------------------------------|
| A. They introduce bends that hinder tight packing | B. They remove all hydrophobicity |
| C. They convert the fatty acid into a sterol | D. They create peptide bonds |

Answers: 1 B · 2 A · 3 A

Fatty acids, triacylglycerols and ketones

Lippincott 6e, Chapter 16 · teacher slides

Fatty-acid activation and the carnitine shuttle

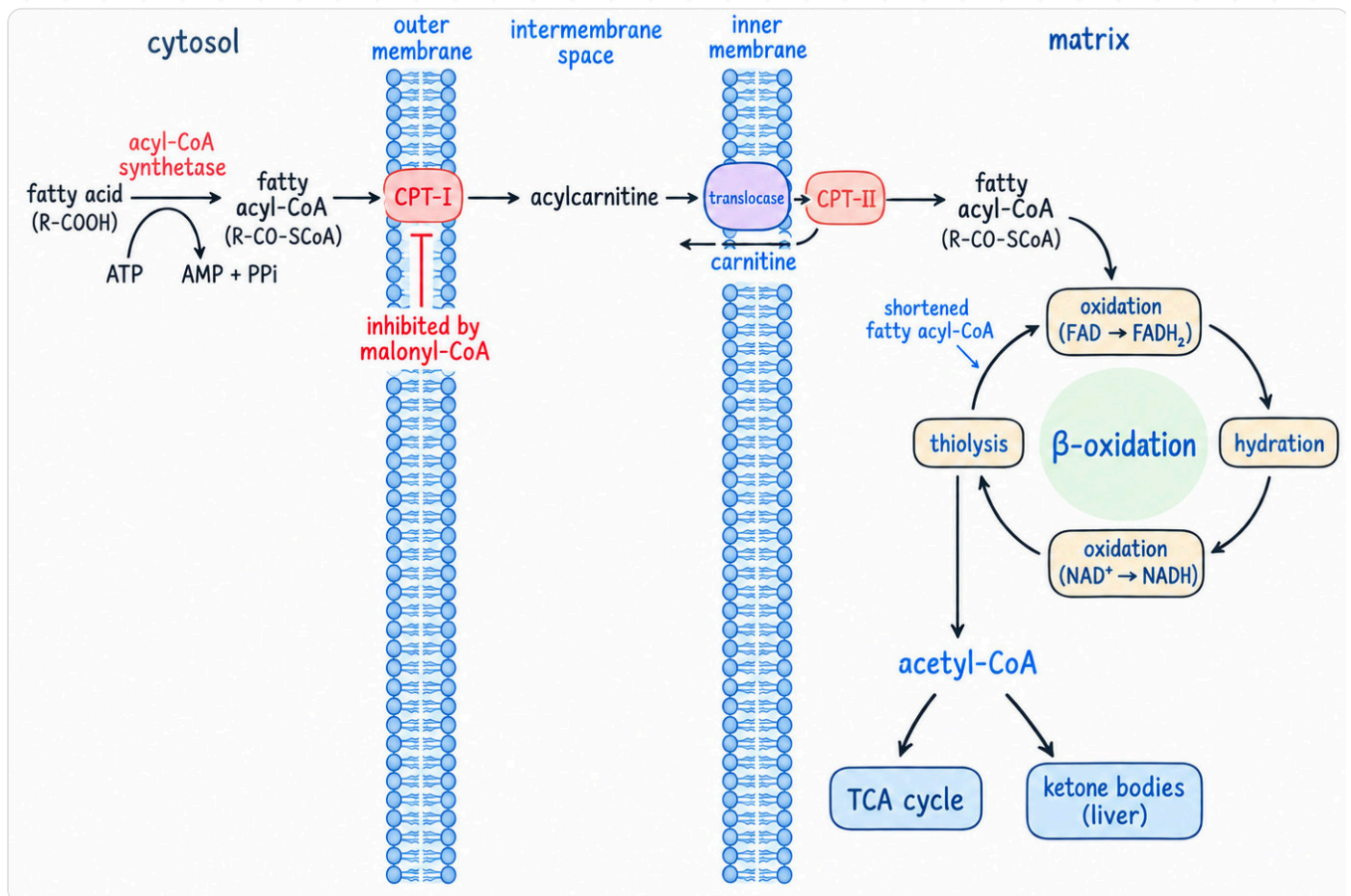


Fig. 16 The carnitine shuttle transports long-chain fatty acyl groups into the mitochondrial matrix for β-oxidation.

- Activation forms fatty acyl-CoA
- CPT-I is the regulated entry step
- Carnitine carries long-chain acyl groups
- Malonyl-CoA inhibits CPT-I
- Primary carnitine deficiency

Mitochondrial beta-oxidation and odd-chain fatty acids

- Occurs mainly in mitochondrial matrix
- Four recurring reactions
- Produces NADH, $FADH_2$, and acetyl-CoA
- Propionyl-CoA is glucogenic through succinyl-CoA
- Methylmalonic acidemia

Ketogenesis and ketone-body utilization

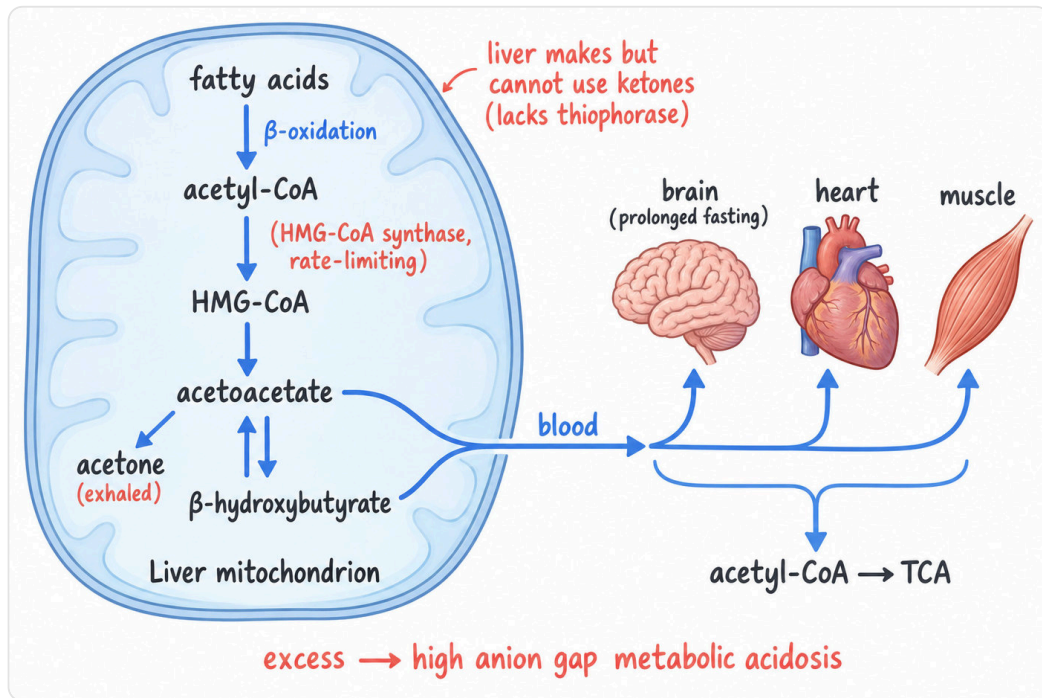


Fig. 17 Ketone body synthesis in liver mitochondria and use by extrahepatic tissues.

Fatty-acid synthesis and reciprocal control

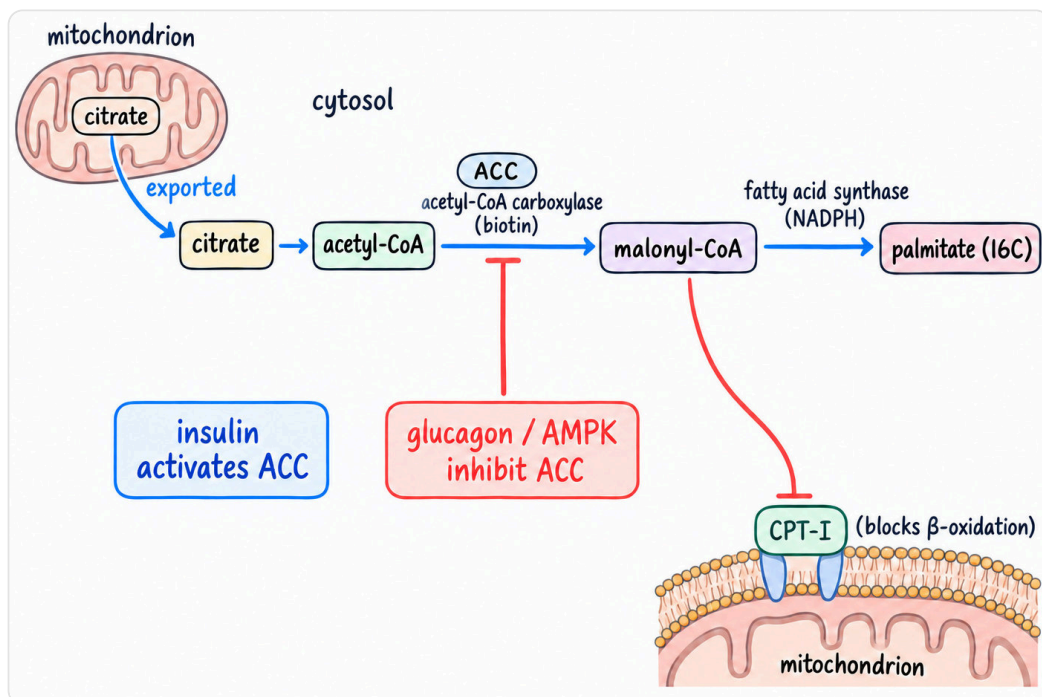


Fig. 18 Cytosolic fatty acid synthesis and regulation of acetyl-CoA carboxylase.

Triacylglycerol synthesis and adipose lipolysis

- Glycerol 3-phosphate supplies the TAG backbone
- Insulin promotes adipose TAG storage
- Fasting activates lipolysis
- Released fatty acids bind albumin
- Glycerol is handled mainly by liver

→ Fasting hyperlipolysis

Fatty-acid oxidation disorders and hypoketotic hypoglycemia

- Low glucose plus low ketones suggests FA-oxidation failure
- MCAD is autosomal recessive
- Avoid fasting
- Dicarboxylic acids reflect omega oxidation

→ MCAD deficiency

Carnitine synthesis from lysine and methionine

- Lysine provides carnitine's carbon skeleton.
- Methionine, through SAM, supplies methyl groups.
- Liver and kidney synthesize carnitine; muscle depends on uptake.
- The carnitine shuttle transfers long-chain acyl groups into mitochondria.

→ A defective carnitine transporter can impair cardiac and skeletal-muscle fatty-acid oxidation.

Citrate export, malic enzyme, and the acyl carrier arm

- Citrate crosses the inner membrane; acetyl-CoA does not diffuse across it.
- ATP-citrate lyase produces cytosolic acetyl-CoA.
- Malic enzyme produces pyruvate, CO₂, and NADPH from malate.
- The ACP phosphopantetheine arm derives from vitamin B5.
- Citrate activates ACC; malonyl-CoA supplies FAS and inhibits CPT-I.

Smooth-ER desaturation and essential fatty acids

- The principal site of long-chain fatty-acid desaturation is smooth ER.
- Delta numbering begins at the carboxyl carbon.
- Humans lack delta-12 and delta-15 desaturases.
- Linoleic acid is omega-6; alpha-linolenic acid is omega-3.

Beta-oxidation reaction accounting and ATP conventions

- A standard cycle gives one NADH, one FADH₂, and an acetyl-CoA; final cleavage gives two acetyl-CoA.
- Enoyl-CoA hydratase consumes water.
- Fatty-acid activation to acyl-CoA costs two ATP equivalents.
- Modern palmitate calculation: $8 \times 10 + 7 \times 2.5 + 7 \times 1.5 - 2 = 106$ ATP.
- The sixth-edition legacy calculation gives 129 ATP net.
- A pre-existing double bond can reduce FADH₂ production during oxidation.

odd-chain carbon fate and limits of energy comparisons

- Pentadecanoate (C15:0) yields six acetyl-CoA plus one propionyl-CoA.
- Propionyl-CoA carboxylase uses biotin and ATP.
- Methylmalonyl-CoA mutase uses adenosylcobalamin (B12).
- Succinyl-CoA can contribute carbon to gluconeogenesis.
- Energy accounting depends on whether residual carbon stops at oxaloacetate or is fully oxidized.
- Even-chain acetyl-CoA cannot give net glucose: PDH is irreversible and the TCA cycle has no net carbon gain.

ATGL initiates adipose lipolysis

- ATGL: triacylglycerol to diacylglycerol.
- HSL preferentially hydrolyzes diacylglycerol.
- Monoacylglycerol lipase completes glycerol release.
- Catecholamines promote cAMP/PKA signaling; insulin opposes lipolysis.
- Plasma nonesterified fatty acids are albumin-bound.

ketone-body identities and the fasting hormonal signal

- Ketone bodies: acetoacetate, beta-hydroxybutyrate, and acetone.
- Fasting favors ketogenesis through a low insulin-to-glucagon ratio.
- Acetoacetate is reduced to beta-hydroxybutyrate using NADH.
- Spontaneous acetoacetate decarboxylation produces acetone.
- Mitochondrial HMG-CoA synthase is the rate-limiting enzyme.
- Hepatocytes lack thiophorase (SCOT); RBCs lack the mitochondria needed for ketone oxidation.

→ Excessive ketone acid production contributes to ketoacidosis.

past papers asked

question	answer and why
Which of the following is an essential fatty acid?	Linoleic acid is the best answer. Linoleic and α -linolenic acids are essential because humans cannot introduce the required distal double bonds.
Which of the following fatty acids has the lowest melting point?	Eicosapentaenoic acid is the best answer. More cis double bonds introduce more kinks and lower fatty-acid melting temperature.
Which of the following fatty acid has 16 carbon atoms?	Palmitic acid is the best answer. Palmitic acid is the 16-carbon saturated fatty acid 16:0.
Which of the following statements concerning fatty acids is correct?	One of them is a precursor of prostaglandins is the best answer. Eicosanoids are local signals derived mainly from arachidonic acid.

Test yourself

1. Primary carnitine deficiency: cause, effect, and treatment.

A. A CPT-I catalytic defect that is always completely corrected by ordinary carnitine uptake

B. A transport defect lowers tissue carnitine, impairing long-chain fatty-acid oxidation; carnitine replacement can help

C. A defect in pancreatic colipase that prevents formation of plasma carnitine

D. An MCAD enzyme defect with normal carnitine transport

2. After a carbohydrate-rich meal, increased malonyl-CoA helps prevent newly made long-chain fatty acids from immediately being oxidized by inhibiting what?

A. Fatty-acid synthase

B. Carnitine palmitoyltransferase I

C. Acetyl-CoA carboxylase

D. Citrate lyase

3. Which final product of odd-chain fatty-acid oxidation can enter metabolism as succinyl-CoA?

A. Propionyl-CoA

B. Acetoacetate

C. Malonyl-CoA

D. Palmitoyl-CoA

Answers: 1 B · 2 B · 3 A

complex lipids and eicosanoids

Lippincott 6e, Chapter 17 · teacher slides

Phospholipid structure, membranes, and PIP₂ signaling

- Polar head and hydrophobic tails
- Phospholipids form bilayers
- PLC cleaves PIP₂
- IP₃ raises cytosolic Ca²⁺
- DAG activates PKC

→ Signal-transduction disorders

DPPC, pulmonary surfactant, and neonatal respiratory distress

- DPPC is the major surface-active phospholipid
- Made by type II pneumocytes
- Reduces surface tension
- Deficiency causes neonatal RDS

→ Prematurity

Sphingolipid architecture and sphingolipidoses

- Ceramide is the common sphingolipid core
- Tay-Sachs: GM2 and hexosaminidase A
- Niemann-Pick A/B: sphingomyelinase
- Gaucher: glucocerebrosidase

→ Cherry-red macula

Arachidonic acid, cyclooxygenase products, and NSAIDs

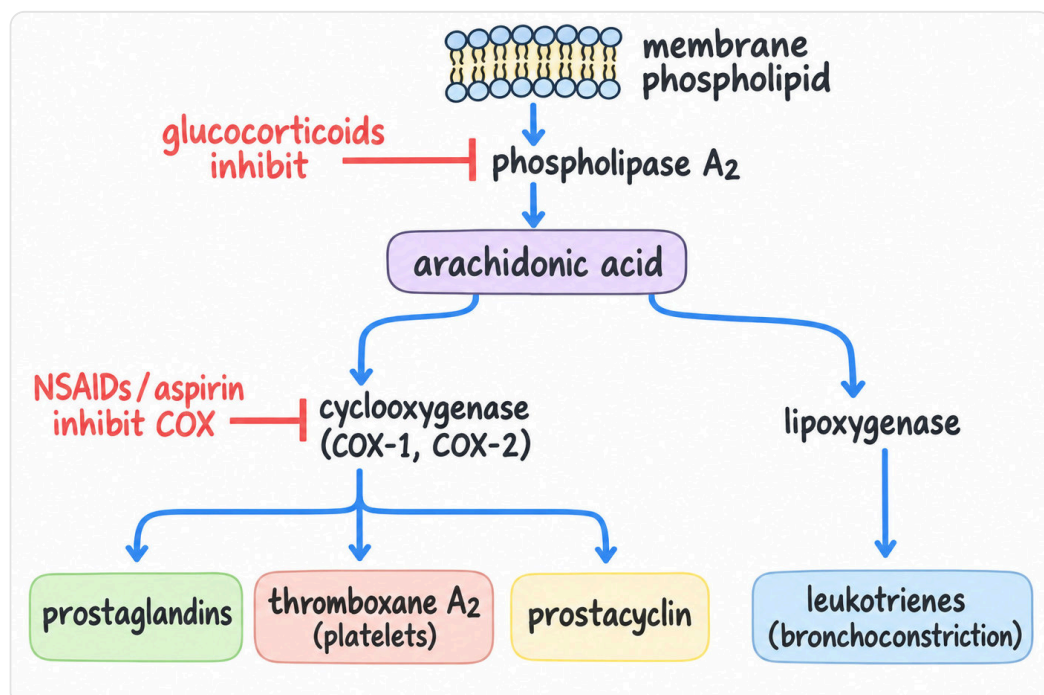


Fig. 19 Arachidonic acid pathways and the targets of glucocorticoids, NSAIDs and aspirin.

Lipoxygenase, leukotrienes, and aspirin-exacerbated asthma

- 5-LOX produces leukotrienes
- Leukotrienes are not blocked by NSAIDs
- Cysteinyl leukotrienes constrict bronchi
- Montelukast blocks their receptor

→ Asthma

Past papers asked

question	answer and why
An example of glycerophospholipid involved in cell signaling is	Phosphatidylinositol is the best answer. Phosphatidylinositol derivatives generate the IP3 and DAG second-messenger pathway. Phosphatidylinositol derivatives participate in signaling; phosphatidic acid is also a signaling lipid. Both choices are valid under this broad stem.
Dipalmitoyl lecithin acts as	Lung surfactant is the best answer. Dipalmitoyl phosphatidylcholine is the major surface-tension-lowering phospholipid of pulmonary surfactant.
Which of the following belong to glycolipids?	Cerebrosides is the best answer. Cerebrosides are glycosphingolipids consisting of ceramide plus one sugar.
What is the starting compound for the synthesis of sphingolipids?	Ceramide is the best answer. Ceramide is the structural core of sphingolipids. Ceramide is the common branch-point precursor for complex sphingolipids. De novo synthesis begins earlier with serine and palmitoyl-CoA; do not mistake the offered intermediate for the first substrates.
Which of the following phospholipids is the main component of inner mitochondrial membrane?	Cardiolipin is the best answer. Cardiolipin is the signature phospholipid of the inner mitochondrial membrane. Cardiolipin is the characteristic, functionally important inner-mitochondrial phospholipid intended by the exam. It is not necessarily the most abundant phospholipid by mass; "main component" is imprecise.
Which membrane fatty acids serve as precursors for eicosanoids?	Eicosanoids are generated from long-chain polyunsaturated fatty acids released from membrane phospholipids; arachidonic acid is the major C20 precursor. This paper uses eicosanoid loosely for lipid mediators from C20/C22 PUFAs. Strictly, eicosanoids derive from C20 precursors; C22 DHA gives docosanoids. The offered D is the intended broad PUFA choice, not the precise definition.
The backbone of phospholipids is	L-glycerol 3-phosphate is the best answer. Mammalian glycerophospholipids use the sn-glycerol-3-phosphate backbone. The intended answer describes glycerophospholipids: sn-glycerol-3-phosphate (historically L-glycerol-3-phosphate). Not all phospholipids have glycerol: sphingomyelin has a sphingosine backbone.
Non-steroidal anti-inflammatory drugs (NSAIDs) like aspirin and ibuprofen act by blocking the production of	Prostaglandins is the best answer. Aspirin and ibuprofen inhibit cyclooxygenase and therefore reduce prostaglandin synthesis.
Sphingosine is not a component of....	Cardiolipin is the best answer. Cardiolipin is a glycerophospholipid; ceramide, sphingomyelin, and gangliosides contain a sphingosine-derived backbone.
Which of the following contains an ether-linked alkyl group?	Platelet-activating factor is the best answer. Platelet-activating factor is an ether glycerophospholipid.

Test yourself

1. Why does deficient surfactant particularly increase collapse of small alveoli?

- A. Surface tension creates greater collapsing pressure at smaller radius
- B. Collapsing pressure depends only on radius, never surface tension
- C. Small alveoli have intrinsically zero surface tension without surfactant
- D. Surfactant normally raises alveolar surface tension

2. PIP₂ cleavage products and what each does.

- A. IP₃ activates PKC directly; DAG opens the ER IP₃ receptor
- B. IP₃ releases stored intracellular Ca²⁺; DAG helps activate protein kinase C
- C. Both products diffuse freely into the nucleus as transcription factors
- D. IP₃ lowers cytosolic calcium; DAG inhibits PKC as the normal acute response

3. Which combination forms ceramide?

- A. Cholesterol plus a phosphate
- B. Glycerol plus three ester-linked fatty acids
- C. Sphingosine plus an amide-linked fatty acid
- D. Glucose plus galactose

Answers: 1. A · 2. B · 3. C

cholesterol, lipoproteins and steroids

Lippincott 6e, Chapter 18 · teacher slides

Lipoprotein traffic

Particle	Main role	Recognition point
Chylomicron	Dietary TAG from intestine to tissues	ApoB-48; apoC-II activates LPL; apoE permits remnant uptake.
VLDL → IDL → LDL	Hepatic TAG export, then cholesterol delivery	ApoB-100; LDL receptor recognizes apoB-100/apoE.
HDL	Cholesterol transport and exchange	Nascent HDL is formed in liver AND intestine; apoA-I activates LCAT.
Albumin-bound fatty acids	Nonesterified fatty-acid transport	Not a lipoprotein particle; especially relevant during fasting.

cholesterol synthesis, HMG-CoA reductase, and statins

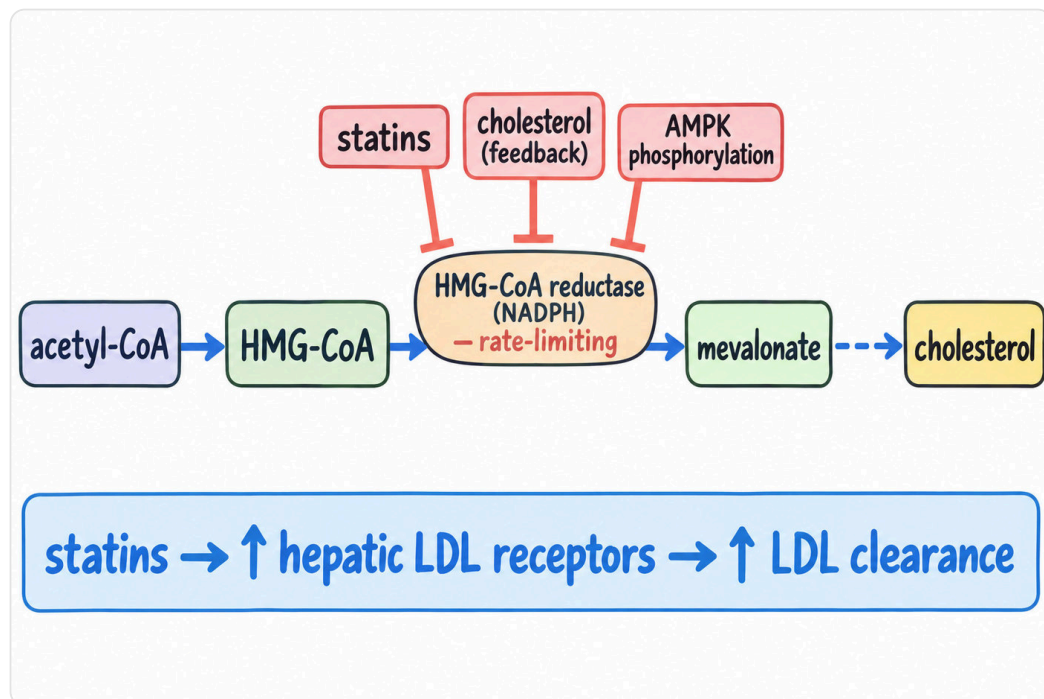


fig. 20 Cholesterol synthesis, reductase inhibition and the effect of statins on LDL clearance.

Bile-acid synthesis and cholesterol elimination

- Biliary secretion is the net elimination route
 - 7- α -hydroxylase is rate limiting
 - Bile salts are amphipathic
 - Most bile salts are recycled
- Cholestasis

Lipoprotein pathways and apolipoprotein map

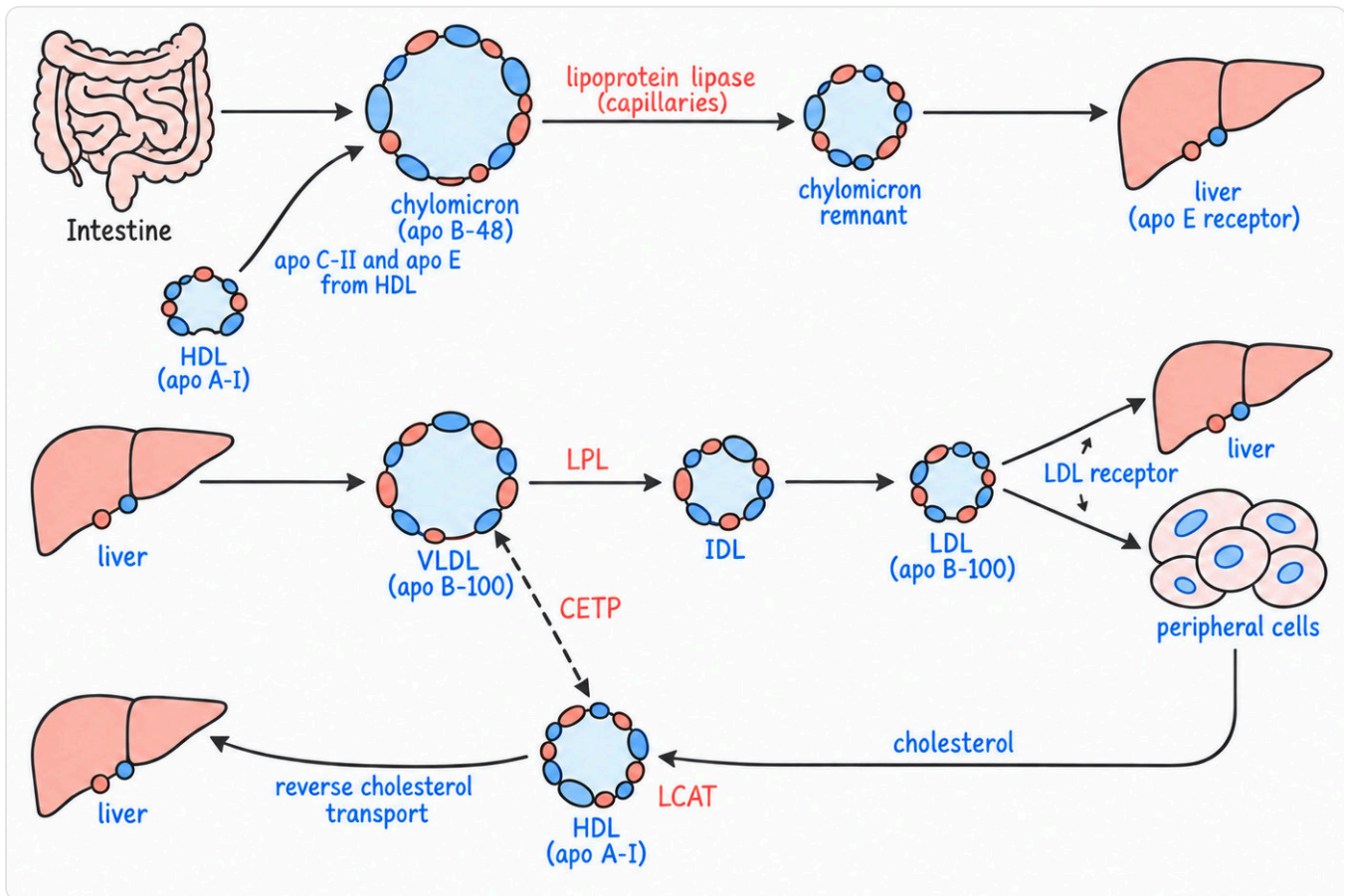


Fig. 21 Exogenous and endogenous lipoprotein transport, receptor uptake, and HDL-mediated reverse cholesterol transport.

- ApoB-48: chylomicron structure
 - ApoB-100: VLDL/LDL structure and LDL-receptor ligand
 - ApoC-II activates LPL
 - ApoE clears remnants
 - ApoA-I activates LCAT
- Familial chylomicronemia

LDL-receptor uptake and atherosclerotic foam cells

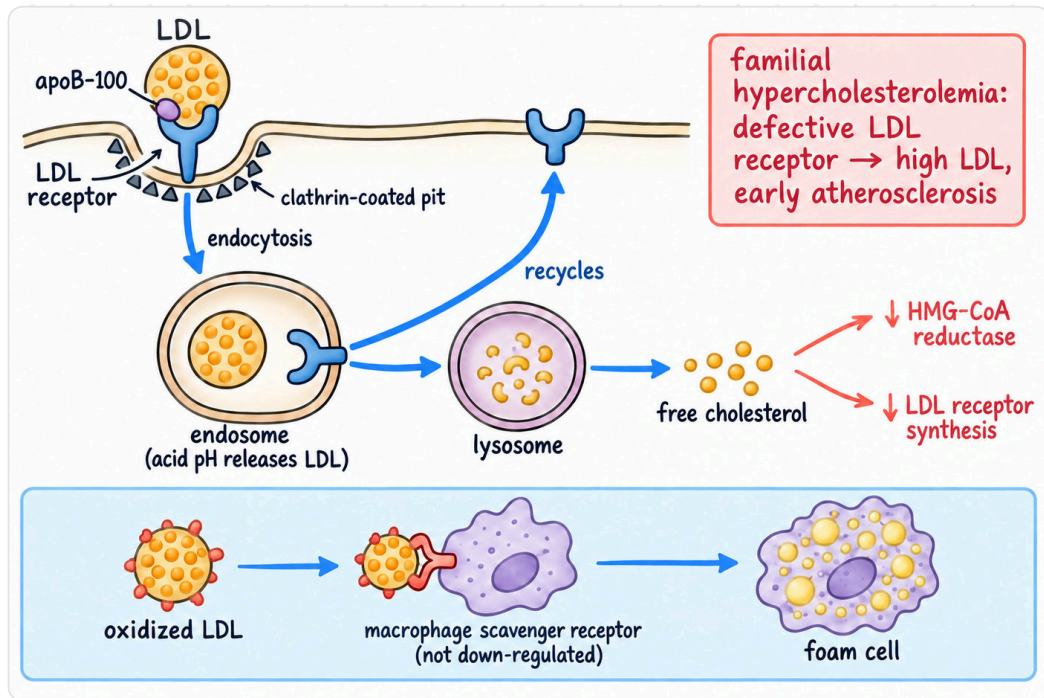


fig. 22 LDL receptor recycling, cholesterol feedback and unregulated scavenger uptake.

HDL maturation and reverse cholesterol transport

- ABCA1 loads cholesterol onto apoA-I
- LCAT esterifies HDL cholesterol
- SR-B1 supports selective hepatic uptake
- CETP exchanges HDL cholesteryl ester for TAG

$\rightarrow$ Tangier disease

cholesterol as steroid precursor and acute StAR control

- Cholesterol is the common precursor
- StAR controls mitochondrial entry
- CYP11A1 makes pregnenolone
- Steroid receptors regulate transcription

$\rightarrow$ Congenital adrenal hyperplasia

LPL, hepatic lipase, and apoB-100/apoE receptors

- LPL hydrolyzes TAG in chylomicrons and VLDL.
- LPL or apoC-II deficiency causes familial chylomicronemia.
- Hepatic lipase helps remodel IDL into LDL.
- LDL-receptor ligands include apoB-100 and apoE.
- ApoB-48 does not supply the apoB-100 LDL-receptor binding site.

$\rightarrow$ Severe chylomicronemia increases pancreatitis risk.

Bile-acid feedback and bile-acid sequestrants

- Bile acids suppress cholesterol 7- α -hydroxylase.
- Terminal-ileal reabsorption maintains the enterohepatic bile-salt pool.
- Sequestrants prevent intestinal bile-acid reabsorption.
- Increased conversion of hepatic cholesterol to bile acids helps lower plasma LDL.

→ Cholestyramine is a bile-acid sequestrant.

Past papers asked

question	answer and why
Dietary fats after absorption appear in the circulation as	Chylomicron is the best answer. Enterocytes package absorbed dietary lipids into chylomicrons.
HDL is synthesized in	Liver and intestine is the best answer. Nascent HDL is produced by both liver and intestine.
The lipoprotein richest in cholesterol is ...	LDL is the best answer. LDL is the most cholesterol-rich major plasma lipoprotein.
Which of the following lipoproteins is produced via the exogenous lipoprotein pathway?	Chylomicrons is the best answer. Chylomicrons transport dietary lipid through the exogenous pathway.
Which one of the following is assembled and secreted by the liver?	Very Low-Density Lipoprotein (VLDL) is the best answer. The liver packages endogenous triacylglycerol into VLDL for secretion. The liver secretes VLDL and also nascent HDL. VLDL is the intended triglyceride-export answer, but the unrestricted stem permits both HDL and VLDL.

Test yourself

- Tendon xanthomas and markedly elevated LDL from childhood most strongly suggest defective function of what pathway?
 - LDL-receptor-mediated clearance
 - Chylomicron lipolysis only
 - Ketone-body synthesis
 - HDL ApoA-I production only
- How does low cellular sterol activate HMG-CoA reductase gene expression?
 - Low sterol retains SREBP-SCAP in ER through stronger INSIG retention
 - SCAP carries SREBP to the Golgi for cleavage; released SREBP activates transcription
 - SREBP binds DNA while permanently anchored to ER membrane
 - Low sterol accelerates proteasomal destruction of every HMG-CoA reductase molecule
- Which apoprotein mediates hepatic uptake of both chylomicron remnants and much of IDL?
 - ApoC-II
 - ApoB-48 only
 - ApoE
 - ApoA-I

Answers: 1 A · 2 B · 3 C

Insulin and glucagon

Lippincott 6e, Chapter 23 · teacher slides

Insulin biosynthesis and glucose-stimulated secretion

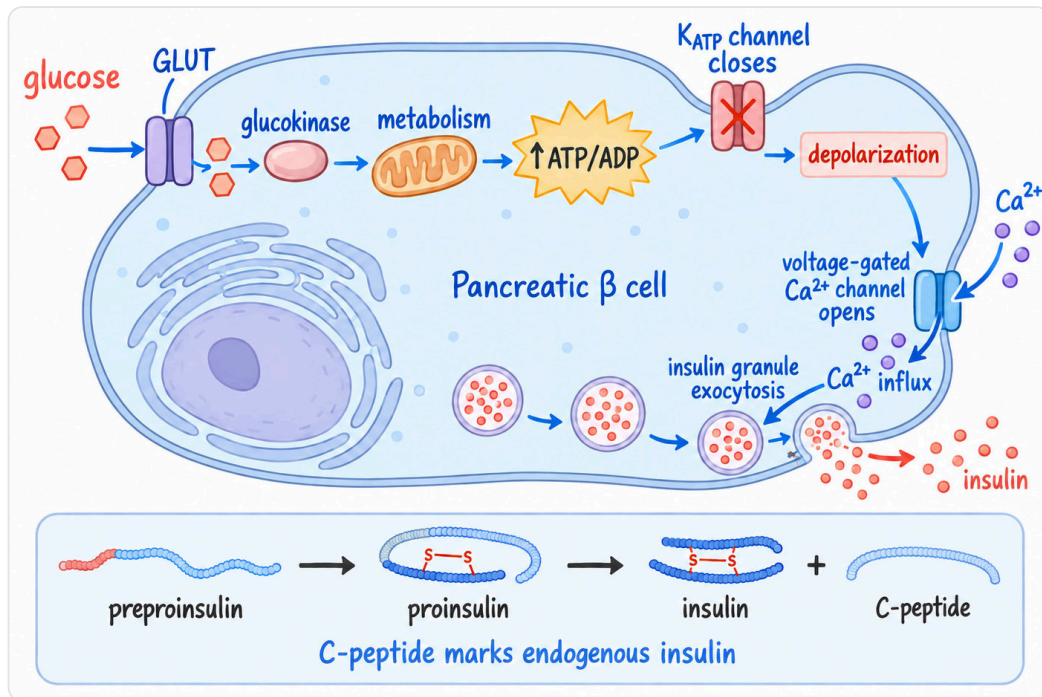


fig. 23 Glucose-stimulated insulin secretion and proinsulin processing.

Insulin receptor signaling and metabolic effects

- Insulin receptor is a receptor tyrosine kinase
- GLUT4 is in muscle and adipose tissue
- Insulin promotes storage and synthesis
- Insulin suppresses hepatic glucose output

→ Insulin resistance

Glucagon signaling and counterregulatory hormones

- Glucagon uses Gs-cAMP-PKA
- Primary target is liver
- No direct glucagon-driven muscle glycogenolysis
- Counterregulation protects against hypoglycemia

→ Glucagon rescue for severe hypoglycemia

Hypoglycemia recognition, mechanism, and biochemical diagnosis

- Autonomic symptoms are early warning signs
- Neuroglycopenia causes confusion, seizures, and coma
- Ethanol inhibits gluconeogenesis through a high-NADH redox shift
- C-peptide distinguishes endogenous from injected insulin
- Interpret insulin and C-peptide only in the context of documented low glucose

→ Insulinoma versus exogenous-insulin exposure

Test yourself

1. During symptomatic hypoglycemia, which laboratory pattern supports an insulin-secreting tumor rather than injected insulin?

- | | |
|--|-------------------------------------|
| A. Normal glucose with high ketones only | B. High insulin with high C-peptide |
| C. High insulin with absent C-peptide | D. Low insulin with low C-peptide |

2. Which metabolic pattern is most consistent with insulin action after a meal?

- | | |
|--|---|
| A. Increased hepatic gluconeogenesis | B. Increased glycogen, fatty-acid, triacylglycerol, and protein synthesis |
| C. Increased hepatic ketogenesis and proteolysis | D. Increased adipose hormone-sensitive lipase |

3. The insulin receptor belongs to which receptor class?

- | | |
|----------------------------------|-----------------------------------|
| A. Receptor tyrosine kinase | B. G-protein-coupled receptor |
| C. Ligand-gated chloride channel | D. Intracellular steroid receptor |

Answers: 1 B · 2 B · 3 A

Fed-fast integration

Lippincott 6e, Chapter 24 · teacher slides

Absorptive-state metabolism by organ

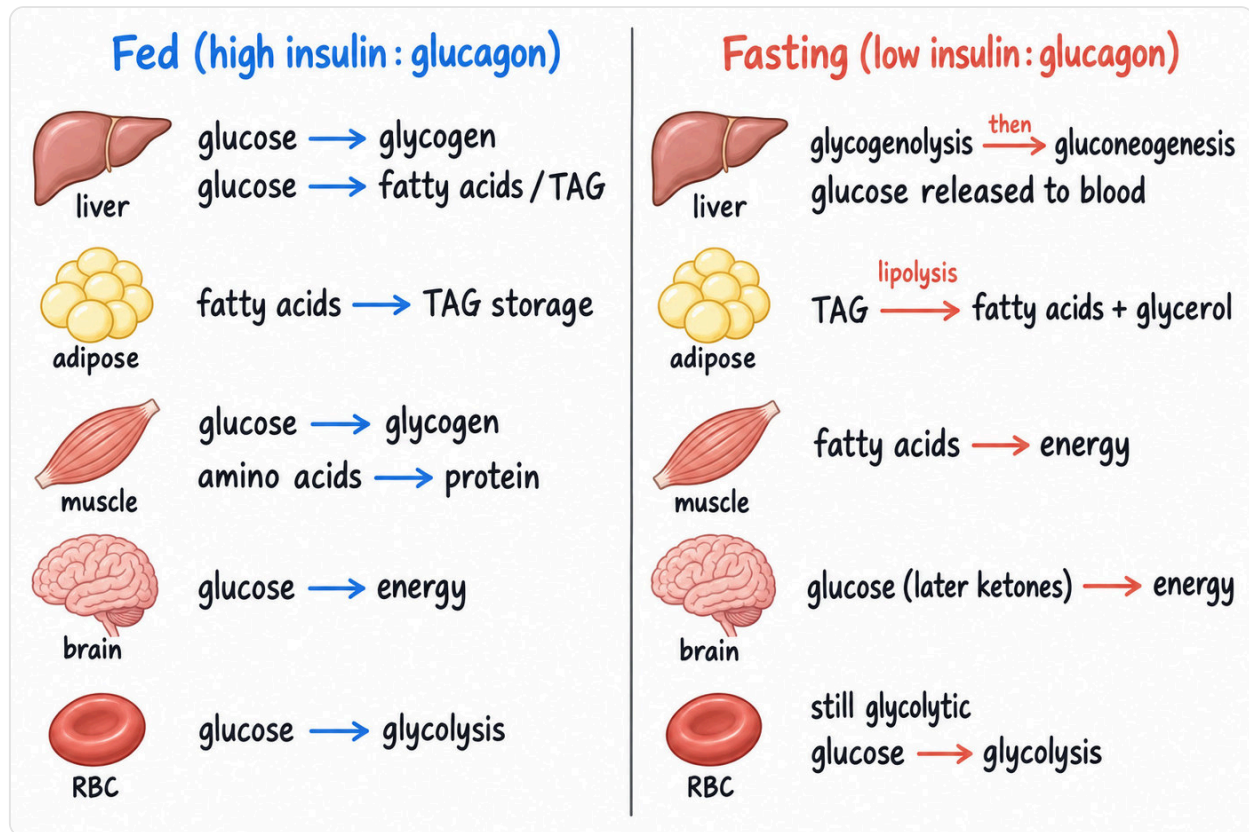


fig. 24 Organ fuel handling in fed and fasting states.

Early fasting: glycogenolysis and gluconeogenesis

- Glycogenolysis dominates first
- Gluconeogenesis rises with fasting duration
- Lipolysis supplies fatty acids and glycerol
- Liver preserves glucose for dependent tissues

→ Fasting hypoglycemia

Prolonged fasting and protein-sparing ketone adaptation

- Brain progressively uses ketones
- Ketones spare muscle protein
- RBCs cannot use ketones
- Liver makes but does not consume ketones

→ Starvation

Tissue-specific fuel constraints

- RBC: glucose only for ATP
- Liver: produces ketones
- Brain: glucose, then more ketones in fasting
- Muscle: flexible fuel use

→ Hypoglycemic neurologic symptoms

Late starvation shifts muscle away from ketones

- Resting muscle favors fatty acids during prolonged starvation.
- Late reduction in muscle ketone use helps preserve ketones for the brain.
- Brain ketone use reduces the need for muscle-protein breakdown.
- These resting-starvation adaptations do not describe all exercise intensities.

Test yourself

1. Why does the liver export ketone bodies rather than oxidize them efficiently itself?

- | | |
|---|------------------------------|
| A. It lacks the succinyl-CoA:acetoacetate CoA-transferase needed for ketone utilization | B. It has no mitochondria |
| C. It cannot transport any ketones across membranes | D. It cannot form acetyl-CoA |

2. Why does skeletal-muscle glycogen primarily support local energy needs rather than directly maintain blood glucose?

- | | |
|--|---|
| A. Muscle lacks glucose 6-phosphatase | B. Muscle has no glycogen phosphorylase |
| C. Muscle glycogen has a different non-glucose monomer | D. Muscle cannot form glucose 6-phosphate |

3. Which tissues can oxidize liver-derived ketone bodies during fasting?

- | | |
|--------------------------------|--|
| A. Adipose lipid droplets only | B. Extrahepatic tissues with mitochondria, including muscle and eventually brain |
| C. Liver itself only | D. Mature red blood cells |

Answers: 1 A · 2 A · 3 B

Diabetes mellitus

Lippincott 6e, Chapter 25 · teacher slides

Type 1 versus type 2 diabetes pathophysiology

- Type 1: absolute insulin deficiency
- Type 2: insulin resistance plus beta-cell dysfunction
- Both cause hyperglycemia
- Type 1 has greater DKA risk

→ Autoimmune type 1 diabetes

Diagnosis and glycemic monitoring

- Repeat abnormal testing when symptoms are absent
- HbA1c reflects recent months
- RBC lifespan affects HbA1c
- Diagnosis and treatment targets are not identical

→ Diabetes screening

DKA versus hyperosmolar hyperglycemic state

- DKA has significant ketonemia and acidosis
- HHS has more extreme hyperosmolality
- Osmotic diuresis drives dehydration
- Potassium balance requires close monitoring

→ Diabetic ketoacidosis

Biochemical basis of chronic diabetic complications

- AGEs alter proteins and vessels
- Sorbitol creates osmotic stress
- Microvascular disease affects retina, kidney, and nerves
- Cardiovascular risk is increased

→ Diabetic retinopathy

Test yourself

1. What is the fundamental defect in type 1 diabetes mellitus?

- A. Defective glucagon receptors only
- B. Autoimmune destruction of pancreatic beta cells causing absolute insulin deficiency
- C. Primary insulin resistance with high insulin forever
- D. Excessive beta-cell mass

2. HbA1c most closely reflects average glycemia over approximately what interval?

- A. Only the current fasting hour
- B. The preceding 2 to 3 months, weighted toward recent weeks
- C. The preceding 10 minutes
- D. Exactly one year

3. A person has insulin resistance but normal fasting glucose and elevated insulin. Which mechanism best explains preserved glucose levels?

- A. Beta-cell insulin release has fallen despite increased tissue demand
- B. Hepatic glucose release has become permanently zero
- C. All insulin-independent tissues have switched to GLUT4
- D. Beta cells have increased secretion to compensate

Answers: 1 B · 2 B · 3 D

Obesity and energy balance

Lippincott 6e, Chapter 26 · teacher slides

Obesity measurement and metabolic risk

- BMI = kg/m^2
- BMI 30 or greater conventionally indicates obesity
- Visceral fat carries higher metabolic risk
- Interpret BMI in clinical context

→ Metabolic syndrome

Leptin, ghrelin, and hypothalamic appetite circuits

- Leptin reflects adipose stores
- Common obesity has leptin resistance
- Ghrelin stimulates appetite
- NPY/AgRP neurons are orexigenic

→ Adaptive hunger after weight loss

Adipose inflammation, adiponectin, and insulin resistance

- Adipose tissue secretes adipokines
- Adiponectin falls in obesity
- Inflammation promotes insulin resistance
- Free-fatty-acid flux worsens ectopic lipid storage

→ Type 2 diabetes

Test yourself

1. Ignoring adaptive changes for this calculation, an adult consumes 2400 kcal and expends 2100 kcal daily for 10 days. What cumulative energy balance results?

- | | |
|---------------|----------------|
| A. -3000 kcal | B. +300 kcal |
| C. +3000 kcal | D. +45000 kcal |

2. What leptin pattern is typical of common obesity?

- | | |
|--|---|
| A. Absent leptin in nearly every patient | B. Low leptin caused by no adipose tissue |
| C. Leptin unrelated to fat mass | D. Elevated leptin with central leptin resistance |

3. Why is upper-body visceral obesity metabolically more hazardous than predominantly gluteofemoral fat?

- | | |
|--|--|
| A. Only visceral fat stores vitamins | B. Visceral fat has high lipolytic and inflammatory activity and drains toward the liver |
| C. Visceral fat contains no triglyceride | D. Gluteofemoral fat always causes diabetes |

Answers: 1 C · 2 D · 3 B

Nutrition

Lippincott 6e, Chapter 27 · teacher slides

EAR, RDA, AI, and tolerable upper intake level

- EAR covers about half of a group
- RDA covers nearly all healthy individuals
- AI substitutes when evidence is insufficient
- UL addresses toxicity risk

→ Dietary counseling

Energy density and total energy expenditure

- Fat: 9 kcal/g
- Alcohol: 7 kcal/g
- Carbohydrate/protein: 4 kcal/g
- Basal metabolism is usually the largest component
- Protein has high thermic effect

→ Weight-management counseling

Essential fatty acids, limiting amino acids, and nitrogen balance

- Linoleic acid is omega-6
- Alpha-linolenic acid is omega-3
- Essential amino acids must come from diet
- Nitrogen balance reflects protein gain or loss

→ Growth and pregnancy

Kwashiorkor versus marasmus

- Kwashiorkor: edema and fatty liver
- Marasmus: severe wasting
- Low albumin contributes to edema
- Infection commonly complicates both

→ Childhood malnutrition

Test yourself

1. When is an Adequate Intake value used?

- A. Only for energy intake
- B. Whenever everyone has the same requirement
- C. When evidence is insufficient to establish an EAR and RDA
- D. Only when a nutrient is toxic

2. Why does processing dietary protein generally have a greater thermic effect than processing dietary fat?

- A. Dietary fat requires greater nitrogen disposal than protein
- B. Protein handling includes energetically costly amino-acid processing and nitrogen disposal
- C. Absorbed amino acids are stored directly in a large inert protein-free depot
- D. Protein contains no usable chemical energy

3. In a sedentary adult, which component usually accounts for the largest share of total daily energy expenditure?

- A. Voluntary exercise in every person
- B. Urinary glucose loss
- C. Basal or resting energy expenditure
- D. Thermic effect of food only

Answers: 1 C · 2 B · 3 C

Vitamins

Lippincott 6e, Chapter 28 · teacher slides

vitamins: function first, then deficiency

vitamin / active form	central role	Exam association
B1 / TPP	Oxidative decarboxylation; transketolase	Beriberi, Wernicke-Korsakoff; PDH, α -KGDH and branched-chain α -ketoacid DH.
B2 / FMN, FAD	Redox electron transfer	Cheilosis, glossitis; light-sensitive riboflavin can be depleted during phototherapy.
B3 / NAD ⁺ , NADP ⁺	Redox and other NAD-dependent reactions	Pellagra: dermatitis, diarrhea, dementia. Tryptophan-to-niacin synthesis requires B6.
B5 / coenzyme A, ACP	Acyl-group transfer	CoA contains pantothenate; not TPP or PLP.
B6 / PLP	Transamination, decarboxylation and other amino-acid reactions	Sideroblastic anemia, neuropathy/seizures; excess B6 can also cause neuropathy.
B7 / biotin	Carboxylation	Pyruvate, acetyl-CoA, propionyl-CoA and methylcrotonyl-CoA carboxylases; avidin binds biotin.
B9 / THF derivatives	One-carbon transfer; DNA synthesis	Megaloblastic anemia; homocysteine increases, methylmalonate does not.
B12 / methyl- and adenosylcobalamin	Methionine synthase; methylmalonyl-CoA mutase	Megaloblastic anemia plus possible neurologic disease; homocysteine and methylmalonate increase.
C / ascorbate	Collagen hydroxylation, antioxidant, iron absorption	Scurvy; deficient hydroxylation weakens collagen.
A / retinal, retinoic acid	Vision and gene regulation	Night blindness / xerophthalmia; excess preformed vitamin A is toxic and teratogenic.
D / calcitriol	Calcium/phosphate regulation	25-OH-D is the major circulating status marker; 1,25-(OH) ₂ -D is the active hormone.
E / tocopherols	Lipid-phase antioxidant	Hemolysis and neurologic dysfunction in deficiency.
K / reduced vitamin K	γ -carboxylation of glutamate residues	Clotting factors II, VII, IX, X and proteins C/S; warfarin blocks vitamin-K recycling.

vitamin solubility, coenzymes, storage, and toxicity

- B and C are water soluble
- A, D, E, and K are fat soluble
- Coenzymes may be loosely or tightly bound
- Fat malabsorption threatens A/D/E/K
- Storage increases toxicity potential

→ Cholestatic vitamin deficiency

Thiamine (B1): TPP reactions and neurologic deficiency

- Active form is TPP
- Required by three major dehydrogenase complexes
- Required by transketolase
- Deficiency causes Wernicke-Korsakoff and beriberi

→ Alcohol-use disorder

Riboflavin (B2) and niacin (B3) in redox metabolism

- B2 gives FMN and FAD
- B3 gives NAD and NADP
- NADH favors energy production
- NADPH favors reductive work
- Pellagra is niacin deficiency

→ Pellagra

Pantothenate, pyridoxine, and biotin reaction patterns

- B5 supports acyl transfer
- PLP supports amino-acid reactions
- Isoniazid can cause B6 deficiency
- Biotin supports carboxylation
- Avidin binds biotin

→ Isoniazid neuropathy prevention

Folate and B12 one-carbon metabolism

- Folate supports nucleotide synthesis
- B12 requires intrinsic factor
- Both can cause megaloblastic anemia
- Neurologic deficits and methylmalonate favor B12 deficiency
- Folate prevents neural-tube defects

→ Pernicious anemia

vitamins A and D: vision, transcription, and calcium balance

- 11-cis-retinal supports vision
- Retinoic acid controls transcription
- Vitamin D activation uses skin, liver, kidney
- Calcitriol increases calcium/phosphate availability
- A and D can be toxic in excess

→ Night blindness

vitamins E, K, and C:
antioxidant, coagulation,
and collagen roles

- Vitamin E protects membranes
- Vitamin K enables gamma-carboxylation
- PT rises early in vitamin-K deficiency
- Vitamin C supports collagen hydroxylation
- Fat malabsorption threatens E and K

→ Hemolysis in vitamin-E deficiency

calcitriol increases
intestinal absorption and
renal calcium conservation

- Calcitriol is active vitamin D, not a B-vitamin coenzyme.
- The kidney performs the final activating hydroxylation.
- Calcitriol increases intestinal calcium uptake and renal calcium reabsorption.
- Deficiency causes rickets in children and osteomalacia in adults.

Past papers asked

question	answer and why
Which of the following vitamins provides the coenzyme for γ -carboxylation reactions?	Menaquinone is the best answer. Fat-soluble vitamins A, D, E, and K have distinct stored functions and deficiency patterns.
How many vitamins are classified as fat-soluble?	4 is the best answer. There are four fat-soluble vitamins: A, D, E, and K.
Which of the following is correctly matched with the vitamin it is derived from?	Pyridoxal phosphate-Vitamin B6 is the best answer. Pyridoxal phosphate is the active coenzyme form of vitamin B6.
Which vitamin is a methyl group acceptor?	Vitamin B12 is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.
Which of the following is not a clinical outcome of niacin deficiency?	Night blindness is the best answer. Niacin deficiency causes pellagra—dermatitis, diarrhea, and dementia—not night blindness.
Which of these is a vitamin A precursor?	Beta-carotene is the best answer. β -Carotene can be cleaved to form vitamin A.
Which of the following is the active form of pantothenic acid?	Coenzyme A is the best answer. Pantothenate is incorporated into coenzyme A.
Compounds that are needed for enzymes to function properly are	Vitamins is the best answer. Many vitamins are precursors of coenzymes required for enzyme function.
A 60-year old female presents with severe back pain for the past week, she is diagnosed with osteoporosis, a condition resulting from calcium depletion in bones. As treatment which of the following vitamins can be prescribed?	Vit D is the best answer. Vitamin D supports calcium and phosphate absorption and normal bone mineralization.
Which vitamin(s) play important roles in the metabolism of branched-chain amino acids and odd-carbon fatty acids?	Biotin supports propionyl-CoA carboxylase, shared by odd-chain fatty-acid and several amino-acid catabolic routes. B6 and B1 participate elsewhere in branched-chain amino-acid breakdown; they do not supply this shared carboxylase. B12 is also important downstream but is not offered.
Which vitamin deficiency is seen in infants with jaundice who undergo phototherapy?	B2 is the best answer. Phototherapy can degrade riboflavin, making vitamin B2 deficiency a recognized risk.
Which of the following vitamins is involved in the structure of coenzyme A?	Pantothenate (B5) is the best answer. Pantothenate, vitamin B5, is a structural component of coenzyme A.

question

answer and why

In vitamin B6 deficiency, which other vitamin deficiency may also be observed?	B3 is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.
In which reaction does biotin participate?	Carboxylation is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.
Which one of the following is the main form of vitamin D in plasma?	25-Hydroxycholecalciferol (25 (OH)-D3) is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.
Which of the following vitamins is directly involved in the conversion of homocysteine to methionine?	B12 (Cobalamin) is the best answer. Vitamin B12 is the cofactor for methionine synthase.
Which vitamin deficiency causes hemolytic anemia?	E is the best answer. Vitamin E deficiency permits oxidative membrane damage and can cause hemolytic anemia.
Which vitamin has thiazole and pyrimidine ring in its structure?	B1 (Thiamine) is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.
Which vitamin is involved in the synthesis of collagen protein?	C is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.
Deficiency of which one of these vitamins is the major cause of blindness?	Vitamin A is the best answer. Vitamin A is required for retinal function and epithelial integrity; deficiency can progress from night blindness to xerophthalmia.
Which vitamin is a major lipid-soluble antioxidant in cell membrane?	Vitamin E is the best answer. Vitamin E is the principal lipid-phase antioxidant protecting membranes from peroxidation.
All of the following vitamins provide the cofactor for pyruvate dehydrogenase, except...	Folate is the best answer. Pyruvate dehydrogenase requires TPP, lipoate, CoA, FAD, and NAD ⁺ ; folate is not one of its cofactors.
Which vitamin requires cobalt for its activity?	Vitamin B12 is the best answer. Cobalamin is vitamin B12; cobalt is the central metal in its corrin ring.
Which is incorrect statement about vitamin C?	It can be synthesized in the body is the best answer. Humans lack gulonolactone oxidase and therefore cannot synthesize vitamin C.
Deficiency of which one of these vitamins may lead to megaloblastic anemia?	Vitamin B12 is the best answer. Vitamin B12 or folate deficiency impairs DNA synthesis and causes megaloblastic anemia.
In scurvy, which amino acid that is normally a part of collagen is not synthesized?	Hydroxyproline is the best answer. Vitamin C deficiency impairs hydroxylation of proline and lysine in collagen.
Which of the following vitamins provides the cofactor for transamination of amino acids?	Pyridoxine is the best answer. Pyridoxal phosphate, derived from vitamin B6, is the cofactor for aminotransferases.
Thiamine deficiency leads to_____?	Beri-beri is the best answer. Thiamine deficiency causes beriberi and may cause Wernicke-Korsakoff syndrome.

Test yourself

1. Which vitamin supports collagen hydroxylation and enhances intestinal iron absorption?
A. Vitamin B12
B. Vitamin C
C. Vitamin K
D. Vitamin A
2. Which vitamin A form is the chromophore precursor required for vision?
A. Retinal
B. Retinoic acid only
C. Calcitriol
D. Tocopherol
3. Which vitamin is particularly important for protecting membrane lipids against radical chain reactions?
A. Vitamin B1
B. Vitamin K
C. Vitamin B12
D. Vitamin E

Answers: 1 B · 2 A · 3 D

DNA structure, replication and repair

Lippincott 6e, Chapter 29 · teacher slides

DNA repair: do not mix organisms

Mechanism	What is repaired	Key distinction
Base-excision repair	Small base lesions, e.g. uracil in DNA	DNA glycosylase removes the base first; AP processing, resynthesis, ligation follow.
Nucleotide-excision repair	Bulky helix-distorting lesions	Human XP commonly involves XPA-XPG; bacterial UvrABC is not a human gene. XP variant: POLH.
Mismatch repair	Replication mismatches missed by proofreading	Recognizes the newly synthesized strand; human deficiency underlies Lynch syndrome.
Polymerase proofreading	Wrong newly incorporated nucleotide	3' → 5' exonuclease; DNA synthesis itself is 5' → 3'.
Homologous recombination / NHEJ	Double-strand breaks	HR uses homologous template; NHEJ directly rejoins ends and may introduce sequence changes.

Nucleosides, nucleotides, and the phosphodiester backbone

- Nucleoside = base + sugar; nucleotide = nucleoside + phosphate.
- Ribose has a 2'-OH, whereas deoxyribose has 2'-H.
- A and G are purines; C, T, and U are pyrimidines.
- The backbone linkage is 3' → 5' phosphodiester, and sequences are written 5' → 3'.

→ Nucleoside and nucleotide analogs are used as antiviral and anticancer drugs because they can inhibit polymerases or terminate chains.

complementary antiparallel DNA, helix forms, and melting

- The strands are complementary and antiparallel.
- A-T has two hydrogen bonds; G-C has three.
- Base stacking also makes a major contribution to duplex stability.
- B-DNA is physiologic and right-handed; Z-DNA is left-handed.
- Denaturation is strand separation without backbone hydrolysis and can be reversed by controlled annealing.

→ Hybridization-based diagnostics and PCR primer design depend on complementarity and melting temperature.

Nucleosomes, histone H1, and chromatin accessibility

- Core octamer = $2 \times$ H2A, H2B, H3, and H4.
 - Histones bind negatively charged DNA because they are rich in lysine and arginine.
 - H1 is the linker histone.
 - Open euchromatin is generally more transcriptionally active than compact heterochromatin.
 - Histone-tail modifications can recruit proteins that activate or repress genes.
- Histone deacetylase inhibitors are used in selected malignancies because chromatin state can alter gene expression.

Semiconservative, bidirectional replication from origins

- Each parental strand is a template for a complementary daughter strand.
 - An origin opens into a replication bubble with two forks.
 - Bacterial chromosomes usually use one main origin; eukaryotic chromosomes use many.
 - Origin firing is restricted so each segment is copied once per cell cycle.
- Replication stress and defective origin control contribute to genomic instability in cancer.

Helicase, single-strand binding proteins, and topoisomerases

- DnaB is the principal bacterial replicative helicase.
 - SSB prevents reannealing and protects exposed single-stranded DNA.
 - Topoisomerase I changes linking number in steps of one.
 - Topoisomerase II changes linking number in steps of two and includes bacterial DNA gyrase.
- Fluoroquinolones inhibit bacterial gyrase/topoisomerase IV; several anticancer drugs target human topoisomerases.

5' → 3' synthesis, primers, and leading versus lagging strands

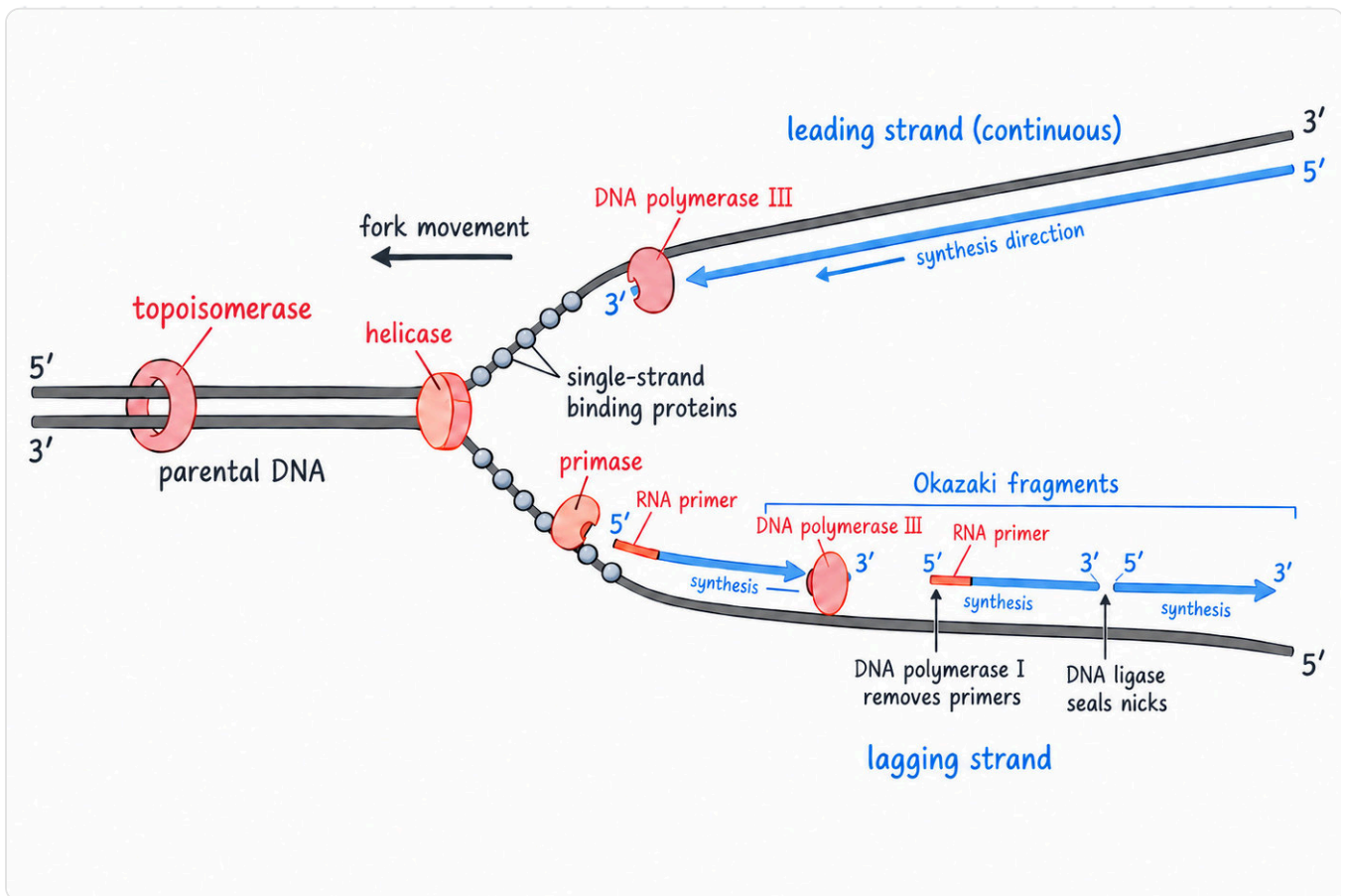


fig. 25 A bacterial replication fork showing continuous leading-strand synthesis and discontinuous lagging-strand synthesis, both in the 5' to 3' direction.

- Polymerase reads template 3' → 5' while synthesizing 5' → 3'.
- A primer supplies the initial paired 3'-OH.
- Leading synthesis is continuous; lagging synthesis is discontinuous.
- Okazaki fragments arise from antiparallel template geometry.

Polymerase specialization, proofreading, primer removal, and ligation

- Bacterial DNA polymerase III performs most fork elongation.
- Proofreading exonuclease is 3' → 5'.
- Bacterial DNA polymerase I removes primers with 5' → 3' exonuclease.
- DNA ligase seals, but does not fill, a nick.
- PCNA is the eukaryotic sliding clamp.

→ Polymerase proofreading defects produce mutator phenotypes and increase cancer susceptibility.

Eukaryotic polymerases and the telomere end-replication problem

- Polymerase α /primase initiates; polymerases ϵ and δ perform most nuclear elongation.
 - Polymerase γ replicates mitochondrial DNA.
 - Telomerase contains RNA plus telomerase reverse transcriptase.
 - Telomerase is active in germ/stem cells and commonly reactivated in cancer.
 - Progressive telomere shortening contributes to replicative senescence.
- Telomerase reactivation supports unlimited proliferation in many cancers.

Mismatch, nucleotide-excision, and base-excision repair

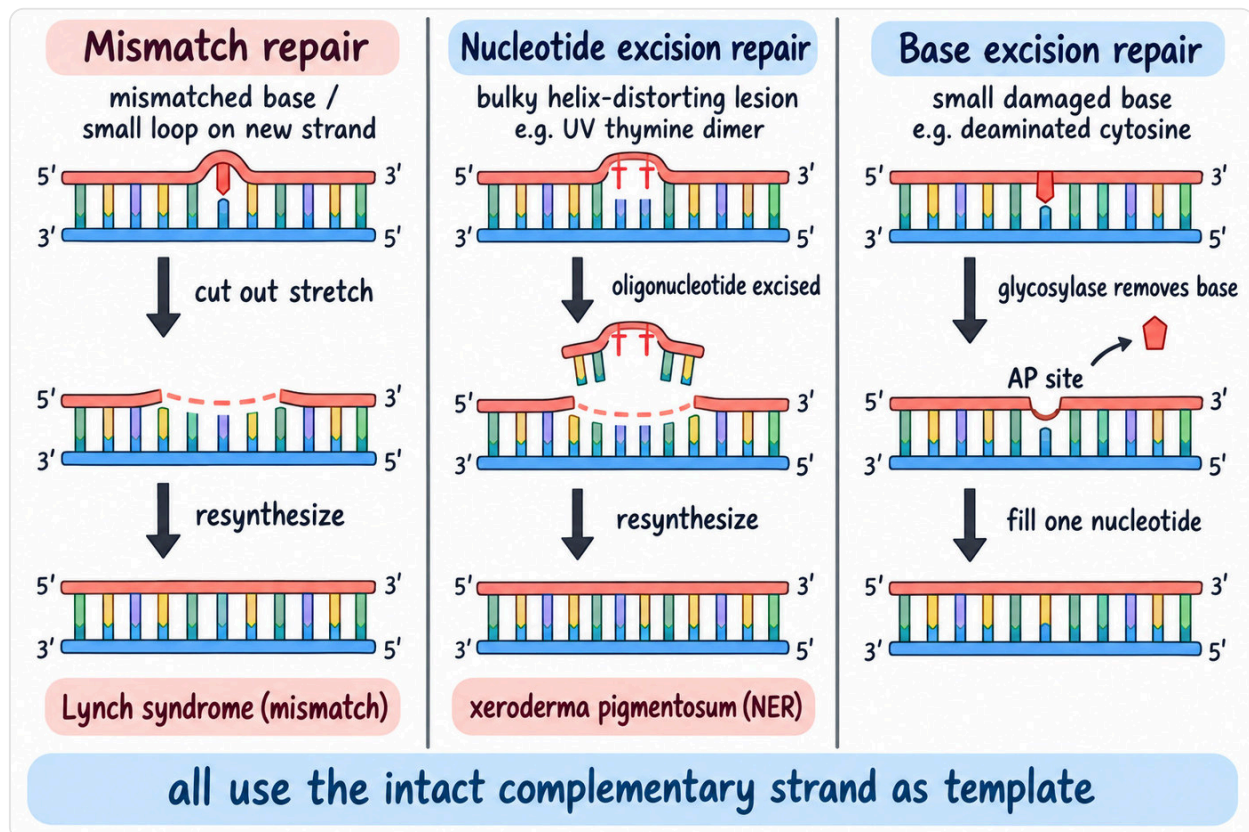


fig. 26 Mismatch, nucleotide excision and base excision repair compared.

Homologous recombination versus nonhomologous end joining

- HR is template directed and favored in S/G2 when a sister chromatid is available.
 - NHEJ is available more broadly but is intrinsically error prone.
 - BRCA1 and BRCA2 participate in homologous recombination.
 - End-joining errors can create deletions and translocations.
- BRCA1/2 loss increases breast, ovarian, prostate, and pancreatic cancer risk.

Past papers asked

question

answer and why

Which of the following enzymes is required for end to end joining of DNA?

DNA ligase is the best answer. Replication requires directional synthesis, proofreading, and coordinated fork enzymes.

question

answer and why

Which statement about DNA structure and denaturation is true?	GC-rich DNA has greater thermal stability and a higher melting temperature than AT-rich DNA because GC base pairs provide three hydrogen bonds and stronger stacking interactions.
Xeroderma pigmentosum results from a defect in	Editorial repair: the source offered bacterial UvrABC as the human xeroderma pigmentosum defect. Human XP usually involves NER proteins XPA-XPG; the XP variant involves POLH/translesion synthesis. UvrABC is bacterial, not a human gene.
Which one of the following nucleic acids has a left-handed helix?	Z-DNA is the best answer. Z-DNA is a left-handed helix.
Okazaki fragments are formed during the synthesis of	DNA is the best answer. Okazaki fragments are short DNA segments synthesized on the lagging strand.
Ciprofloxacin inhibits.....	DNA gyrase is the best answer. Ciprofloxacin inhibits bacterial DNA gyrase/topoisomerase II, disrupting DNA topology and replication.
Which of the following mechanisms will remove uracil and incorporate the correct base?	Base excision repair is the best answer. DNA repair pathways recognize a lesion, remove it, resynthesize DNA, and seal the strand.
According to Chargaff's rule, in a DNA molecule the percentage of	T+G=A+C is the best answer. Because A=T and G=C in double-stranded DNA, T+G equals A+C.
The enzyme responsible for the removal of supercoiling in replicating DNA, ahead of the replication fork, is	Topoisomerase is the best answer. Topoisomerase relieves torsional stress ahead of the replication fork.
Which of the following reactions is required for proofreading, during DNA replication, by DNA polymerase III?	3' to 5' exonuclease activity is the best answer. Replication requires directional synthesis, proofreading, and coordinated fork enzymes.
Which of the following leads to disruption of nucleosome structure?	Acetylation is the best answer. Histone acetylation weakens histone-DNA attraction and opens chromatin.
Which is an example of RNA-dependent DNA polymerase?	Reverse transcriptase is the best answer. Reverse transcriptase synthesizes DNA from an RNA template.
The proofreading function of DNA polymerase involves all of the following except	Reversal of the polymerization reaction is the best answer. Proofreading detects a mismatch and removes it with 3'→5' exonuclease activity; it is not reversal of polymerization.
When double-stranded DNA is heated at neutral pH, which change does not occur?	The covalent N-glycosidic bond between the base and the pentose breaks is the best answer. Heat disrupts hydrogen bonding and base stacking, but does not cleave the covalent base-sugar N-glycosidic bond.
Histones are _____ that are usually associated with	Basic proteins - DNA is the best answer. Histones are basic, lysine/arginine-rich proteins that bind negatively charged DNA.
The compound that consists of ribose linked by an N-glycosidic bond to N-9 of adenine is	Adenosine is the best answer. Adenine plus ribose is the nucleoside adenosine; a phosphate would make it a nucleotide.
In base-excision repair, the first enzyme to act is	DNA glycosylase is the best answer. A DNA glycosylase first removes the damaged base, creating an abasic site.
When a DNA molecule is described as replicating bidirectionally, it means that it has two ...	Replication forks is the best answer. Bidirectional replication produces two forks moving away from an origin.

Test yourself

1. A nucleoside consists of which components?

- A. A base plus sugar plus phosphate
- B. A phosphate plus fatty acid
- C. A nitrogenous base plus a pentose sugar
- D. A sugar plus amino acid

2. How can histone-tail acetylation favor gene expression?

- A. It increases positive charge and closes chromatin
- B. It neutralizes positive charge and loosens histone-DNA interaction
- C. It cleaves DNA into Okazaki fragments
- D. It methylates every cytosine

3. What is nucleotide excision repair (NER) best known for fixing (in this chapter)?

- A. Single-base uracil lesions removed by uracil-DNA glycosylase
- B. Replication insertion-deletion loops recognized by mismatch-repair proteins
- C. Bulky helix-distorting lesions such as UV-induced pyrimidine dimers
- D. Chromosome-end shortening corrected by telomerase

Answers: 1 C · 2 B · 3 C

RNA, transcription and processing

Lippincott 6e, Chapter 30 · teacher slides

RNA types and bacterial versus eukaryotic mRNA architecture

- mRNA is the translated RNA; tRNA decodes; rRNA forms the catalytic ribosome.
- RNA uses ribose and uracil and is usually single stranded.
- Bacterial transcription and translation can be coupled.
- Eukaryotic pre-mRNA is processed in the nucleus before translation.

Primer-independent bacterial transcription initiation

- RNA polymerase synthesizes RNA 5' → 3' without a primer.
- Sigma confers promoter specificity but is not the elongating catalytic core.
- Canonical bacterial promoter elements are near -35 and -10.
- Promoter sequences are not themselves transcribed.

→ Rifampicin selectively inhibits bacterial RNA polymerase and is central to tuberculosis therapy.

Transcript direction, coding/template strands, and bacterial termination

- RNA grows 5' → 3' and the template is read 3' → 5'.
- Coding-strand sequence corresponds to RNA with T replaced by U.
- Sigma commonly dissociates after promoter clearance.
- Intrinsic termination requires a hairpin plus a U-rich tract.
- Rho binds RNA and catches a paused polymerase.

Division of labor among RNA polymerases I, II, and III

- Pol I: 45S pre-rRNA → 18S, 5.8S, 28S.
- Pol II: pre-mRNA and selected sn/miRNAs.
- Pol III: tRNA and 5S rRNA.
- Pol I acts in the nucleolus; Pol II and III act in the nucleoplasm.

→ α -Amanitin strongly inhibits RNA polymerase II and, at higher doses, polymerase III.

RNA polymerase II promoters and general transcription factors

- TATA/Hogness box is a core promoter element near -25 in many genes.
- TBP is part of TFIID.
- General factors position Pol II and create an open complex.
- TFIID has helicase and kinase activities.

→ Defects in TFIID-associated nucleotide-excision repair/transcription proteins can cause xeroderma pigmentosum or trichothiodystrophy.

5' capping and 3' polyadenylation of pre-mRNA

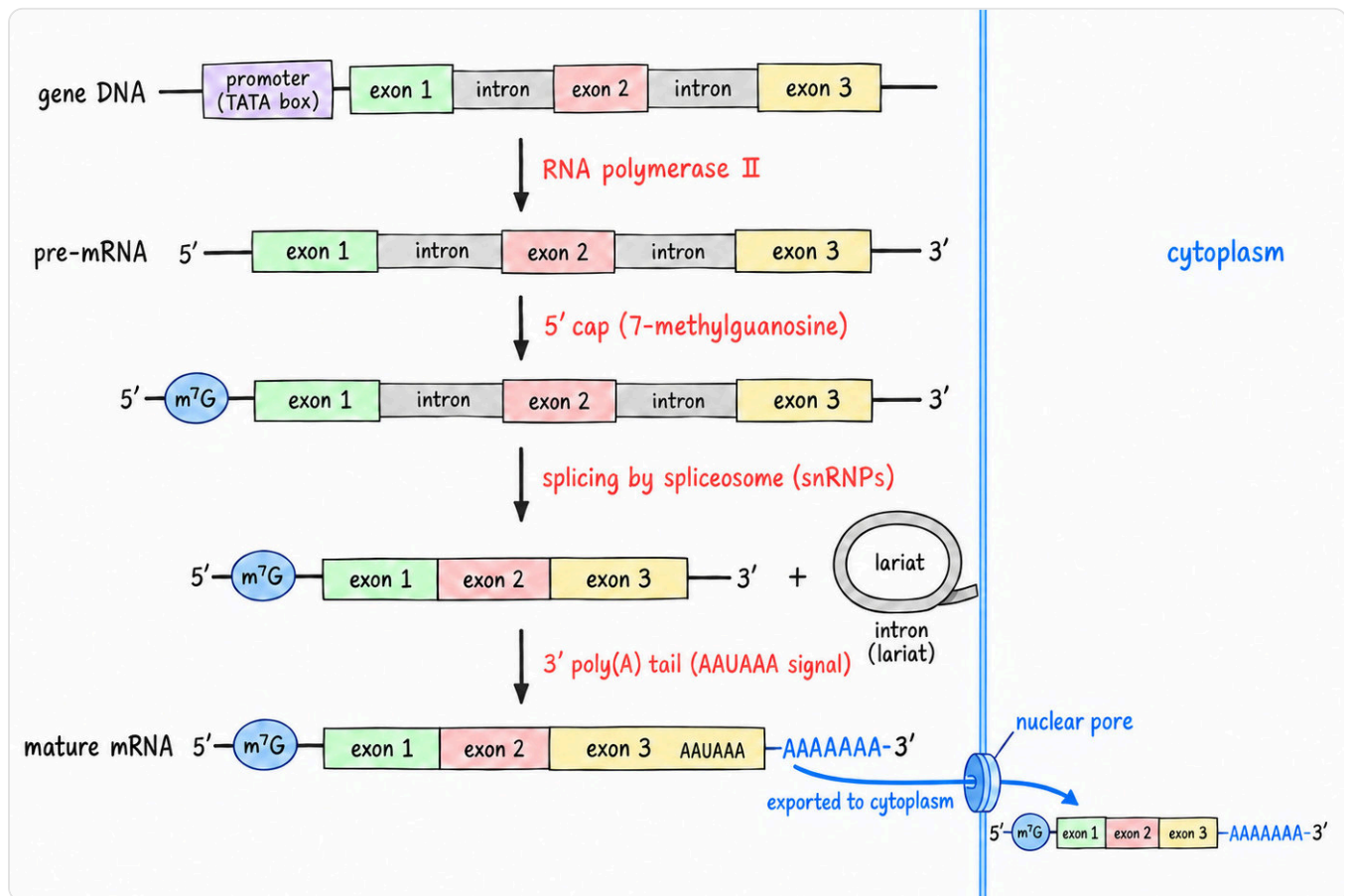


fig. 27 Eukaryotic pre-mRNA receives a 5' cap, undergoes intron splicing, and gains a 3' poly(A) tail before nuclear export.

- The cap is 7-methylguanosine linked 5'-5'.
 - The cap supports stability, export, and ribosome recruitment.
 - AAUAAA directs cleavage/polyadenylation.
 - Poly(A) polymerase adds the non-templated tail.
 - Histone mRNAs are an important poly(A)-tail exception.
- Mutations in polyadenylation signals can reduce normal mRNA production, including in globin disorders.

Spliceosome mechanism, splice-site disease, and alternative splicing

- Introns are removed and exons are ligated by the spliceosome.
- The branch-point A forms a 2'-5' bond in the lariat.
- Consensus boundaries are usually GU at the 5' end and AG at the 3' end.
- Alternative splicing expands the proteome without adding genes.
- Splice-site mutations can alter the reading frame or introduce premature stops.

→ β -thalassemia can result from abnormal β -globin pre-mRNA splicing.

Transcription inhibitors as mechanism probes and drugs

- Actinomycin D targets DNA and inhibits transcription broadly.
- Rifampicin selectively targets bacterial RNA polymerase.
- α -Amanitin is most potent against Pol II.
- Selectivity depends on whether the target is DNA itself or a domain-specific polymerase.

→ Rifampicin is a key antimycobacterial drug and a strong hepatic enzyme inducer.

past papers asked

question	answer and why
The initiation site for transcription is recognized by?	Sigma factor is the best answer. Bacterial sigma factor gives RNA polymerase promoter specificity.
Splice sites are present in?	Eukaryotic hnRNA is the best answer. Splice sites are present in eukaryotic hnRNA/pre-mRNA before introns are removed.
mRNA is complementary to the nucleotide sequence of?	Template strand is the best answer. mRNA is complementary to the DNA template strand and matches the coding strand except U replaces T.
The structure of tRNA appears like a?	Clover leaf is the best answer. The classic two-dimensional structure of tRNA is a cloverleaf.
Binding of the prokaryotic RNA polymerase to promoter site is inhibited by?	Rifampicin is the best answer. RNA polymerase and its factors copy a DNA template into RNA. Rifampicin is the intended drug. More precisely, it binds the bacterial RNA-polymerase beta subunit and blocks early RNA-chain extension, rather than simply preventing promoter recognition.
Actinomycin D is an inhibitor of?	Transcription is the best answer. RNA polymerase and its factors copy a DNA template into RNA.
What is the role of sigma factor in bacterial RNA polymerase?	Positioning RNA polymerase correctly on the DNA template is the best answer. Sigma positions bacterial RNA polymerase at the correct promoter.
Which of the following is an incorrect statement about mRNA?	B and C are both incorrect. Replication-dependent histone mRNAs are capped but usually lack poly-A tails. Splicing removes introns and joins exons. Polyadenylation is typical, not universal.
RNA polymerase from E. coli (core enzyme alone) has all of the following properties except that it	Recognizes specific start signals in DNA is the best answer. The bacterial core RNA polymerase can elongate RNA but needs sigma factor to recognize specific promoters.
Which of the followings inhibit the DNA-dependent RNA polymerase in prokaryotes?	Rifampicin is the best answer. Rifampicin inhibits bacterial DNA-dependent RNA polymerase.

Test yourself

1. Poly-A tail: list 3 functions.

- A. It supports mRNA stability, nuclear export and efficient translation
- B. It serves as the obligatory coding sequence for a long lysine tail
- C. It replaces every intron with deoxyribose
- D. It supplies heme iron directly to the nascent protein

2. What converts the core E. coli RNA polymerase into a promoter-recognizing holoenzyme?

- A. Binding of DNA ligase
- B. Addition of histone H1
- C. Binding of a ribosomal subunit
- D. Binding of a sigma factor

3. Which RNA species serves as the adaptor between an mRNA codon and an amino acid?

- A. rRNA
- B. snRNA
- C. DNA
- D. tRNA

Answers: 1 A · 2 D · 3 D

Translation

Lippincott 6e, Chapter 31 · teacher slides

Inhibitors: target and process

Agent	Target / action	Process
Rifampicin	Bacterial RNA-polymerase β subunit; early RNA extension	Transcription, not translation
Actinomycin D	DNA intercalation; blocks RNA synthesis	Transcription
Tetracyclines	Prevent aminoacyl-tRNA access to bacterial A site	Translation, 30S
Aminoglycosides	Initiation disruption / misreading	Translation, 30S
Chloramphenicol	Peptidyl-transferase inhibition	Translation, 50S
Puromycin	Aminoacyl-tRNA mimic, premature chain termination	Translation
Ricin	Depurinates 28S rRNA in the 60S subunit	Eukaryotic ribosome inactivation
Diphtheria toxin	ADP-ribosylates eEF2	Eukaryotic translocation
Cycloheximide	Inhibits eukaryotic elongation	Translation, 60S

Genetic code properties, reading frames, and mutation consequences

- Degenerate does not mean ambiguous.
 - AUG usually initiates translation and also encodes methionine.
 - UAA, UAG, and UGA are stop codons.
 - Missense changes an amino acid; nonsense creates a stop; frameshift changes downstream codon grouping.
 - Silent substitutions can still affect splicing or translation efficiency.
- Nonsense and frameshift variants often cause severe loss-of-function genetic disease.

Aminoacyl-tRNA synthetases, proofreading, and wobble

- Each amino acid has a corresponding synthetase specificity system.
 - Charging consumes ATP to AMP, equivalent to two high-energy phosphate bonds.
 - The ribosome checks codon-anticodon pairing but cannot verify the attached amino acid.
 - Wobble explains why fewer tRNAs than sense codons are required.
- Some antimicrobial drugs exploit differences in microbial aminoacyl-tRNA synthetases.

Ribosomal A/P/E sites, peptidyl transferase, and directionality

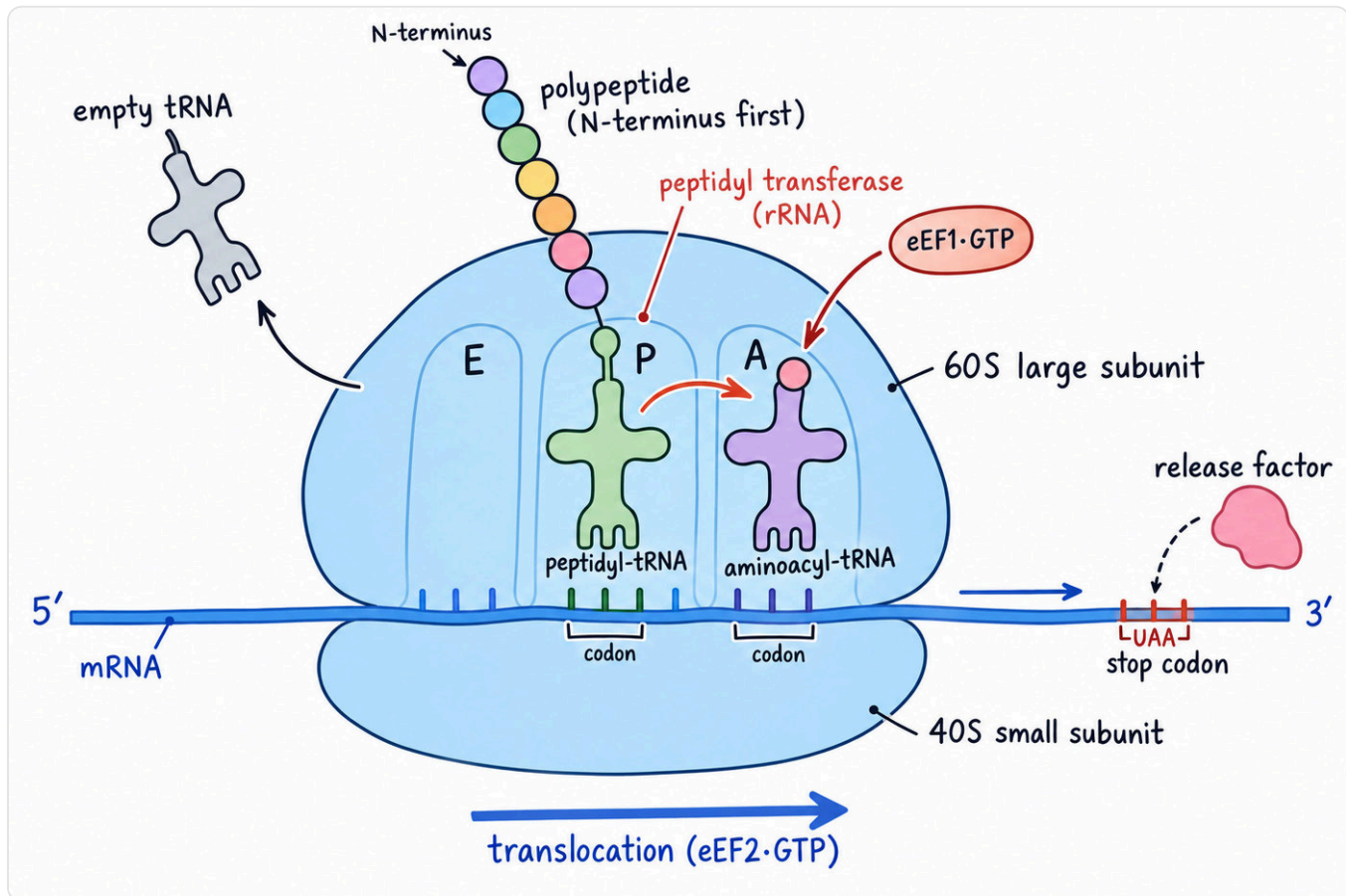


fig. 28 Eukaryotic translation uses the 40S and 60S ribosomal subunits, with tRNAs moving through A, P, and E sites as the ribosome reads mRNA 5' to 3'.

- A = aminoacyl entry, P = peptidyl chain, E = exit.
- Protein elongates N→C.
- Peptide-bond energy comes from the high-energy aminoacyl-tRNA ester bond.
- Bacterial 23S rRNA is the peptidyl-transferase catalyst.
→ Many antibiotics selectively bind bacterial ribosomal sites or the peptidyl-transferase center.

Prokaryotic Shine-Dalgarno alignment versus eukaryotic cap scanning

- Shine-Dalgarno is bacterial; Kozak/cap scanning is eukaryotic.
- Initiator tRNA starts in the P site, not the A site.
- Bacterial initiation uses fMet-tRNA.
- Initiation factors and GTP assemble the complete ribosome only after correct start recognition.

Elongation-factor cycle and translocation

- EF-Tu delivers charged tRNA; EF-Ts reloads EF-Tu with GTP.
- Peptidyl transfer precedes translocation.
- EF-G is the bacterial translocase.
- Eukaryotic homologs include eEF1A and eEF2.
- At least two GTP molecules are hydrolyzed per elongation cycle.
→ Diphtheria toxin and Pseudomonas exotoxin A inactivate eEF2 and stop translocation.

Stop-codon termination, ribosome recycling, and polysomes

- Stop codons are recognized by release factors, not suppressor tRNAs under normal conditions.
 - Termination hydrolyzes the peptidyl-tRNA ester bond.
 - Ribosomal subunits, mRNA, and tRNA are recycled.
 - A polysome is many ribosomes translating one mRNA.
- Premature termination codons can trigger nonsense-mediated mRNA decay.

Signal sequences direct proteins to ER, nucleus, and mitochondria

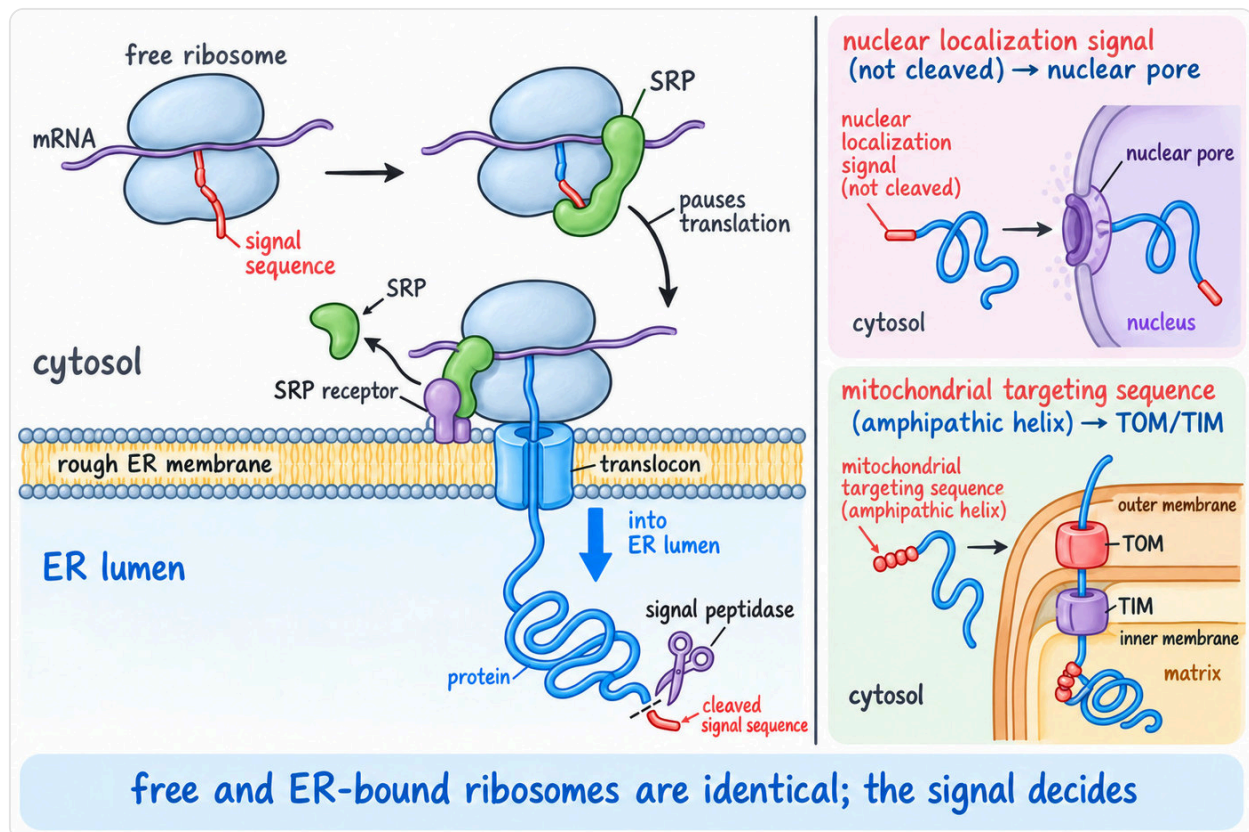


fig. 29 Signal-directed targeting to the ER, nucleus and mitochondrion.

Posttranslational modification, folding, and ubiquitin-proteasome quality control

- Modification can activate a precursor, target a protein, or change stability.
 - N-linked glycosylation is on Asn; O-linked is commonly on Ser/Thr.
 - Collagen proline/lysine hydroxylation requires vitamin C.
 - Vitamin K supports γ -carboxylation of glutamate in clotting proteins.
 - Polyubiquitin commonly marks proteins for proteasomal degradation.
- $\Delta F508$ CFTR is often degraded by quality control before reaching the plasma membrane.

Antibiotics and toxins mapped to translation steps

- Tetracycline: blocks A-site entry on bacterial ribosomes.
- Macrolides: 50S binding and impaired translocation.
- Chloramphenicol: inhibits bacterial/mitochondrial peptidyl transferase.
- Puromycin: aminoacyl-tRNA analog causing premature termination.
- Diphtheria toxin targets eEF2; ricin targets 28S rRNA.

→ Mitochondrial ribosomes partly resemble bacterial ribosomes, explaining selected antimicrobial toxicities.

Past papers asked

question	answer and why
Tetracycline prevents synthesis of polypeptide by:	Preventing binding of aminoacyl tRNA is the best answer. Tetracycline binds the bacterial 30S subunit and blocks aminoacyl-tRNA entry into the A site.
Which of the following inactivates the 60S subunit of eukaryotic ribosomes?	Ricin is the best answer. Ricin depurinates 28S rRNA in the eukaryotic 60S subunit and inactivates it. Ricin irreversibly depurinates 28S rRNA in the 60S subunit. Cycloheximide also inhibits eukaryotic elongation at the large subunit; the intended distinction is ricin-mediated ribosome damage.
Which of the following mediates carrying genetic information from DNA to protein synthesis?	m-RNA is the best answer. mRNA carries coding information from DNA to ribosomes.
What is the main function of t-RNA?	Identifies amino acids and transports them to ribosomes is the best answer. Ribosomes read mRNA through tRNA-mediated amino-acid delivery.
In bacteria the elongation stage of protein synthesis does not involve	IF-2 is the best answer. IF-2 is an initiation factor; elongation uses EF-Tu, GTP, and peptidyl-transferase activity.
Regarding protein synthesis in eukaryotes, all proteins are initially synthesized with?	Methionine at their N-terminus is the best answer. Eukaryotic translation initiates with methionine at the N-terminus.
Which one of the following antibiotics does not function by interfering with the translational process?	Penicillin is the best answer. Penicillin inhibits bacterial cell-wall cross-linking, not translation.

Test yourself

- Which antibiotic blocks bacterial aminoacyl-tRNA entry into the A site?
 - Rifampicin
 - Actinomycin D
 - Alpha-amanitin
 - Tetracycline
- Which coding-sequence deletion is most likely to shift all downstream reading-frame assignments?
 - Deletion of three nucleotides
 - Deletion of one nucleotide
 - Deletion of nine nucleotides
 - Deletion of six nucleotides
- Termination trigger: what event starts termination?
 - A leucine codon enters the A site
 - The poly(A) tail binds the P-site tRNA as a peptide
 - A stop codon reaches the A site and is recognized by a release factor
 - The ribosome encounters a promoter on DNA

Answers: 1 D · 2 B · 3 C

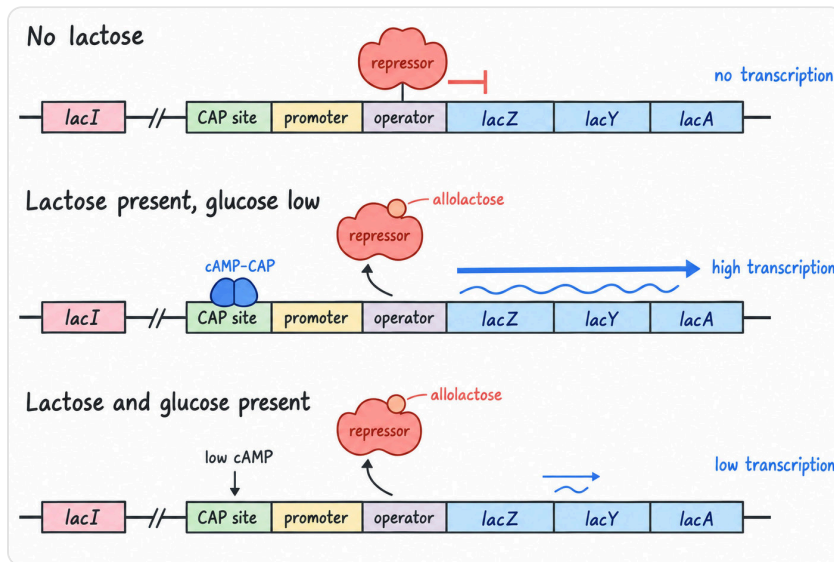
Gene regulation

Lippincott 6e, Chapter 32 · teacher slides

constitutive versus regulated genes, cis elements, trans factors, and operons

- Cis = local DNA sequence; trans = diffusible regulator.
 - Promoters, operators, enhancers, and response elements are cis elements.
 - Repressors, activators, and many transcription factors act in trans.
 - Operons coordinate related bacterial genes efficiently.
- Pathogenic variants can affect either a coding trans factor or a noncoding cis-regulatory element.

Lac operon integrates lactose availability with glucose status



- Allolactose is the inducer that inactivates lac repressor.
- Low glucose → high cAMP → active CAP.
- Lactose present removes negative control; low glucose provides positive control.
- Maximal lacZYA transcription: lactose present, glucose absent.
- Glucose-mediated suppression is catabolite repression.

fig. 30 The lac operon is repressed without lactose, strongly expressed when lactose is present and glucose is low, and weakly expressed when both sugars are present.

Trp operon repression and attenuation sense amino-acid supply

- Lac is inducible and catabolic; trp is repressible and anabolic.
- Tryptophan activates its repressor as a corepressor.
- Attenuation depends on coupled transcription and translation.
- High Trp favors termination; low Trp permits transcription.

Enhancers, specific transcription factors, and combinatorial control

- Enhancers are cis DNA elements; activators are trans proteins.
 - Enhancer action is often distance and orientation independent.
 - Combinatorial control lets a limited factor set specify many expression patterns.
 - Shared consensus elements coordinate unlinked genes.
- Enhancer mutations and enhancer hijacking can activate oncogenes without altering their coding sequence.

Nuclear receptors and cAMP-CREB connect hormones to transcription

- Nuclear receptors are ligand-activated transcription factors.
- HRE is cis acting; hormone-receptor complex is trans acting.
- Glucagon → GPCR → cAMP → PKA → CREB → CRE.
- Coactivator versus corepressor recruitment determines transcriptional direction.

→ Glucocorticoid and thyroid-hormone therapies alter transcription through nuclear receptors.

Histone modifications, chromatin remodeling, and DNA methylation

- Epigenetic change alters expression without changing base sequence.
- Histone acetylation commonly opens chromatin.
- DNA methyltransferases use S-adenosylmethionine as methyl donor.
- Promoter CpG-island methylation usually represses transcription.
- Chromatin remodeling repositions or ejects nucleosomes.

→ Tumor-suppressor promoters can be silenced by aberrant hypermethylation.

Alternative processing, RNA editing, and mRNA stability

- One pre-mRNA can yield different isoforms through alternative processing.
- RNA editing changes transcript information without changing genomic DNA.
- ApoB-48 is intestinal; ApoB-100 is hepatic.
- Longer mRNA half-life generally permits more protein synthesis.
- Iron-response elements provide a classic stability/translation control system.

→ ApoB-100 is required for LDL-receptor recognition, whereas ApoB-48 packages dietary lipid in chylomicrons.

RNA interference and regulation of translation initiation

- Dicer helps generate short RNA duplexes; Argonaute is the RISC effector.
- Partial pairing usually represses translation; extensive pairing favors cleavage.
- siRNA can be experimentally or therapeutically supplied.
- Phosphorylated eIF2 inhibits translation initiation during stress.

→ Approved siRNA drugs silence pathogenic hepatic transcripts.

Gene rearrangement, amplification, and transposable elements

- Somatic V(D)J recombination permanently rearranges lymphocyte DNA.
- Copy-number increase can drive overexpression and drug resistance.
- Transposons use transposase; retrotransposons use an RNA intermediate.
- Mobile elements can cause insertional mutation or move resistance genes.

→ Defective V(D)J machinery causes severe immunodeficiency.

Past papers asked

question

answer and why

The gene of lac operon which has constitutive expression is.....

i is the best answer. lacI is expressed constitutively to supply the repressor protein.

question

answer and why

Which of these operons is anabolic?

The trp operon encodes enzymes for tryptophan biosynthesis, so it controls an anabolic pathway and is repressible by its end product. Both tryptophan and phenylalanine biosynthetic operons are anabolic. The original intended trp answer is not uniquely correct when phe is also offered.

Where does a repressor bind to an operon?

Operator is the best answer. Operators, repressors, activators, and feedback control gene expression.

Lactose utilization by E. coli requires which lac gene products?

LacY permease imports lactose and LacZ beta-galactosidase hydrolyzes it. Both A and B are valid for the unrestricted original question.

The operator region normally can be bound by

Repressor is the best answer. A repressor binds the operator and alters transcription of the adjacent structural genes.

The tryptophan operon of E. coli is repressed by tryptophan added to the growth medium. The tryptophan repressor probably

Binds to the trp operator in the presence of tryptophan is the best answer. Tryptophan acts as a corepressor, enabling the repressor to bind the trp operator.

Test yourself

1. Which mechanism is an important posttranscriptional control of eukaryotic gene expression?

- A. Alternative mRNA splicing
- C. Tus-Ter binding

- B. Changing the bacterial -10 promoter
- D. Okazaki-fragment ligation

2. An E.coli culture has lactose available but abundant glucose. Why is lac transcription less than its fully induced level?

- A. The glucose signal keeps cAMP-CAP at maximum activation
- C. Allolactose necessarily strengthens repressor binding

- B. Lactose can induce the operon only after all DNA replication stops
- D. Low cAMP limits CAP-mediated positive activation despite reduced repressor binding

3. Define attenuation (active recall).

- A. Premature termination of a transcript after initiation in response to a regulatory signal
- C. Removal of all repressors from every bacterial chromosome

- B. Inhibition of every DNA polymerase before replication begins
- D. Translation of the operon from its final codon backward

Answers: 1 A · 2 D · 3 A

Biotechnology

Lippincott 6e, Chapter 33 · teacher slides

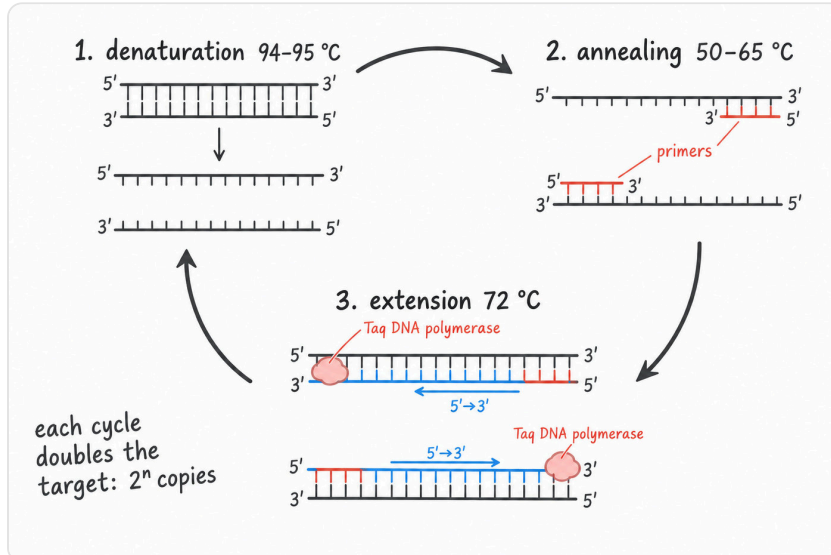
Restriction enzymes, sticky ends, and plasmid cloning vectors

- Restriction enzymes cut sequence-specific sites.
 - Compatible sticky ends improve directional assembly efficiency.
 - DNA ligase covalently seals vector and insert.
 - Origin, selectable marker, and cloning site are core vector features.
- Recombinant plasmids enable production of therapeutic proteins such as human insulin.

cDNA and complementary nucleic-acid probes

- cDNA reflects the transcripts present in the sampled cell.
 - Mature-mRNA-derived cDNA lacks introns.
 - Reverse transcriptase makes the initial DNA copy.
 - Probe specificity depends on complementarity, temperature, salt, and mismatch tolerance.
- cDNA and probes are used to measure gene expression and detect infectious or inherited sequences.

PCR amplification and reverse-transcription PCR



- Cycle order: denature → anneal → extend.
- Primers determine specificity and amplicon boundaries.
- Thermostable polymerase survives repeated heating.
- RT-PCR begins with reverse transcription of RNA.
- Quantitative PCR follows amplification using fluorescence.

→ PCR is used for pathogen detection, genotyping, minimal-residual-disease testing, and viral-load assays.

fig. 31 A PCR cycle denatures DNA, anneals two inward-facing primers, and extends new strands with Taq DNA polymerase.

Southern, Northern, and Western blot targets

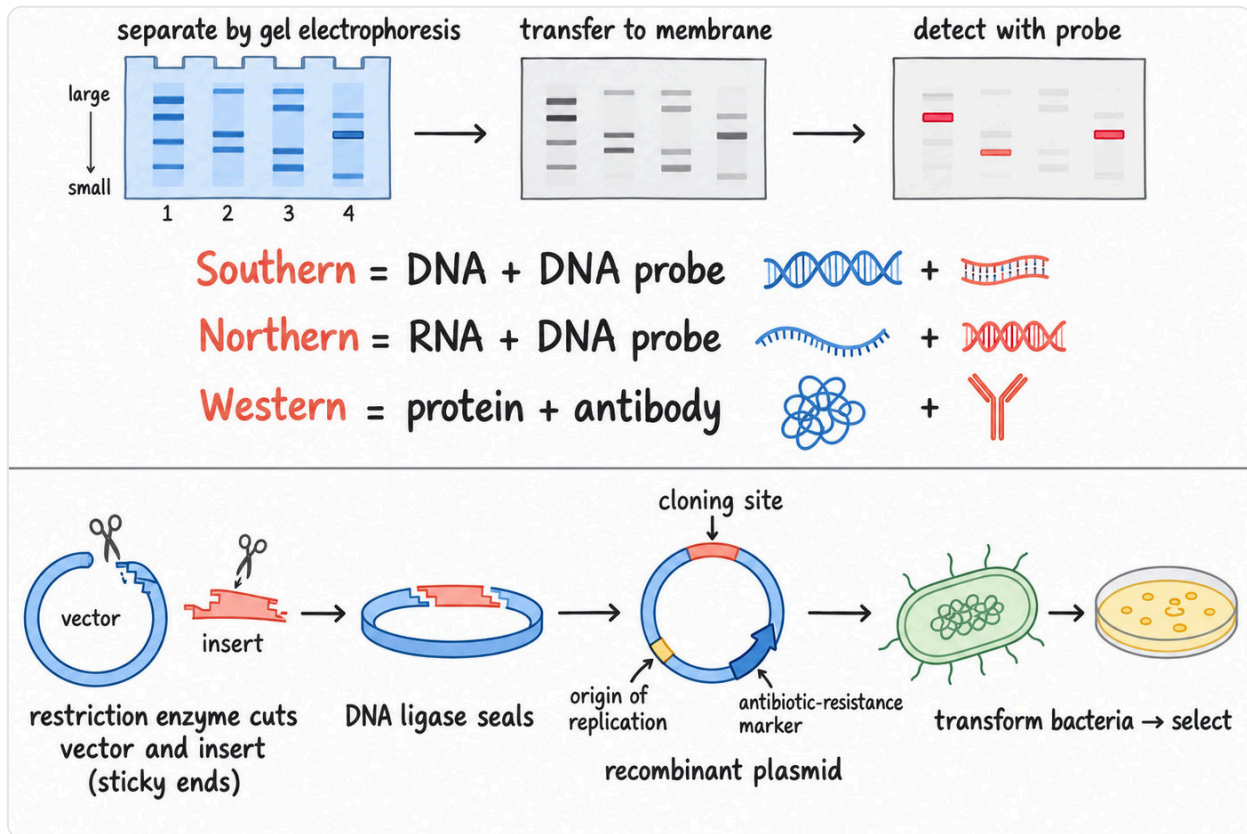


fig. 32 Blotting methods and the plasmid cloning workflow.

Restriction-fragment polymorphism and Sanger sequencing

- RFLP detects sequence-dependent fragment-length differences.
- ddNTP incorporation prevents the next phosphodiester bond.
- Sanger products differ by one nucleotide in length.
- Sequence is read from the order of labeled terminated fragments.

→ Sanger sequencing remains a common confirmatory method for targeted variants.

Expression microarrays and allele-specific hybridization

- Microarray signal estimates relative mRNA abundance through cDNA.
- Allele-specific probes depend on mismatch-sensitive stringency.
- Hybridization results require normalization and suitable controls.
- Expression change does not prove a causal functional change.

→ Expression signatures can classify selected tumors and predict biologic behavior.

Test yourself

1. Which features are essential in a basic plasmid cloning vector?

- A. A ribosome and a mitochondrion
- B. Only a poly-A tail
- C. A complete human chromosome
- D. An origin of replication, a selectable marker, and a cloning site

2. For PCR amplification across a target region, how should the two primers be oriented?

- A. Their 5' ends must be covalently joined before the first cycle
- B. Both must bind the exact same sequence on the same strand
- C. Both 3' ends point away from the target
- D. Their 3' ends point toward the region to be copied on opposite strands

3. How do dideoxynucleotides terminate DNA synthesis in Sanger sequencing?

- A. They lack the 3-prime hydroxyl required for the next phosphodiester bond
- B. They remove the template strand
- C. They inhibit all base pairing
- D. They contain ribose instead of deoxyribose only

Answers: 1 D · 2 D · 3 A

Laboratory principles and tests

Teacher slides and laboratory material · teacher slides

Bench-test discrimination

Test / method	Detects or measures	Interpretation caution
Benedict / Fehling	Reducing sugars	Not specific to glucose; sucrose is negative before hydrolysis.
Barfoed	Monosaccharides reduce reagent relatively rapidly	Heating time matters; prolonged heating blurs the distinction.
Seliwanoff	Ketoses react rapidly in acid	A timed distinction; sucrose can yield fructose after hydrolysis.
Iodine	Starch-like helical polysaccharide complexes	Not a general test for all carbohydrates.
Biuret	Peptide-bond copper complex	Free amino acids do not give the protein reaction.
Ninhydrin	Free amino groups	Most amino acids purple; proline gives a different yellow response.
Beer-Lambert law	$A = \epsilon lc$; $A = -\log_{10}(T)$	Blank correctly; use linear range and matching wavelength/path length.
Chromatography	Differential partition/interaction	R_f = solute distance / solvent-front distance in planar chromatography.

General carbohydrate, starch, and reducing-sugar tests

- Molisch: general carbohydrate; violet interfacial ring.
 - Iodine: starch/polysaccharide; blue-black complex.
 - Benedict: reducing sugar; colored Cu_2O precipitate after heating.
 - A positive broad test does not identify the exact carbohydrate.
- Copper-reduction tests are historical screening methods; modern glucose assays are more specific and quantitative.

Use reaction conditions and timing to distinguish carbohydrate classes

- Benedict estimates reducing-sugar amount only roughly by color.
- Barfoed separates rapid monosaccharide reduction from slower disaccharide reduction.
- Seliwanoff favors rapid ketose detection.
- Timing matters: prolonged heating can create misleading late positives.

Ninhydrin, Biuret, and xanthoproteic reaction targets

- Ninhydrin: free amino groups; purple for most amino acids.
- Proline is the yellow ninhydrin exception.
- Biuret: Cu^{2+} complex with multiple peptide bonds in alkali.
- Xanthoproteic: aromatic amino-acid residues; yellow-to-orange product.

→ Modern clinical protein quantification uses standardized colorimetric or immunochemical assays, but Biuret chemistry remains widely used for total protein.

Precision glassware, meniscus reading, and burette delivery

- Volumetric pipette: one precise delivered volume.
- Volumetric flask: one precise contained final volume.
- Read the lower aqueous meniscus at eye level.
- Burette: variable controlled delivery for titration.
- Correct pipette fillers are used; mouth pipetting is never acceptable.

→ Preanalytical volume errors can bias every downstream calculated concentration.

Recognize core equipment and match it to safety or separation function

- Fume hood protects from chemical vapors; it does not guarantee product sterility.
- Analytical balance doors reduce air-current error.
- Centrifugation separates by sedimentation behavior, not pH alone.
- Rotors must be balanced by mass and tube position before spinning.
- Use relative centrifugal force when protocols require cross-instrument reproducibility.

→ Improper centrifugation can hemolyze samples or leave cells in plasma/serum and distort results.

Acid-base titration endpoint and normality calculation

- Titrant is known; analyte is unknown.
- Teacher standard: 0.1 N NaOH.
- Endpoint is persistent pale pink, not deep pink.
- Normality counts reactive equivalents per liter.
- Record initial and final burette readings and use their difference.

→ Titrimetric principles underlie several classical laboratory standardizations and acid-base measurements.

Initial velocity, reagent blanks, and enzyme units

- Use initial velocity in the linear range.
- A blank measures non-analyte/reagent background.
- Activity depends on temperature, pH, substrate concentration, and assay time.
- An enzyme unit is a rate under specified conditions, not a mass.
- Boiling usually destroys activity by denaturing protein structure.

→ Serum ALT, AST, ALP, CK, and lipase are commonly reported as catalytic activity rather than enzyme concentration.

Beer–Lambert law, blanking, and wavelength selection

- $A = \epsilon bc$ within the validated linear range.
- Absorbance is $-\log_{10}$ of fractional transmittance.
- Blank composition should match the sample except for the measured analyte/reaction.
- Use $\lambda_{\max}$ for sensitivity.
- Dilute and apply the dilution factor when absorbance exceeds the calibration range.

→ Many glucose, bilirubin, protein, and enzyme assays convert concentration or reaction rate into absorbance.

DNA extraction by lysis, nuclease inhibition, and alcohol precipitation

- Detergent lyses membranes.
- EDTA chelates Mg^{2+}/Ca^{2+} and suppresses nucleases.
- Protease removes DNA-bound and contaminating proteins.
- Salt plus cold alcohol precipitates DNA.
- Avoid vigorous shearing when intact high-molecular-weight DNA is needed.

→ Poor extraction quality can cause false-negative PCR or failed sequencing.

Chromatographic separation by mobility, charge, size, or affinity

- R_f is unitless and lies between zero and one in a valid planar run.
- Ion exchange separates principally by net charge.
- Affinity uses specific target–ligand recognition.
- Size exclusion: large molecules elute before small molecules.
- Changing pH or ionic strength can alter binding and elution.

→ Chromatographic methods separate hemoglobin variants, drugs, metabolites, and proteins in clinical laboratories.

Atomic emission, calibration, and clinical ions in flame photometry

- Flame photometry measures emitted light, not absorbance.
- Emission wavelength identifies the element; intensity relates to concentration.
- Na^+ and K^+ are classic analytes.
- Standards and blanks establish a valid calibration.
- Never extrapolate far beyond the highest standard.

→ Flame emission was widely used for serum sodium and potassium and remains conceptually important in electrolyte analysis.

Osmolarity, tonicity, RBC behavior, and dialysis

- Hypotonic → cell swelling/hemolysis.
- Hypertonic → cell shrinkage/crenation.
- Isosmotic does not always mean isotonic.
- Penetrating solutes contribute little to sustained tonicity after equilibration.
- Water moves toward higher effective osmolarity across a water-permeable membrane.

→ Incorrect IV-fluid tonicity can cause dangerous cellular swelling or dehydration, especially in the brain.

Test yourself

1. Barfoed's test is primarily used to distinguish which carbohydrates?
 - A. Ketoses from aldoses only
 - B. Monosaccharides from disaccharides by rapid reduction in acidic medium
 - C. Starch from glycogen
 - D. Proteins from lipids
2. The violet color in the Biuret test results from Cu^{2+} complexing with what?
 - A. Free glucose carbonyls
 - B. Aromatic rings after nitration
 - C. Multiple peptide bonds in alkaline solution
 - D. DNA phosphate groups in acid
3. What happens to red blood cells placed in a hypertonic solution?
 - A. They take up water and lyse
 - B. They lose water and crenate
 - C. They synthesize more membrane immediately
 - D. They become isotonic by importing the nonpenetrating solute

Answers: 1 B · 2 C · 3 B

Sources and limits

1. Latest original course-scope reconciliation, 20 August 2026; original syllabus and teacher-material archive.
2. Ferrier, D. R. Lippincott Illustrated Reviews: Biochemistry, 6th edition (2014): confirmed chapters listed above.
3. Teacher foundations: INTRODUCTION.ppt; Water and Buffer 1404; Marks Essentials water/buffers reference.
4. Teacher biomolecules: Amino acids and Proteins Parts 1-2; Lipid structure (Esmaili); Vitamin 2024.
5. Teacher enzymes and genetics: Enzyme Kinetics and Regulation; DNA Structure; DNA Replication; Transcription 2025; Translation 2025; Regulation of Gene Expression 2025.
6. Teacher laboratories: Laboratory Equipment, Titration and Carbohydrate Qualification Test (Esmaili, January 2026), plus the original laboratory scope.
7. Four source-paper biochemistry sections: April 2021 Cell & Molecules report, February 2021, September 2021, Biochemistry 1 Finals 2022. Filename dates are retained; a report date is not proof of the exam sitting date.
8. Source caveats: human DNA repair — <https://www.ncbi.nlm.nih.gov/books/NBK1397/> ; enzyme kinetics — <https://www.ncbi.nlm.nih.gov/books/NBK92007/> ; hepatic HDL/VLDL — <https://www.ncbi.nlm.nih.gov/books/NBK351/> .

Original confirmed Term 1 syllabus: Lippincott Chapters 1-7, 14-18 and 23-33, plus foundations, water/buffers and laboratory-derived theory. Retake-specific exclusions have not been announced here. Chapters 8-13 and 19-22 (carbohydrate and nitrogen metabolism) belong to Term 2 Biochemistry II and are excluded, together with their practice questions. Physiology, histology and the Term 2 metabolism course are excluded. Study keys are not certified university keys.

This review covers the confirmed original biochemistry syllabus and the four imported biochemistry past papers, plus clearly marked practice-bank topics. It does not certify a new retake announcement or predict future questions. Figures marked as diagrams are teaching illustrations; check details against Lippincott.

Sources by topic

01 Biochemical foundations INTRODUCTION.ppt, slides 3-6 (definition, scope, and medical relevance). · biochemistry-scope.md, Primary course files: INTRODUCTION.ppt, slides 9-13 (CHNOPS, cells, biomolecule classes, and central dogma). · INTRODUCTION.ppt, slide 13 (central dogma and information flow). · INTRODUCTION.ppt, slides 16-19 (anabolism, catabolism, and ATP). · INTRODUCTION.ppt, slides 20-22 (major biological functional groups). · INTRODUCTION.ppt, slides 24-31 (ionic/covalent bonds, polarity, disulfides, hydrogen bonds, and hydrophobic interactions).

02 Water, acids, bases and buffers Water and Buffer 1404.ppt, slides 4-5 (body water and compartments). · Marks water/buffer reference, local reference pages 2-4 (compartments, electrolytes, and osmolality). · Water and Buffer 1404.ppt, slides 4 and 6-8 (water polarity, hydrogen bonds, and solubility classes). · Water and Buffer 1404.ppt, slides 9-15 (pH, pOH, clinical pH, acidosis, and alkalosis). · Water and Buffer 1404.ppt, slides 14 and 17-20 (acid-base definitions, conjugate pairs, K_a , and pK_a). · Water and Buffer 1404.ppt, slides 24-28 and teacher MCQs 37-38. · Lippincott 6e, Chapter 1 acid-base section, local PDF pages 18-21. · Water and Buffer 1404.ppt, slides 22-28 (titration, buffer region, range, and capacity). · Water and Buffer 1404.ppt, slides 29-35 (physiological buffers). · Marks water/buffer reference, local reference pages 6-8 (bicarbonate, hemoglobin, chloride shift, and urinary buffering).

03 Amino acids Amino acids and Proteins.Part1.pdf, pages 7-15 (structure and side-chain classes). · Lippincott 6e, Chapter 1, local extraction items 11 and 135. · Amino acids and Proteins.Part1.pdf, pages 10-15 (amino-acid side-chain classes). · Lippincott 6e, Chapter 1, local extraction items 20 and 23. · Amino acids and Proteins.Part1.pdf, pages 12-13 and 31-33. · Lippincott 6e, Chapter 1, local extraction item 35. · Amino acids and Proteins.Part1.pdf, pages 14-15 (acidic/basic residues and histidine). · Amino acids and Proteins.Part1.pdf, page 20 (glycine and L-amino acids). · Lippincott 6e, Chapter 1 stereochemistry section. · Amino acids and Proteins.Part1.pdf, pages 25-26

(isoelectric point). · Lippincott 6e, Chapter 1, local PDF pages 10–29. · Amino acids and Proteins.Part1.pdf, page 21 (aromatic amino acids and absorbance). · Confirmed TUMS practical curriculum: xanthoproteic test.

04 Protein structure and folding Amino acids and Proteins.Part1.pdf, pages 31–33 and 43–49. · Lippincott 6e, Chapter 2, local extraction items 5–6. · Amino acids and Proteins.Part1.pdf, pages 31–33 (peptide bond). · Lippincott 6e, Chapter 2, local PDF pages 30–54. · Amino acids and Proteins.Part1.pdf, pages 36–40 (alpha helix, beta sheet, and beta turn). · Amino acids and Proteins.Part1.pdf, page 43 (protein domains). · Lippincott 6e, Chapter 2, local extraction item 91. · Amino acids and Proteins.Part1.pdf, page 49 (quaternary structure). · Lippincott 6e, Chapter 2, local extraction item 160. · Exam scope update 2026–08–20: complete confirmed Lippincott scaffold includes Chapter 2. · Lippincott 6e, Chapter 2, local extraction items 101, 110, and 115. · Amino acids and Proteins.Part1.pdf, page 48 (protein denaturation). · Lippincott 6e, Chapter 2, local extraction item 107. · Lippincott 6e, Chapter 2, local extraction items 125, 148, and 172.

05 Hemoglobin and globular proteins exam-scope.md, August 25 active biochemistry scope: proteins and hemoglobin; complete Lippincott Chapters 1–7 scaffold. · Lippincott 6e, Chapter 3, local extraction items 8 and 28. · Lippincott 6e, Chapter 3, local extraction items 7, 12–16, and 41–53. · exam-scope.md, complete confirmed Lippincott Chapters 1–7 scaffold. · Lippincott 6e, Chapter 3, local extraction items 39–40, 49, and 53–64. · Water and Buffer 1404.ppt, slides 33–35 (hemoglobin buffering and CO₂ transport). · Lippincott 6e, Chapter 3, local extraction item 65. · Lippincott 6e, Chapter 3, local extraction items 77–91. · Lippincott 6e, Chapter 3, local extraction items 108 and 116–119. · Exam scope update 2026–08–20: complete confirmed Lippincott scaffold includes Chapter 3. · Lippincott 6e, Chapter 3, local extraction items 97–104 and 167–174. · Lippincott 6e, Chapter 3, local extraction items 122–128 and 204. · Lippincott 6e, Chapter 1 local extraction item 24 and Chapter 3 local extraction items 151–158. · Lippincott 6e, Chapter 3, local extraction items 175–193.

06 Collagen, elastin and fibrous proteins exam-scope.md, complete confirmed Lippincott Chapters 1–7 scaffold. · Lippincott 6e, Chapter 4, local PDF pages 87–104. · Lippincott 6e, Chapter 4, local PDF pages 87–104 and extraction items 31–39. · Lippincott 6e, Chapter 4, local PDF pages 87–104 and extraction items 40–47. · Exam scope update 2026–08–20: complete confirmed Lippincott scaffold includes Chapter 4. · Lippincott 6e, Chapter 4, local extraction items 52–53. · Lippincott 6e, Chapter 4, local PDF pages 87–104 and extraction item 58. · Lippincott 6e, Chapter 4, local extraction items 68–74. · Lippincott 6e, Chapter 4, local extraction items 74 and 78–85.

07 Enzymes, kinetics and regulation exam-scope.md, August 25 active biochemistry scope: enzymes and regulation; complete Lippincott Chapters 1–7 scaffold. · Lippincott 6e, Chapter 5, local extraction items 12, 14, and 18. · Lippincott 6e, Chapter 5, local extraction items 29–36. · Vitamin 2024.pdf, pages 4–12 (cofactors, coenzymes, and prosthetic groups). · Lippincott 6e, Chapter 5, local extraction items 20–25 and 114. · Lippincott 6e, Chapter 5, local extraction items 43–48. · Enzyme Kinetics and regulation- Session2 2024.pptx, slides 7 and 9–10. · Enzyme Kinetics and regulation- Session2 2024.pptx, slides 13–14. · Lippincott 6e, Chapter 5, local extraction items 62–65. · Enzyme Kinetics and regulation- Session2 2024.pptx, slides 16–18 and 21. · Lippincott 6e, Chapter 5, local extraction item 112. · Enzyme Kinetics and regulation- Session2 2024.pptx, slides 19–24. · Lippincott 6e, Chapter 5, local extraction item 82. · Enzyme Kinetics and regulation- Session2 2024.pptx, slides 35–38. · Enzyme Kinetics and regulation- Session2 2024.pptx, slides 39, 41, and 43. · Exam scope update 2026–08–20: complete confirmed Lippincott scaffold includes Chapter 5. · Lippincott 6e, Chapter 5, local extraction items 96–110. · Lippincott 6e, Chapter 5, pp. 62–64 (covalent regulation), and Chapter 11, pp. 131–133 (glycogen-enzyme regulation). · Enzyme Kinetics and regulation session 2.pdf, PDF p. 36 (slide footer 37): phosphorylation/dephosphorylation. · Lippincott 6e, Chapter 5, pp. 65–67 (enzyme leakage, tissue specificity, and CK isoenzymes); the ALP and urinary-amylase details are supplemented below. · Mayo Clinic Laboratories, Creatine Kinase (CK), Serum, Clinical Information and Interpretation: <https://www.mayocliniclabs.com/test-catalog/Overview/8336> · Mayo Clinic Laboratories, Alkaline Phosphatase, Serum, Interpretation: <https://www.mayocliniclabs.com/test-catalog/Overview/800001> · Mayo Clinic Laboratories, Amylase, Total, Serum, Interpretation and Cautions: <https://www.mayocliniclabs.com/test-catalog/Overview/8352>

08 Bioenergetics and oxidative phosphorylation Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, bioenergetics and free energy (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026–08–20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33. · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, electron transport (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, oxidative phosphorylation (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, inhibitors and uncoupling (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, shuttles and ROS (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott 6e, Chapter 6, pp. 73–76 (ETC carrier sequence and dehydrogenases), and Chapter 9, pp. 112–113 (succinate dehydrogenase-bound FAD). See also Chapter 28, p. 380 (flavin binding to enzymes). · Lippincott 6e, Chapter 6, pp. 78–79 (mitochondrial ATP/ADP and phosphate transport). · Supplement for atractyloside specificity: Identification of adenine nucleotide translocase 4 inhibitors by molecular docking (2013), Introduction: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3821932/> · Lippincott 6e, Chapter 6, pp. 79–80 (mitochondria and apoptosis). · Supplement for Apaf-1/caspase-9: Cytochrome c and dATP-dependent formation of Apaf-1/caspase-9 complex initiates an apoptotic protease cascade (Cell, 1997): <https://pubmed.ncbi.nlm.nih.gov/9390557/> · Lippincott 6e, Chapter 6, pp. 72–73 (ATP and energy coupling); Chapter 8, pp. 97–103 (glucose phosphorylation and high-energy glycolytic intermediates); Chapter 19, pp. 253–254 (carbamoyl phosphate); Chapter 21, pp. 287–288 (creatine phosphate). · Lippincott 6e, Chapter 6, pp. 77–79; Chapter 8, p. 104; Chapter 9, pp. 113–114 (historical ATP convention and pathway yields). · carbohydrate metabolism MD Kish.pdf, pp. 79–80 (TCA products and 32-ATP glucose calculation). · Hinkle et al., Mechanistic stoichiometry of mitochondrial oxidative phosphorylation, Biochemistry 1991;30:3576–3582, <https://pubmed.ncbi.nlm.nih.gov/2012815/> (experimental basis of approximately 2.5/1.5 P/O estimates).

09 Carbohydrate structure and stereochemistry Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 7, monosaccharide structure (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026–08–20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33. · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 7, glycosidic bonds (textbook-confirmed). · Carbohydrate Qualification Test Esmaili Jan 2026 — 21-page teacher practical deck (teacher-confirmed practical context). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 7, digestion of carbohydrates (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 7, monosaccharide uptake (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott 6e, Chapter 7, pp. 84–85, Figure 7.6 (cyclization and pyranose structure). · Lippincott 6e, Chapter 7, pp. 85–87; Chapter 11, pp. 126–128 (polysaccharides, starch digestion, and branching).

10 Glycoconjugates Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 14, glycosaminoglycans (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026–08–20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33. · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 14, proteoglycans and glycoproteins (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 14, glycoprotein synthesis and lysosomal targeting (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 14, mucopolysaccharidoses (textbook concept; this chapter is confirmed for August 25 by exam-scope.md).

11 Dietary lipids and absorption Lipid structure Esmaeili.pdf, 96-page teacher deck, early fatty-acid and lipid-classification sections. · biochemistry-telegram-scope.md: Lipid structure, classification, fatty acids, complex lipids, and lipoproteins are confirmed first-semester teacher material. · Lipid and Lipoprotein Metabolism 1, Fataneh Esmaeili — pp. 4-21 (teacher-confirmed). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 15 (cross-check). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33. · Lipid and Lipoprotein Metabolism 1, Fataneh Esmaeili — pp. 17-24 (teacher-confirmed). · Lipid and Lipoprotein Metabolism 1, Fataneh Esmaeili — pp. 22-27 (teacher-confirmed). · Lipid and Lipoprotein Metabolism 1, Fataneh Esmaeili — pp. 27-42 (teacher-confirmed). · Lippincott 6e, Chapter 15, pp. 175-178 (lipase inhibition and reduced intestinal cholesterol absorption). · Lipid metabolism 1.pdf, p. 25 (ezetimibe and intestinal cholesterol absorption); p. 14 names acid lipases, but its orlistat wording is superseded here by the textbook gastric/pancreatic targets. · FDA, ZETIA prescribing information, section 12.1 (NPC1L1 target): https://www.accessdata.fda.gov/drugsatfda_docs/label/2011/021445s030lbl.pdf

12 Fatty acids, triacylglycerols and ketones Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33. · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16, fatty-acid oxidation (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16, ketone bodies (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16, TAG metabolism (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16, oxidation disorders (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott 6e, Chapter 16, pp. 190-192; Chapter 20, p. 264. · Supplemental primary isotope evidence for the distinction between carbon skeleton and methyl donors: Cox and Hoppel, Biosynthesis of carnitine and 4-N-trimethylaminobutyrate from lysine, Biochemical Journal 136 (1973), pp. 1075-1082, <https://pubmed.ncbi.nlm.nih.gov/4786529/>. · Lippincott 6e, Chapter 16, pp. 183-187 (citrate export, ACC, fatty-acid synthase, and malic enzyme), and Chapter 28, p. 381 (pantothenate). · Lippincott 6e, Chapter 16, pp. 182-183 and 187-188 (essential fatty acids and desaturation). · Lippincott 6e, Chapter 16, pp. 192-195 (reaction sequence, legacy ATP yield, and unsaturated fatty acids). · Supplement for modern P/O accounting: Hinkle et al., Mechanistic stoichiometry of mitochondrial oxidative phosphorylation, Biochemistry 1991;30:3576-3582, <https://pubmed.ncbi.nlm.nih.gov/2012815/> (experimental basis of approximately 2.5/1.5 P/O estimates; the 106-ATP total is calculated from these ratios). · Lippincott 6e, Chapter 16, pp. 193-195 (odd-chain oxidation); Chapter 9, pp. 112-114 (succinyl-CoA to oxaloacetate reactions). · The four-ATP partial-pathway figure is calculated from the stated reactions using modern P/O ratios, not quoted as the textbook yield. · Lippincott 6e, Chapter 16, pp. 189-191 (mobilization of stored fats and lipase regulation). · Lipid metabolism 1.pdf, pp. 8 and 32-34 (fed/fasted state and fatty-acid/glycerol transport). · Lippincott 6e, Chapter 16, pp. 195-198 (ketone synthesis and use), and Chapter 24, pp. 329-331 (hepatic fasting metabolism).

13 Complex lipids and eicosanoids Lipid Structure Esmaeili — 96-page teacher deck, phospholipid section (teacher-confirmed). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 17 (mechanism cross-check). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33. · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 17, surfactant (confirmed August 25 scope in tracked bank). · Lipid Structure Esmaeili — sphingolipid section (teacher-confirmed class). · Tracked comprehensive bank — Chapter 17, PDF pp. 374-405 (marked confirmed August 25 scope). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 17, eicosanoids (confirmed August 25 scope in tracked bank). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 17, leukotrienes (textbook concept; this chapter is confirmed for August 25 by exam-scope.md).

14 Cholesterol, lipoproteins and steroids Lipid Structure Esmaeili — cholesterol/lipoprotein sections (teacher-confirmed class). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 18 (regulatory cross-check). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33. · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 18, bile acids (textbook-confirmed; teacher emphasis uncertain). · Lipid Structure Esmaeili — 96-page deck, lipoprotein section (teacher-confirmed). · Lipid Structure Esmaeili — lipoprotein section (teacher-confirmed). · Tracked comprehensive bank — Chapter 18 (marked confirmed August 25 scope). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 18, steroid hormones (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · Lippincott 6e, Chapter 18, pp. 228-234 (LPL, remnant processing, and LDL receptor ligands). · Lipid metabolism 1.pdf, pp. 29-31 (LPL action and familial chylomicronemia). · Supplement for the specific IDL-to-LDL role of hepatic lipase: Lipoprotein metabolism in hepatic lipase deficiency (Journal of Lipid Research, 1988;29:1603-1611): <https://pubmed.ncbi.nlm.nih.gov/3244012/>. · Lippincott 6e, Chapter 18, pp. 224-227 (bile-acid synthesis, feedback, and sequestrants).

15 Insulin and glucagon Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 23 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33. · exam-scope.md (updated 2026-08-20): the active August 25 scope explicitly includes insulin, glucagon, and hypoglycemia. · Lippincott's Illustrated Reviews: Biochemistry, 6e, Chapter 23, hypoglycemia mechanisms and evaluation.

16 Fed-fast integration Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 24 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33. · Lippincott 6e, Chapter 24, pp. 331-333 (resting skeletal muscle and brain in fasting).

17 Diabetes mellitus Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 25 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33. · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 25 (textbook/tracked-bank; current thresholds require guideline verification).

18 Obesity and energy balance Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 26 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

19 Nutrition Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 27 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

20 Vitamins Vitamin 2024, TUMS Clinical Biochemistry — pp. 4-12 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33. · Vitamin 2024 — water-soluble vitamin section, pp. 13-45 (teacher-confirmed). · Vitamin 2024 — folate p. 34; B12 pp. 37-39 (teacher-confirmed). · Vitamin 2024 — vitamin A pp. 46-55; vitamin D pp. 56-63 (teacher-confirmed). · Vitamin 2024 — vitamin C pp. 40-45; vitamins E/K pp. 63-70 (teacher-confirmed). · Lippincott 6e, Chapter 28, pp. 385-388 (vitamin D activation and actions). · Vitamin 2024.pdf, PDF pp. 59-61 (vitamin D metabolism and functions).

21 DNA structure, replication and repair Teacher: Deoxyribonucleic Acid(DNA)Structure (2).pdf, pp. 6-14 and 25-28. · Lippincott Illustrated Reviews: Biochemistry, 6e, Chapter 29, DNA primary structure and polarity. · Teacher: Deoxyribonucleic Acid(DNA)Structure (2).pdf, pp. 22-34. · Lippincott 6e, Chapter 29, DNA secondary structure, denaturation, and A/B/Z forms. · Teacher: Deoxyribonucleic Acid(DNA)Structure (2).pdf, pp. 40-43. · Lippincott 6e, Chapter 29, chromatin packaging and nucleosome organization. · Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, slides 3-11 and 53-54. · Lippincott 6e, Chapter 29, semiconservative and bidirectional replication. · Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, slides 15 onward in the fork-machinery block. · Lippincott 6e, Chapter 29, prepriming complex and topoisomerases. · Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, strand-synthesis and Okazaki-fragment slides. · Lippincott 6e, Chapter 29, DNA synthesis directionality and primer requirement. · Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, polymerase, exonuclease, and ligase block. · Lippincott 6e, Chapter 29, polymerases I/III, proofreading, and ligation. · Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, eukaryotic polymerase and telomere sections. · Lippincott 6e, Chapter 29, telomere synthesis and reverse transcriptase. · Lippincott 6e, Chapter 29, DNA repair section: mismatch, nucleotide-excision, and base-excision repair. · The teacher replication deck is titled DNA replication and the textbook supplies the repair mechanisms needed to complete that chapter outcome. · Lippincott 6e, Chapter 29, double-strand break repair by HR and NHEJ. · Lippincott Chapter 29 question extraction includes BRCA and clinical consequences of defective end joining.

22 RNA, transcription and processing Teacher: Transcription 2025.pptx, RNA-structure and RNA-type introduction. · Lippincott 6e, Chapter 30, major RNA types and eukaryotic mRNA architecture. · Teacher: Transcription 2025.pptx, bacterial RNA polymerase and promoter-initiation block. · Lippincott 6e, Chapter 30, bacterial promoter recognition and sigma factor. · Teacher: Transcription 2025.pptx, elongation and termination sections. · Lippincott 6e, Chapter 30, transcript directionality and bacterial termination mechanisms. · Teacher: Transcription 2025.pptx, eukaryotic polymerase section. · Lippincott 6e, Chapter 30, eukaryotic nuclear RNA polymerases. · Teacher: Transcription 2025.pptx, Pol II general-factor section. · Lippincott 6e, Chapter 30, TATA box, TFIID, and TFIIF. · Lippincott 6e, Chapter 30, pre-mRNA capping and polyadenylation. · Teacher: Transcription 2025.pptx includes eukaryotic transcription and RNA processing in its confirmed scope. · Lippincott 6e, Chapter 30, spliceosome, splice-site mutations, and alternative splicing. · Teacher: Transcription 2025.pptx includes eukaryotic RNA processing in the confirmed course block. · Teacher: Transcription 2025.pptx, transcription-inhibitor section. · Lippincott 6e, Chapter 30, rifampin, dactinomycin, and α -amanitin.

23 Translation Teacher: Translation 2025.pptx, slides 6 and 9-10. · Lippincott 6e, Chapter 31, genetic code and mutation effects. · Teacher: Translation 2025.pptx, slides 17-25. · Lippincott 6e, Chapter 31, tRNA charging and wobble. · Teacher: Translation 2025.pptx, slides 26, 28, and 41-42. · Lippincott 6e, Chapter 31, ribosome structure and directionality. · Teacher: Translation 2025.pptx, slides 27, 29-30, and 37-38. · Lippincott 6e, Chapter 31, prokaryotic and eukaryotic initiation. · Teacher: Translation 2025.pptx, slides 39-43. · Lippincott 6e, Chapter 31, elongation cycle and energy use. · Lippincott 6e, Chapter 31, termination, recycling, and polysomes. · Teacher: Translation 2025.pptx covers the complete translation cycle. · Lippincott 6e, Chapter 31, protein targeting to ER, nucleus, and mitochondria. · Teacher translation deck establishes translation machinery; the book supplies the medically essential destination step. · Lippincott 6e, Chapter 31, posttranslational modifications and protein degradation. · Lippincott Chapter 31 clinical examples include collagen, clotting proteins, and CFTR quality control. · Teacher: Translation 2025.pptx, slides 47-50. · Lippincott 6e, Chapter 31, inhibitors of protein synthesis.

24 Gene regulation Teacher: Regulation of Gene Expression 2025.pptx, constitutive/controllable genes, cis/trans factors, and operons. · Lippincott 6e, Chapter 32, universal transcriptional-regulation logic. · Teacher: Regulation of Gene Expression 2025.pptx, lac-operon block. · Lippincott 6e, Chapter 32, lac negative and positive regulation. · Teacher: Regulation of Gene Expression 2025.pptx, trp-operon section. · Lippincott 6e, Chapter 32, trp attenuation mechanism. · Teacher: Regulation of Gene Expression 2025.pptx, eukaryotic regulation section. · Lippincott 6e, Chapter 32, specific transcription factors and coordinate regulation. · Lippincott 6e, Chapter 32, hormone-response elements, nuclear receptors, and CREB. · Teacher regulation deck confirms multi-level eukaryotic gene regulation as exam scope. · Teacher: Regulation of Gene Expression 2025.pptx, DNA-methylation and eukaryotic-regulation content. · Lippincott 6e, Chapter 32, DNA-level and chromatin-level regulation. · Teacher: Regulation of Gene Expression 2025.pptx, posttranscriptional regulation and ApoB editing. · Lippincott 6e, Chapter 32, mRNA processing and stability. · Lippincott 6e, Chapter 32, RNA interference and eIF2-mediated translational control. · Teacher regulation deck confirms posttranscriptional and translational regulation in the course scope. · Teacher: Regulation of Gene Expression 2025.pptx, immunoglobulin rearrangement in the eukaryotic-regulation section. · Lippincott 6e, Chapter 32, gene amplification and transposons.

25 Biotechnology Lippincott Illustrated Reviews: Biochemistry, 6e, Chapter 33, PDF pp. 838-877. · Tracked bank source labels Chapter 33 as confirmed August 25 Cell & Molecules scope. · Lippincott 6e, Chapter 33, cDNA synthesis and nucleic-acid hybridization. · Confirmed August 25 Chapter 33 question bank, PDF pp. 838-877. · Lippincott 6e, Chapter 33, PCR and reverse-transcription PCR. · Lippincott 6e, Chapter 33, blotting methods. · Lippincott 6e, Chapter 33, RFLP analysis and chain-termination sequencing. · Lippincott 6e, Chapter 33, microarray and allele-specific detection methods.

26 Laboratory principles and tests Teacher: Carbohydrate Qualification Test Esmaeili Jan 2026.pdf, pp. 7-15. · Telegram scope audit identifies this 21-page file as confirmed practical biochemistry material. · Teacher: Carbohydrate Qualification Test Esmaeili Jan 2026.pdf, pp. 14-19. · Teacher-derived image spotters confirm the expected Benedict, Barfoed, and Seliwanoff appearances. · Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. · Tracked curriculum-gap bank: Practical amino acid and protein tests. No separate local teacher deck is recorded in the current Telegram scope audit. · Teacher: Laboratory Equipment Esmaeili Jan 2026.pdf, pp. 18-27. · Telegram scope audit confirms the 46-page laboratory-equipment deck. · Teacher: Laboratory Equipment Esmaeili Jan 2026.pdf, pp. 36 and 43-44. · Teacher-derived image spotters identify analytical balance, fume hood, and centrifuges. · Teacher: Titration Esmaeili Jan 2026.pdf, pp. 6-9. · Telegram scope audit confirms this 10-page practical titration file. · Tracked curriculum-gap bank: Practical enzyme assays. No separate local teacher deck is recorded in the current Telegram scope audit. · Tracked curriculum-gap bank: Spectrophotometry. No separate local teacher deck is recorded in the current Telegram scope audit. · Tracked curriculum-gap bank: DNA extraction. No separate local teacher deck is recorded in the current Telegram scope audit. · Tracked curriculum-gap bank: Chromatography. No separate local teacher deck is recorded in the current Telegram scope audit. · Tracked curriculum-gap bank: Flame photometry. No separate local teacher deck is recorded in the current Telegram scope audit. · Tracked curriculum-gap bank: Osmosis practical. No separate local teacher deck is recorded in the current Telegram scope audit.